# GREYBULL
# REV 1.0
# FAB M
# RETAIL

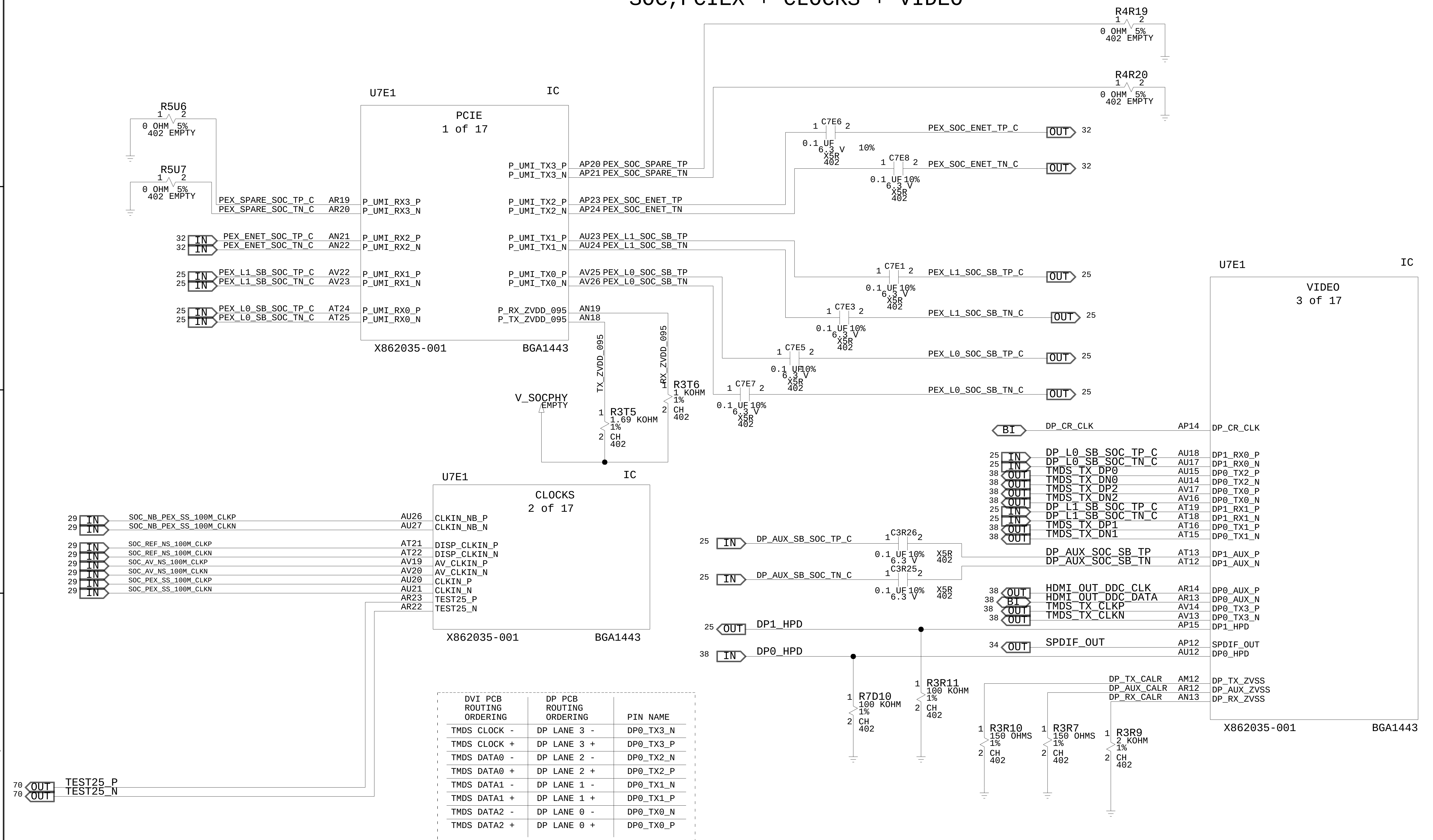


RULES: (APPLIED WHEN POSSIBLE)
1.) MSB TO LSB IS TOP TO BOTTOM
2.) WHEN POSSIBLE: INPUTS ON LEFT, OUTPUTS ON RIGHT
3.) ORDER OF PAGES=CHIP INTERFACES, TERMINATION, POWER, DECOUPLING
4.) AVOID USING OFF PAGE CONNECTORS FOR ON PAGE CONNECTIONS
5.) LANED SIGNALS ARE GROUPED ON SYMBOLS
6.) TRANSIMITTER NAME USED AS PREFIX WITH RX AND TX CONNECTIONS
7.) SUFFIX V_ IS USED FOR VOLTAGE RAIL SIGNAL NAMES
8.) SUFFIX _DP AND _DN ARE USED FOR DIFFERIENTAL PAIRS
9.) UNNAMED NETS ARE NAMED WITH /2 TEXT SIZE
10.) SUFFIX _N FOR ACTIVE LOW OR N JUNCTION
12.) SUFFIX _P FOR P JUNCTION
13.) SUFFIX _EN FOR ENABLE
14.) 'CLK' FOR CLOCKS, 'RST' FOR RESETS
15.) PWRGD FOR POWER GOOD
16.) REV AND FAB ARE SET USING CUSTOM VARIABLES
TOOLS>OPTIONS>VARIABLES

[PAGE_TITLE=COVER PAGE]

| DRAWING | MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|---|
| Fri May 10 13:22:55 2013 | | GREYBULL_RETAIL | 1/72 | 1/72 | M | 1.0 |

# SOC,PCIEX + CLOCKS + VIDEO




| DVI PCB ROUTING ORDERING | DP PCB ROUTING ORDERING | PIN NAME |
|---|---|---|
| TMDS CLOCK - | DP LANE 3 - | DP0_TX3_N |
| TMDS CLOCK + | DP LANE 3 + | DP0_TX3_P |
| TMDS DATA0 - | DP LANE 2 - | DP0_TX2_N |
| TMDS DATA0 + | DP LANE 2 + | DP0_TX2_P |
| TMDS DATA1 - | DP LANE 1 - | DP0_TX1_N |
| TMDS DATA1 + | DP LANE 1 + | DP0_TX1_P |
| TMDS DATA2 - | DP LANE 0 - | DP0_TX0_N |
| TMDS DATA2 + | DP LANE 0 + | DP0_TX0_P |

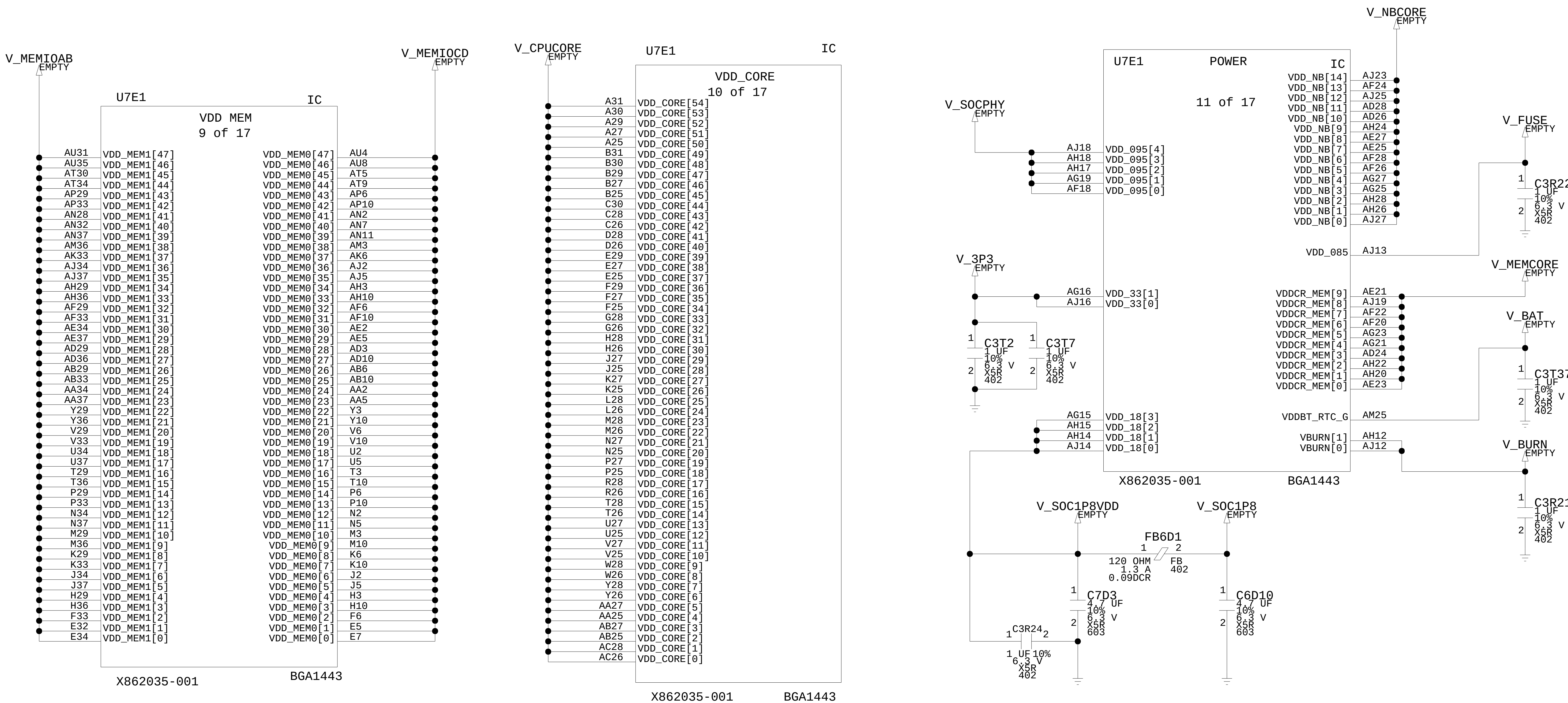
[PAGE_TITLE=SOC PAGE 1]

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U7E1 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

DRAWING
Tue Jun 18 16:42:23 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 2/72 | 2/72 | M | 1.0 |

# SOC POWER, MEM + CPUCORE + MEMCORE + NBCORE + MISC

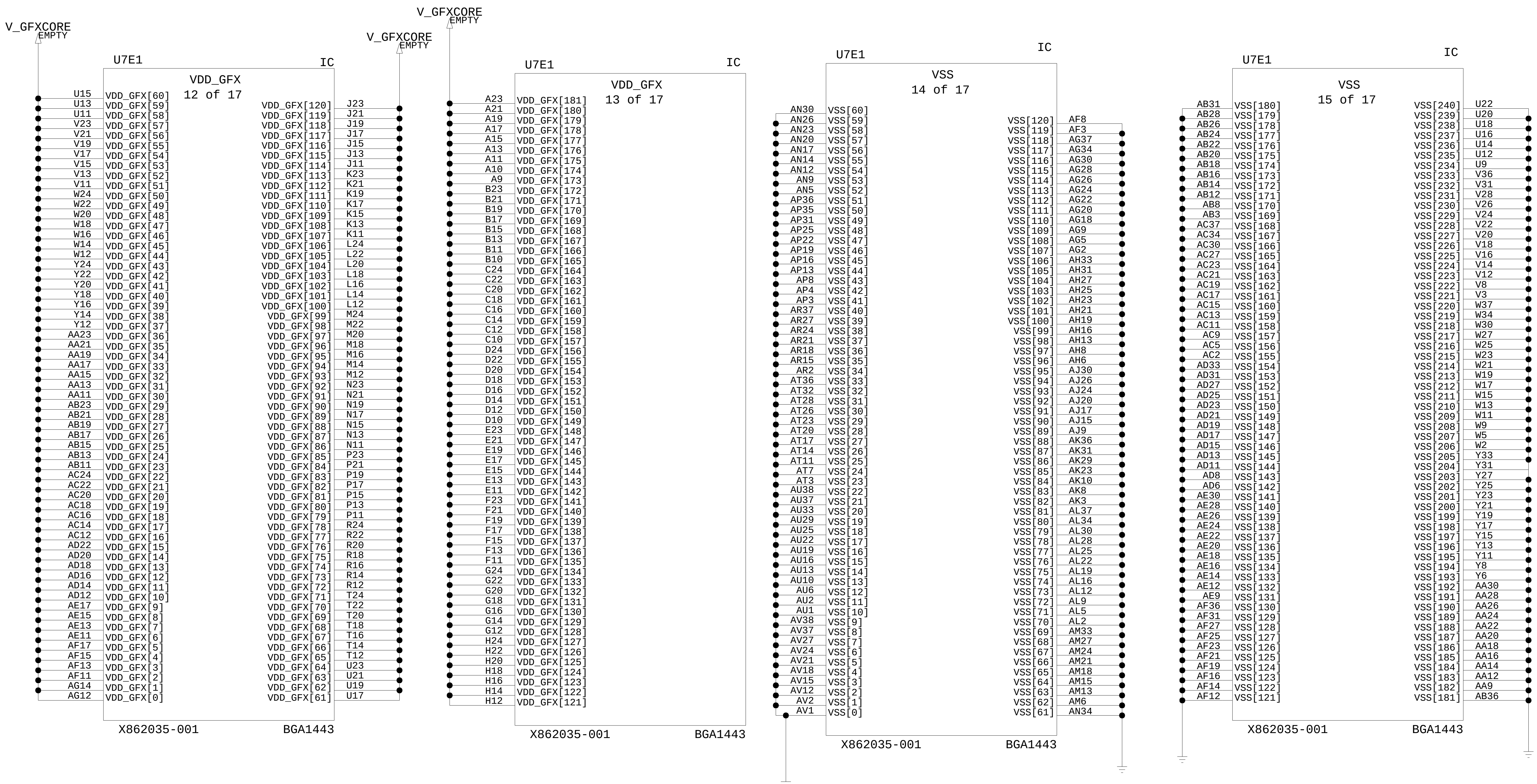



[PAGE_TITLE=SOC PAGE 2]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 3/72 | 3/72 | M | 1.0 |

DRAWING
Tue Jun 18 16:42:23 2013

# SOC POWER, V_GFXCORE + VSS

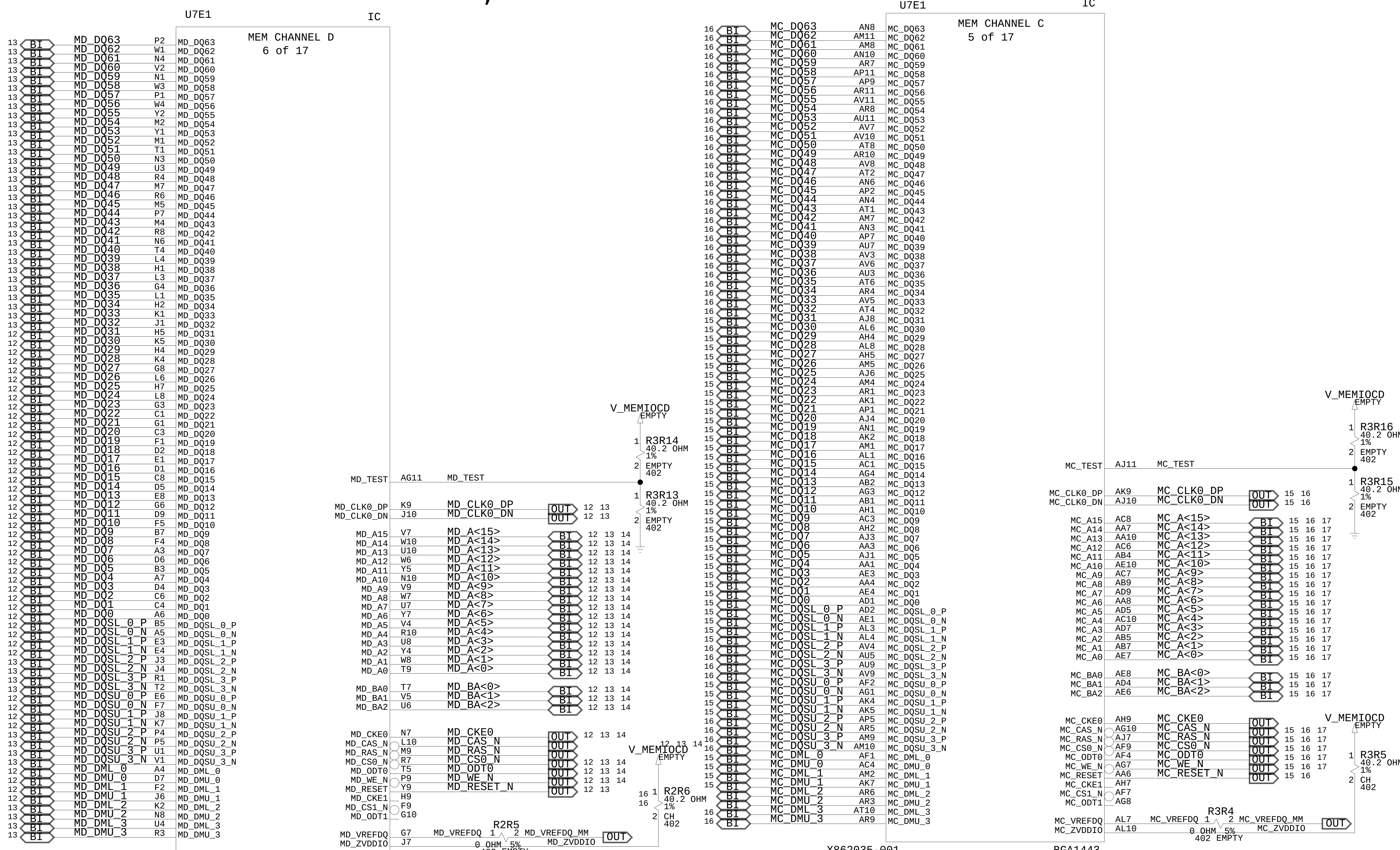



[PAGE_TITLE=SOC PAGE 3]

DRAWING
Tue Jun 18 16:42:24 2003

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 4/72 | 4/72 | M | 1.0 |

# SOC, MEMORY PARTITION C + D

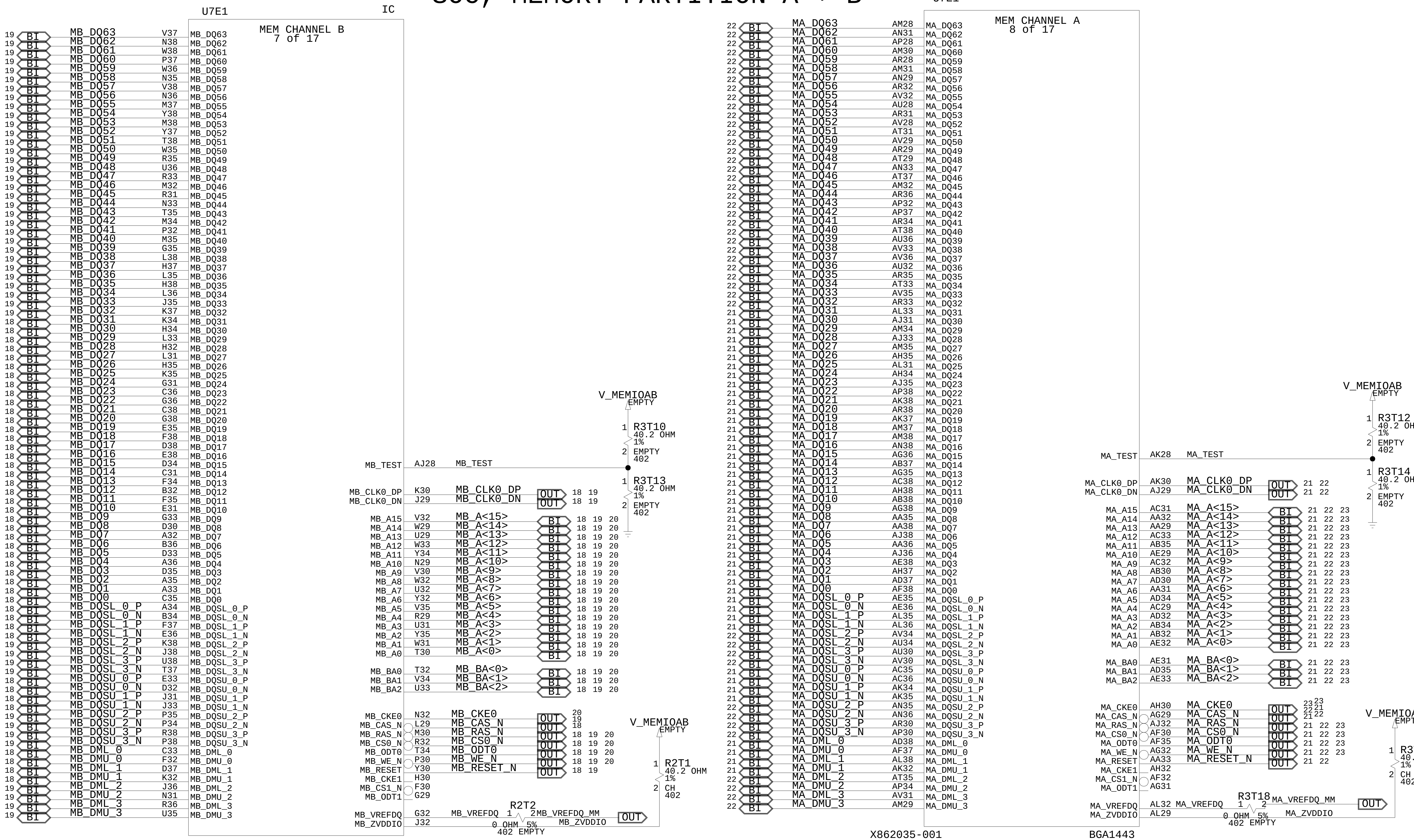



[PAGE_TITLE=SOC MEMORY PARTITION A & B]

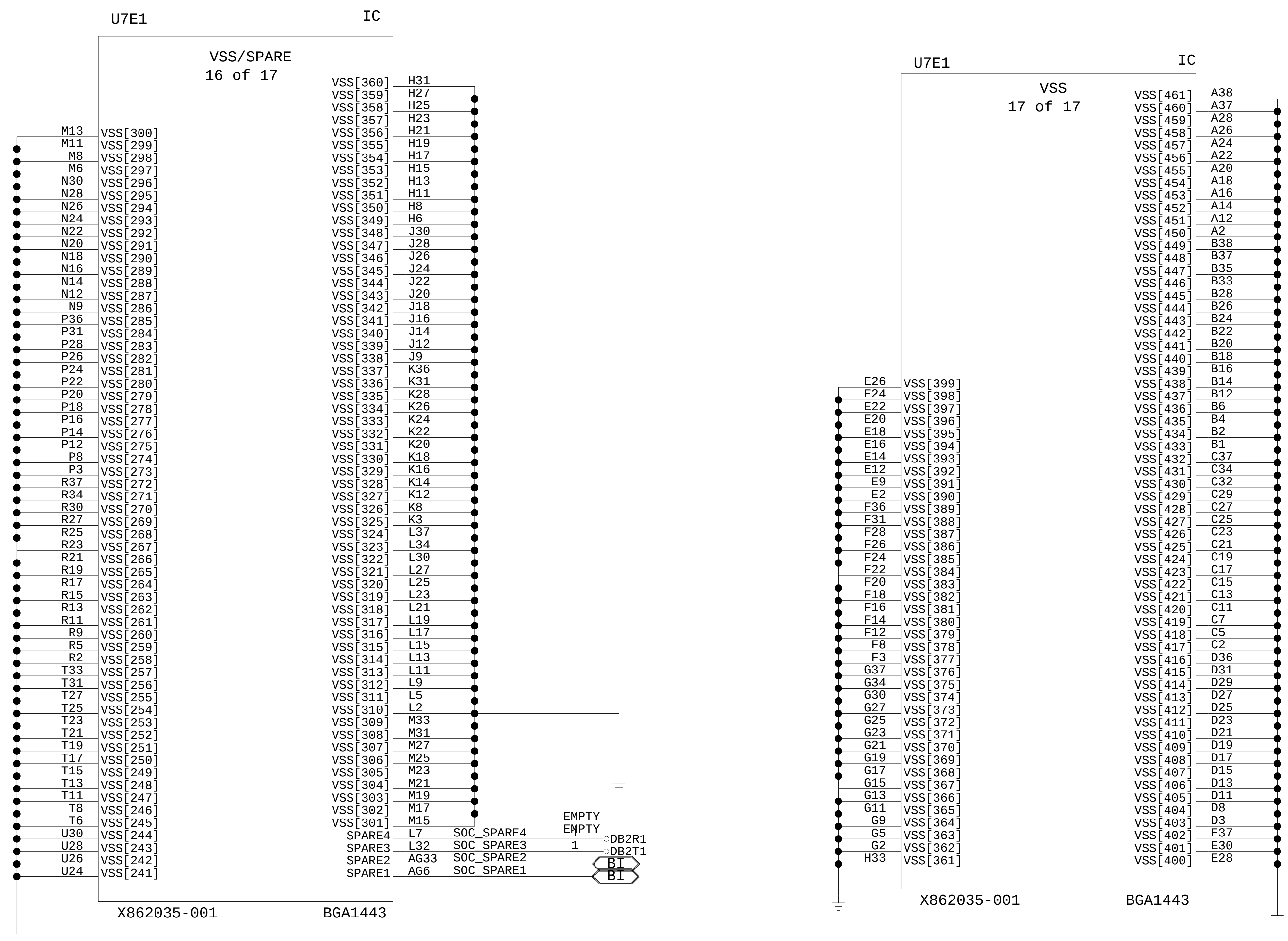
DRAWING
Tue Jun 18 16:42:18 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 5/72 | 5/72 | M | 1.0 |

# SOC, MEMORY PARTITION A + B



# SOC VSS + SPARE

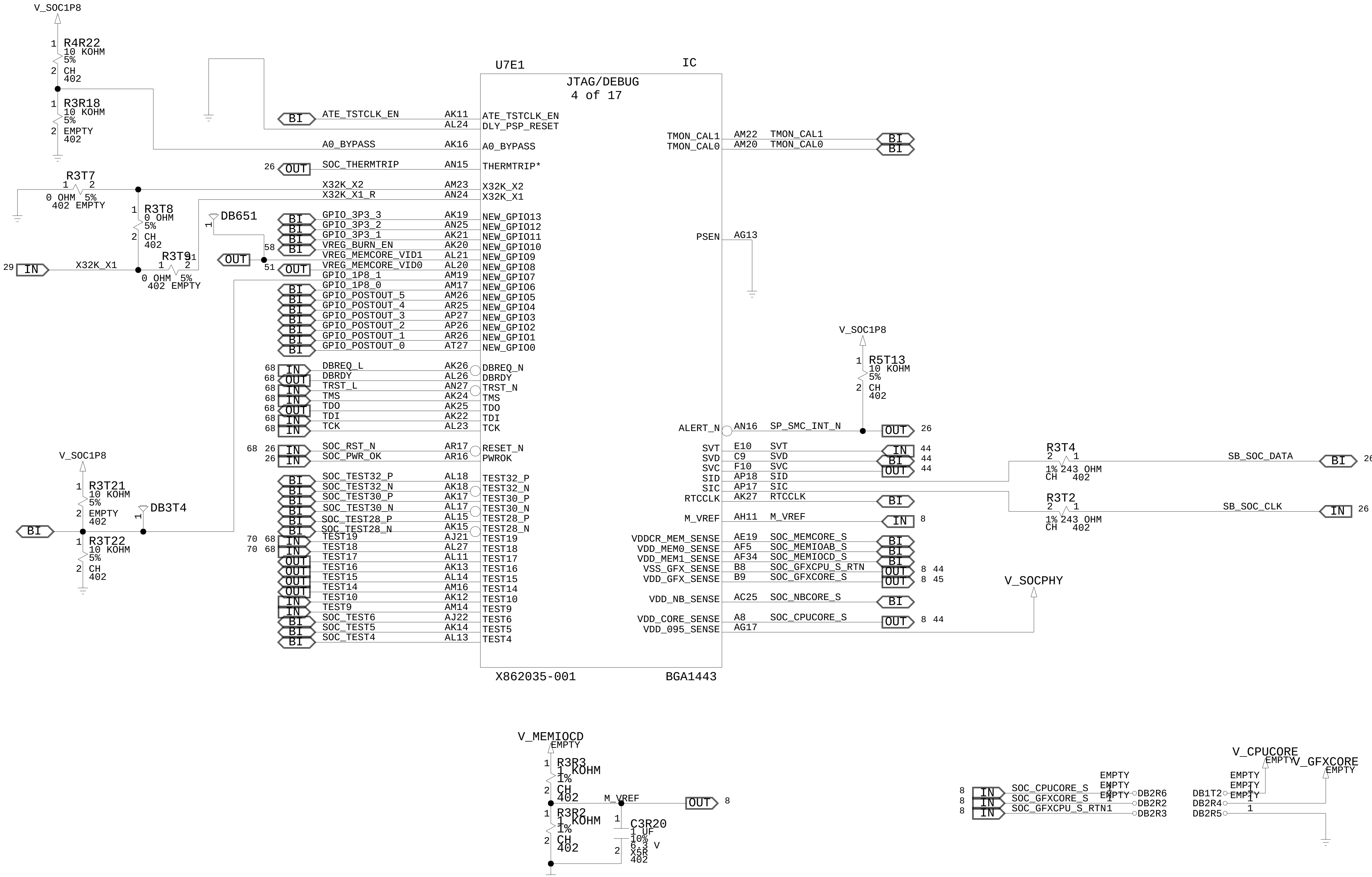



[PAGE_TITLE=SOC PAGE 4]

DRAWING
Tue Jun 18 16:42:24 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 7/72 | 7/72 | M | 1.0 |

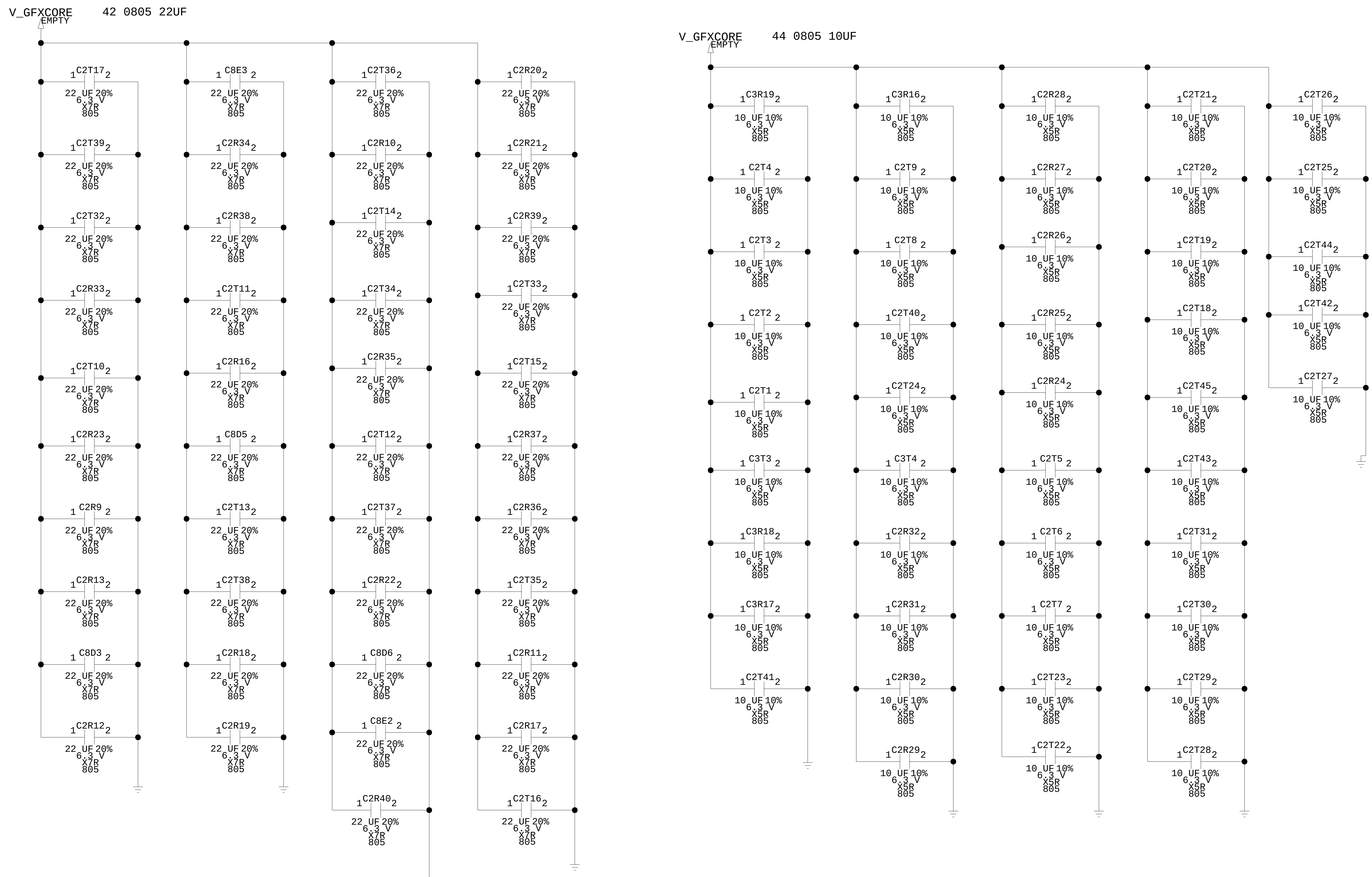
# SOC DEBUG + SB SIGNALS




[PAGE_TITLE=SOC PAGE 4]

| DRAWING | MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|---|
| Tue Jun 18 16:42:29 2013 | | GREYBULL_RETAIL | 8/72 | 8/72 | M | 1.0 |

# SOC, DECOUPLING

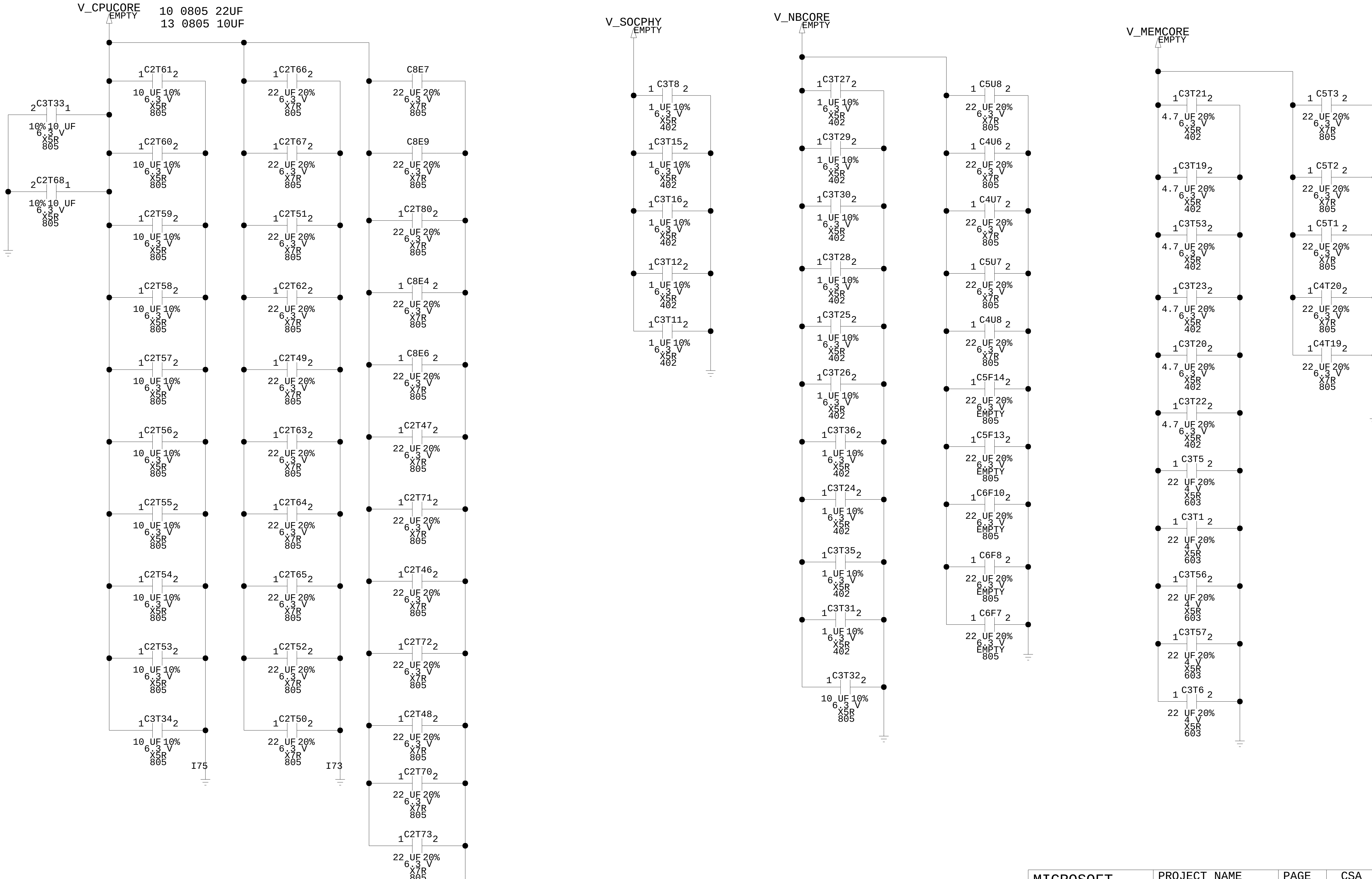



[PAGE_TITLE=GCPU, DECOUPLING]

DRAWING
Tue Jun 18 16:42:26 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 9/72 | 9/72 | M | 1.0 |

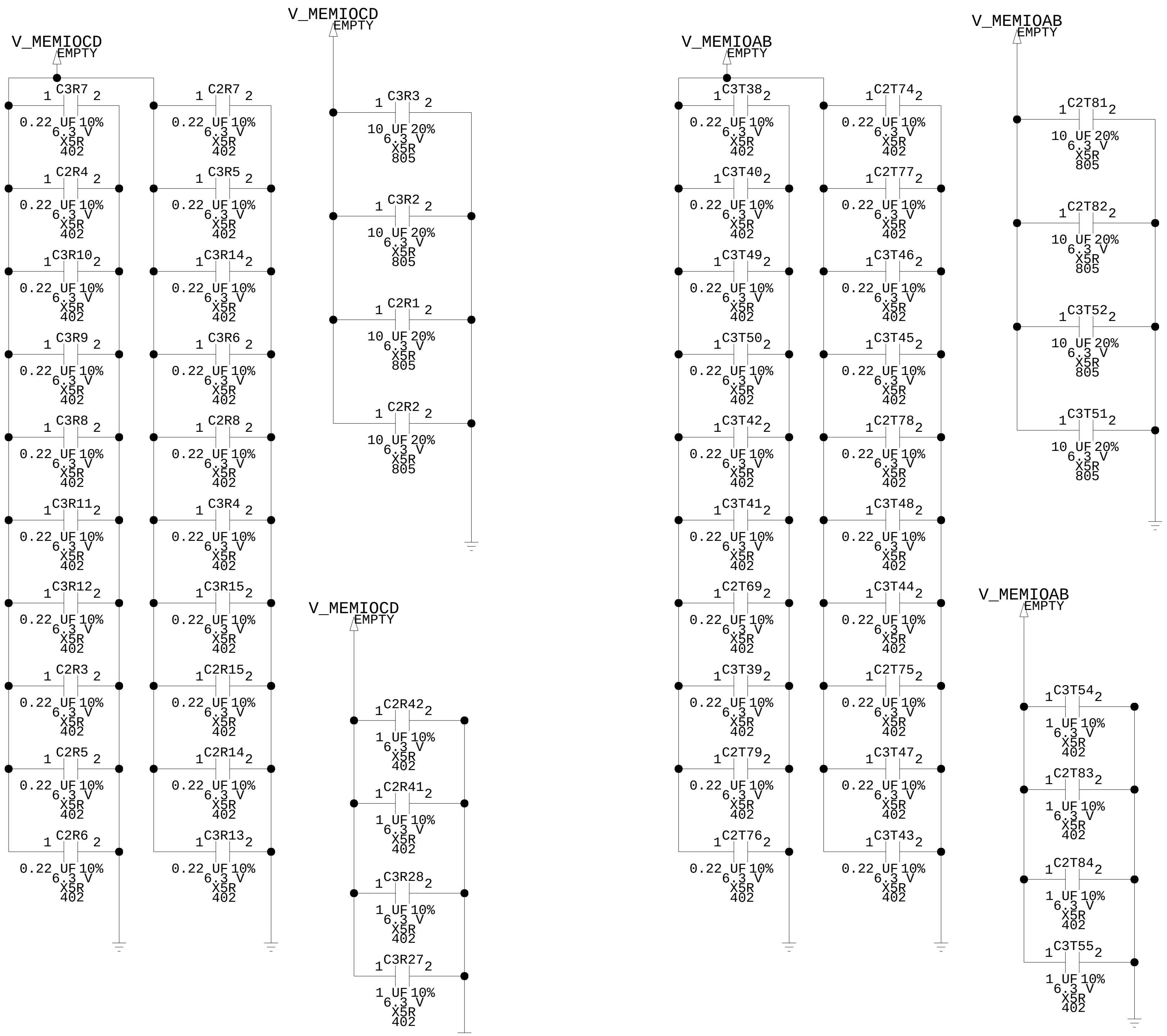
# SOC, DECOUPLING




[PAGE_TITLE=GCPU, DECOUPLING]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 10/72 | 10/72 | M | 1.0 |

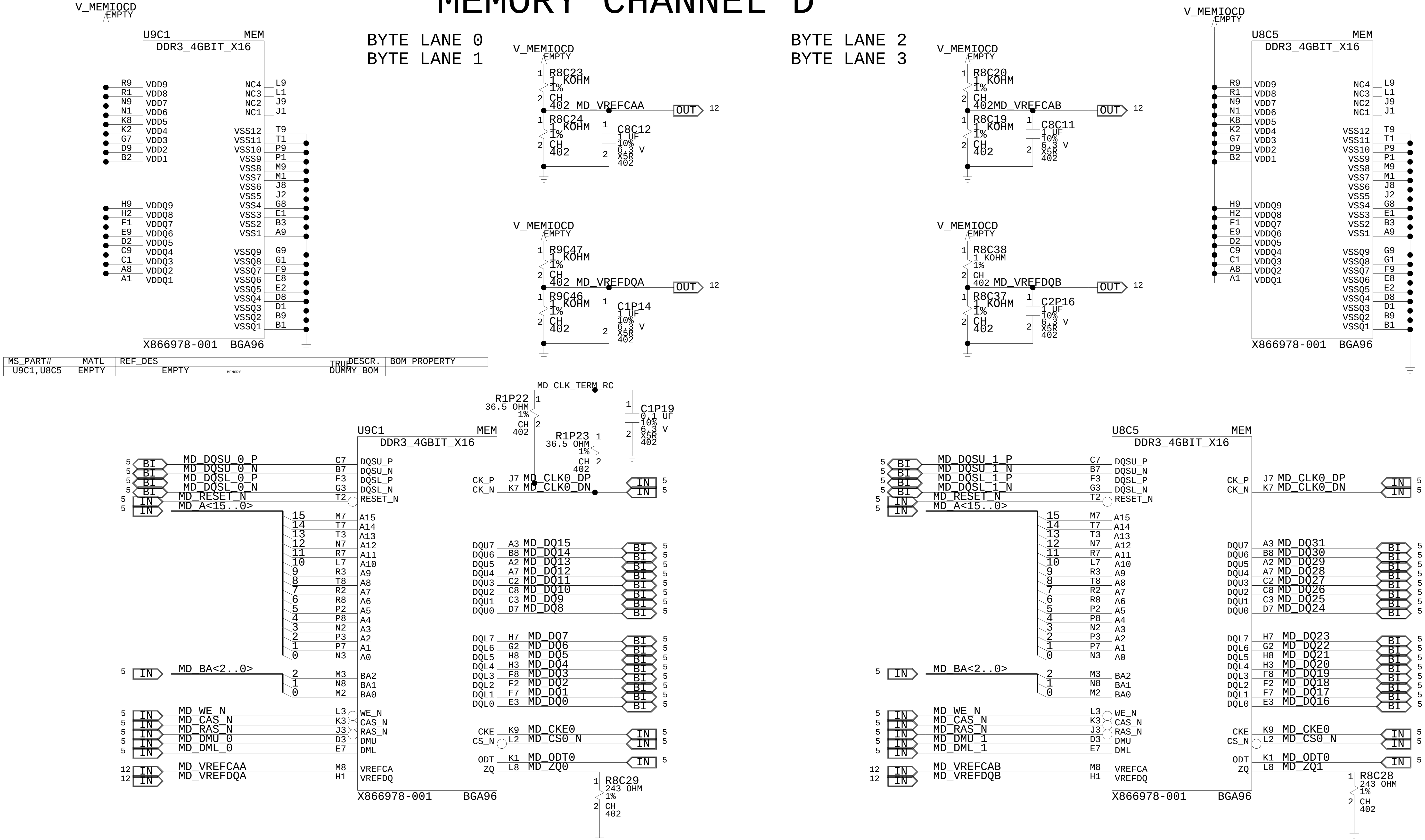
# SOC, DECOUPLING




[PAGE_TITLE=GCPU, DECOUPLING]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 11/72 | 11/72 | M | 1.0 |

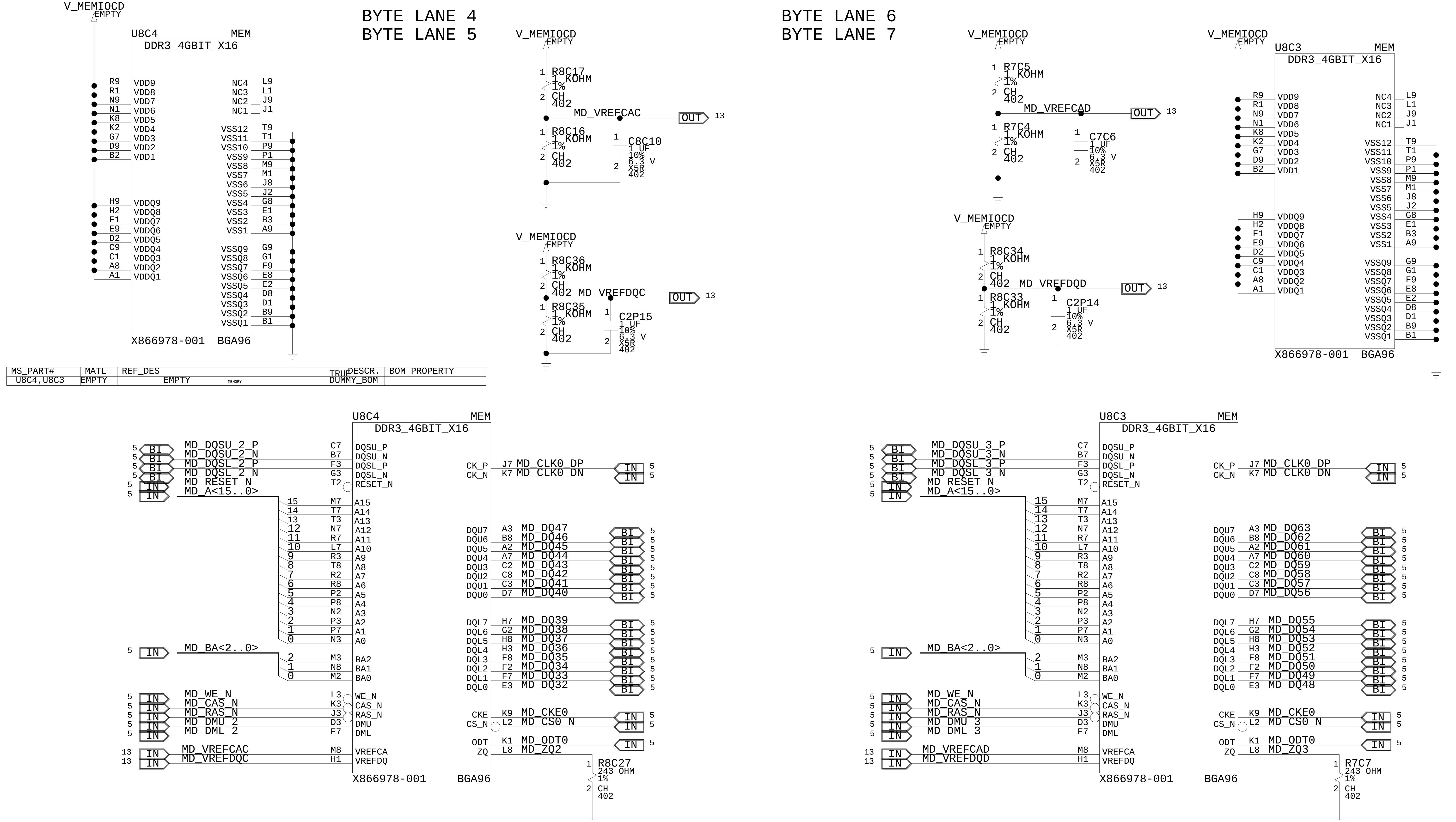
# MEMORY CHANNEL D




[PAGE_TITLE=MEMORY PARTITION A, LOW]

| DRAWING | MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|---|
| Tue Jun 18 16:42:22 2013 | | GREYBULL_RETAIL | 12/72 | 12/72 | M | 1.0 |

# MEMORY CHANNEL D

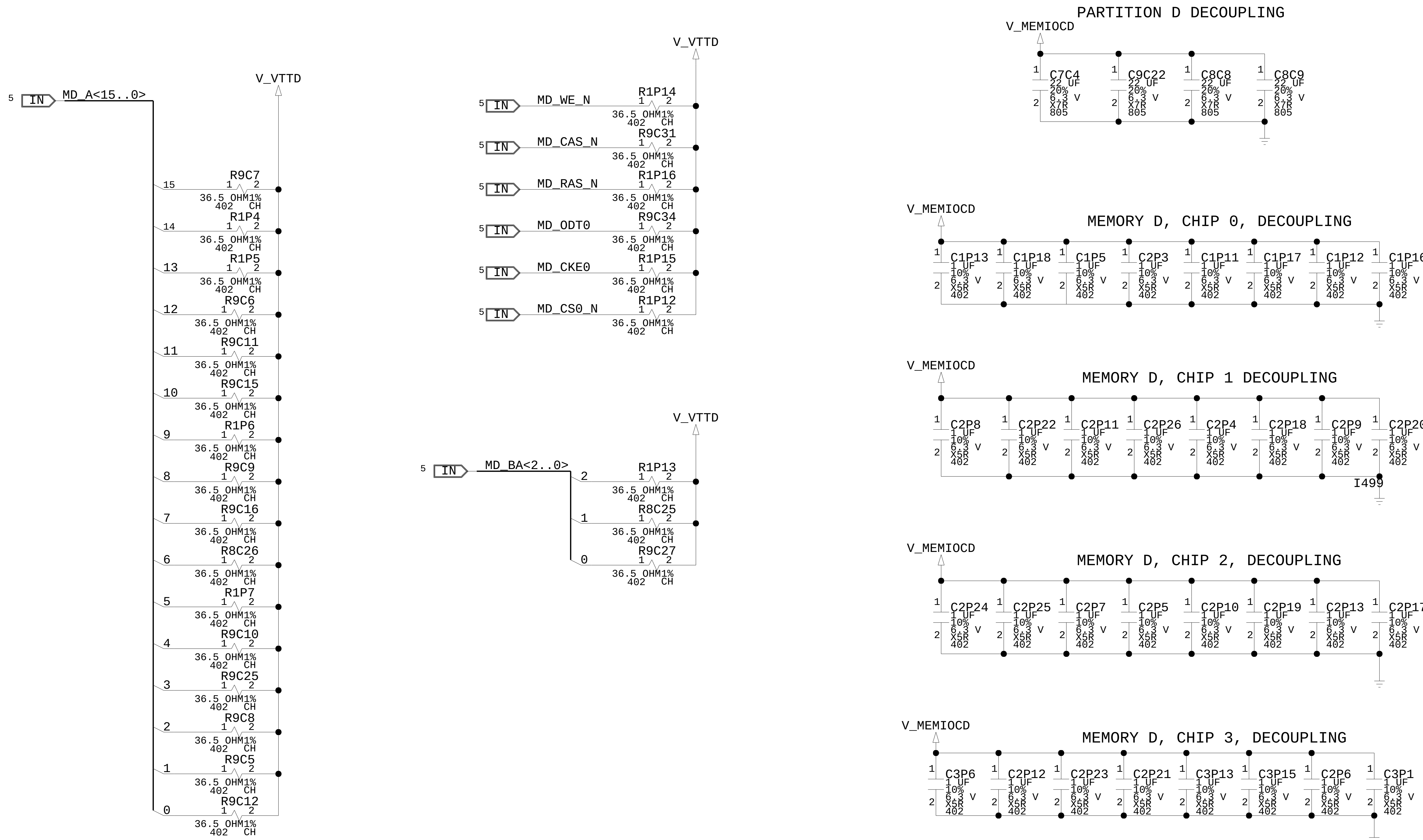



| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U8C4,U8C3 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

[PAGE_TITLE=MEMORY PARTITION A, HIGH]

| DRAWING | MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|---|
| Tue Jun 18 16:42:22 2013 | | GREYBULL_RETAIL | 13/72 | 13/72 | M | 1.0 |

# MEMORY CHANNEL D, DECOUPLING + TERMINATION

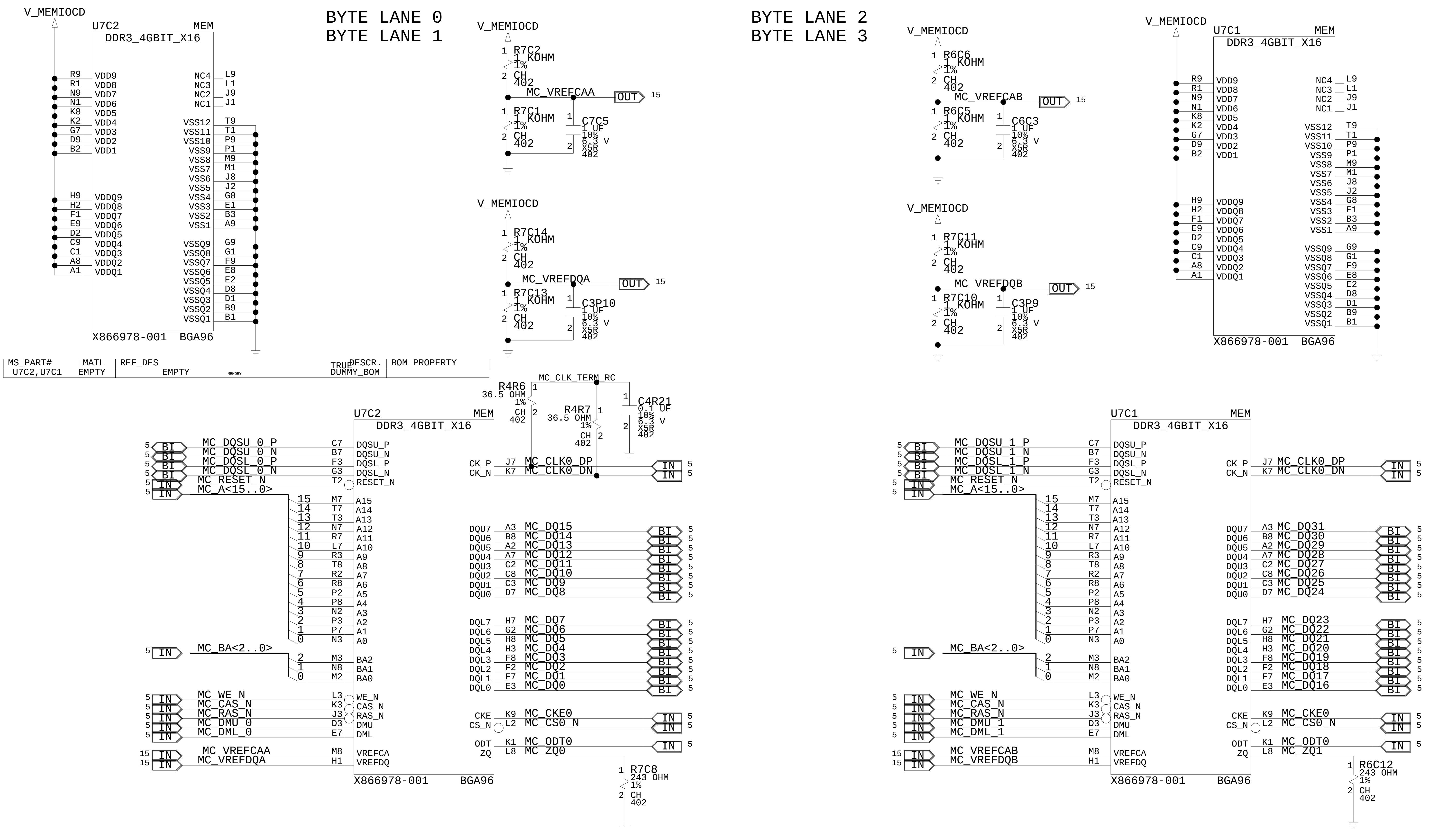



[PAGE_TITLE=MEMORY PARTITION A, DECOUPLING, TERMINATION]

DRAWING
Tue Jun 18 16:42:22 2003

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 14/72 | 14/72 | M | 1.0 |

# MEMORY CHANNEL C




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U7C2,U7C1 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

[PAGE_TITLE=MEMORY PARTITION B, LOW]

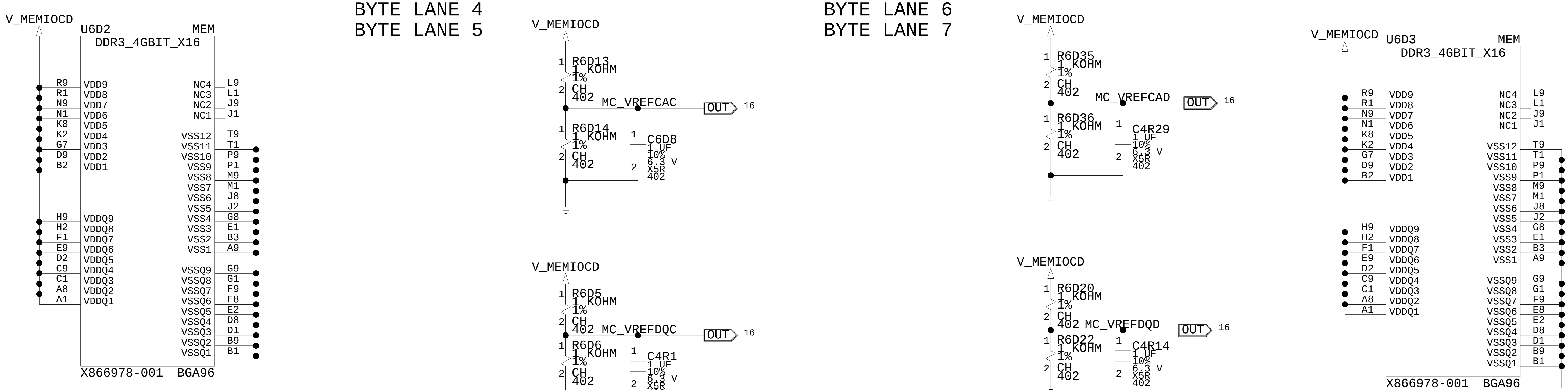
| DRAWING | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| MICROSOFT CONFIDENTIAL | GREYBULL_RETAIL | 15/72 | 15/72 | M | 1.0 |

Tue Jun 18 16:42:23 2013

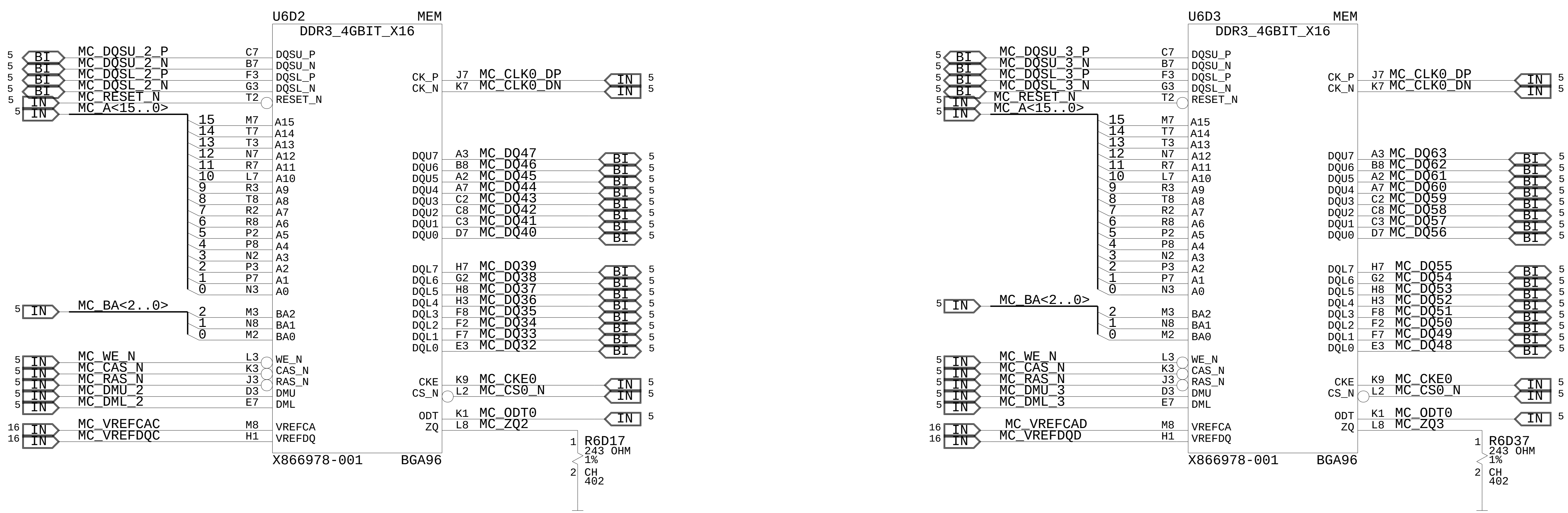
# MEMORY CHANNEL C

BYTE LANE 4
BYTE LANE 5

BYTE LANE 6
BYTE LANE 7




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U6D2,U6D3 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

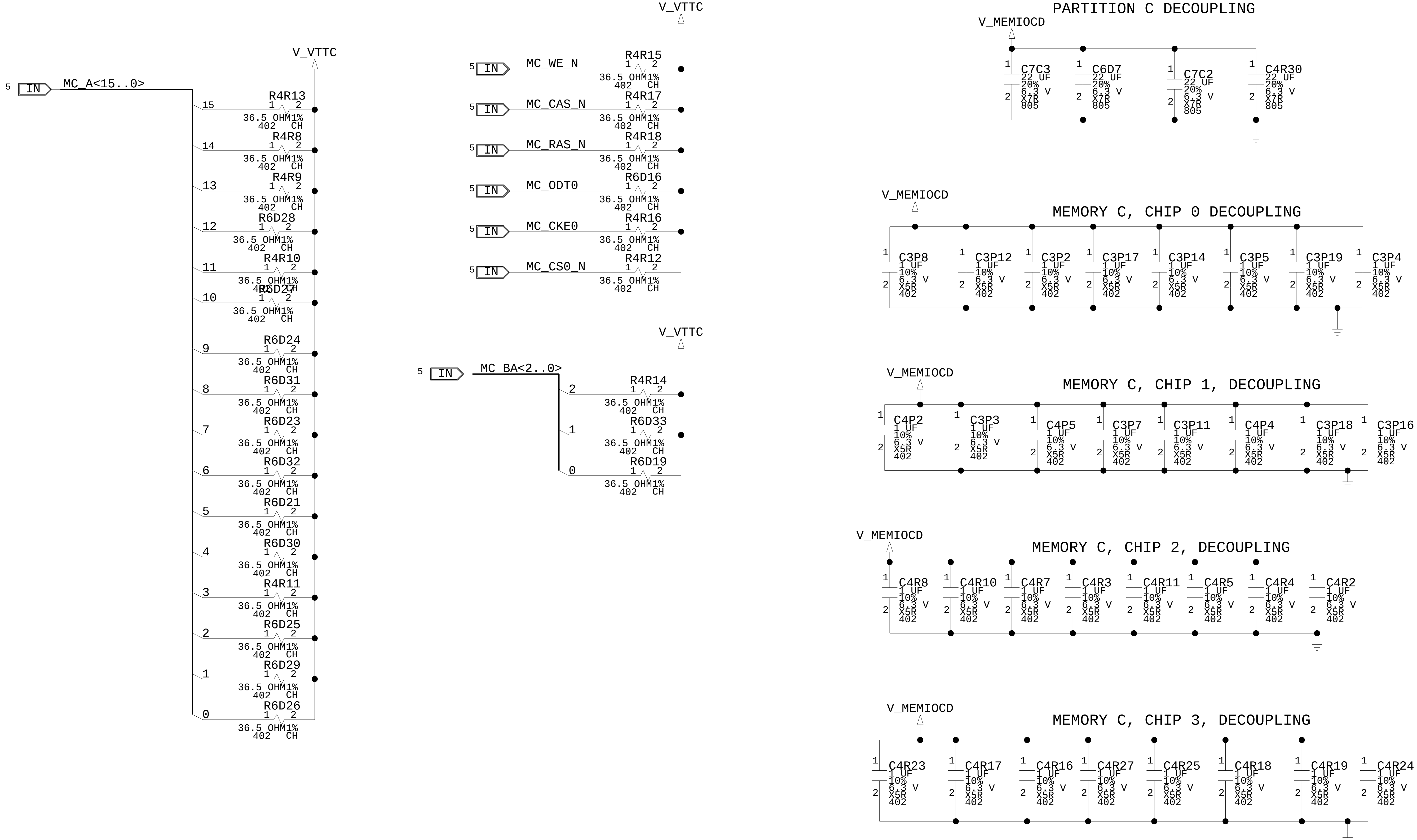



[PAGE_TITLE=MEMORY PARTITION B, HIGH]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 16/72 | 16/72 | M | 1.0 |

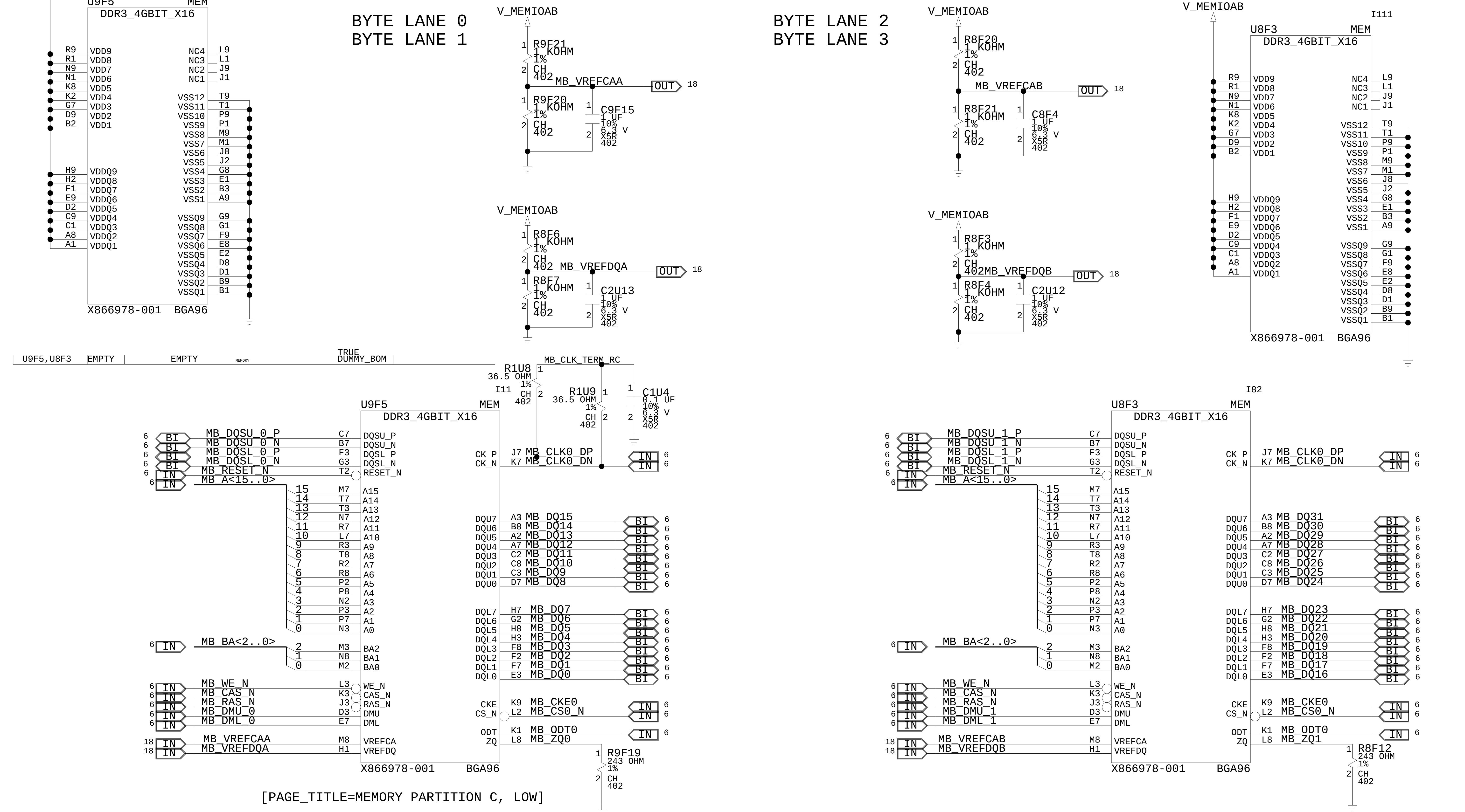
DRAWING
Tue Jun 18 16:42:23 2013

# MEMORY CHANNEL C, DECOUPLING + TERMINATION




[PAGE_TITLE=MEMORY PARTITION B, DECOUPLING, TERMINATION]

# MEMORY CHANNEL B

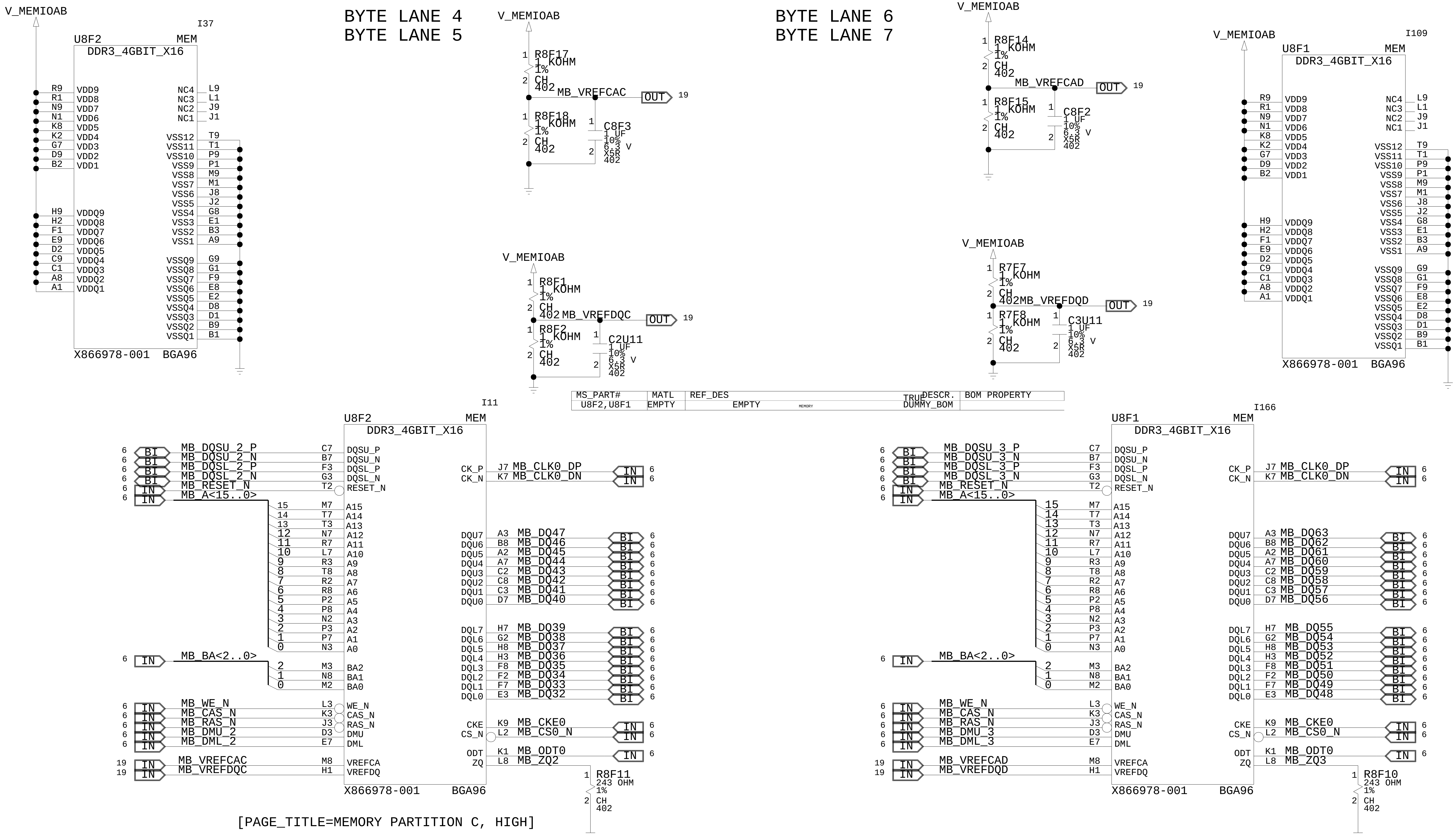



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 18/72 | 18/72 | M | 1.0 |

# MEMORY CHANNEL B

BYTE LANE 4
BYTE LANE 5

BYTE LANE 6
BYTE LANE 7

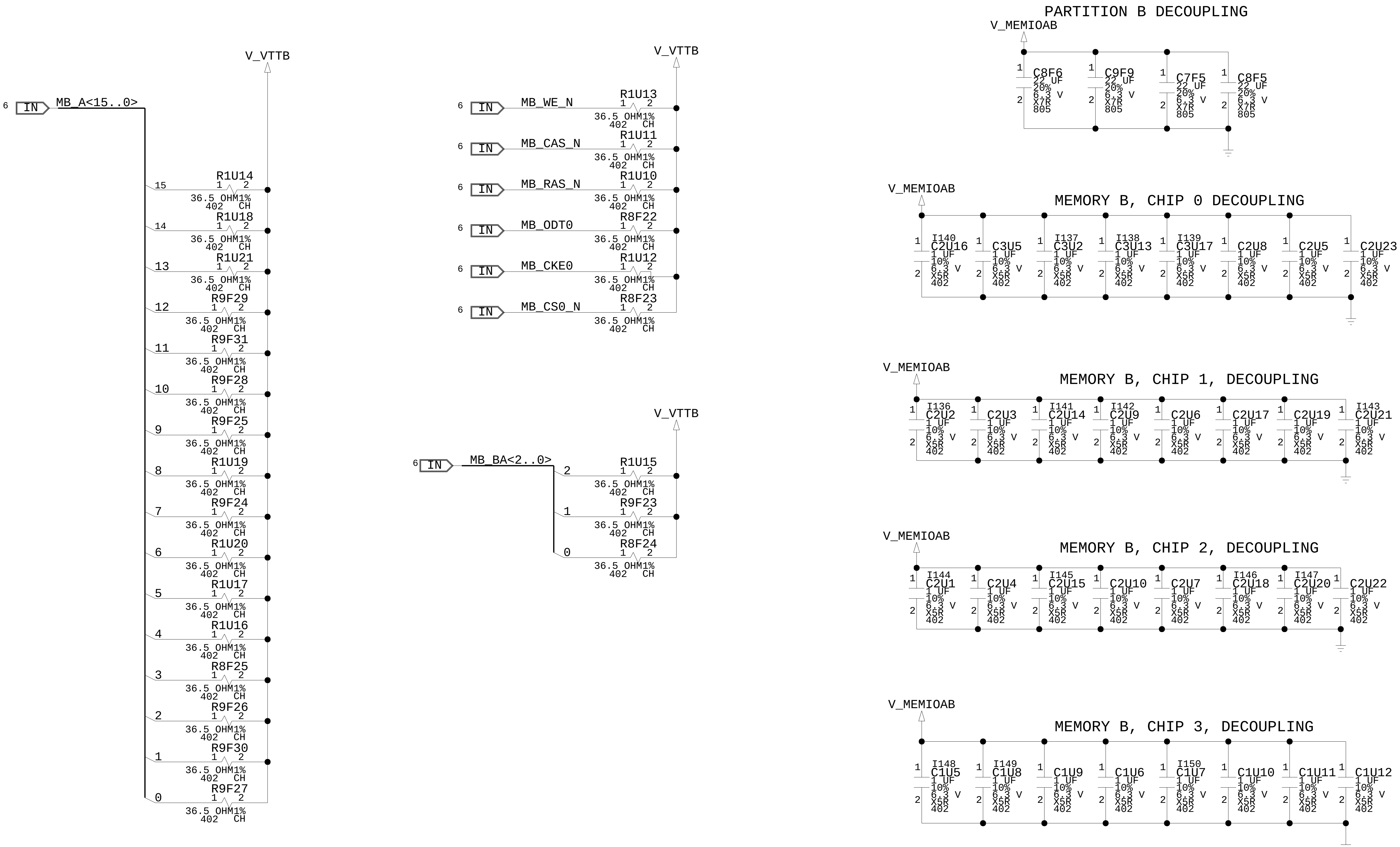



| MS_PART# | MATL | REF_DES | TRUE DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U8F2,U8F1 | EMPTY | EMPTY | DUMMY_BOM | |

[PAGE_TITLE=MEMORY PARTITION C, HIGH]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 19/72 | 19/72 | M | 1.0 |

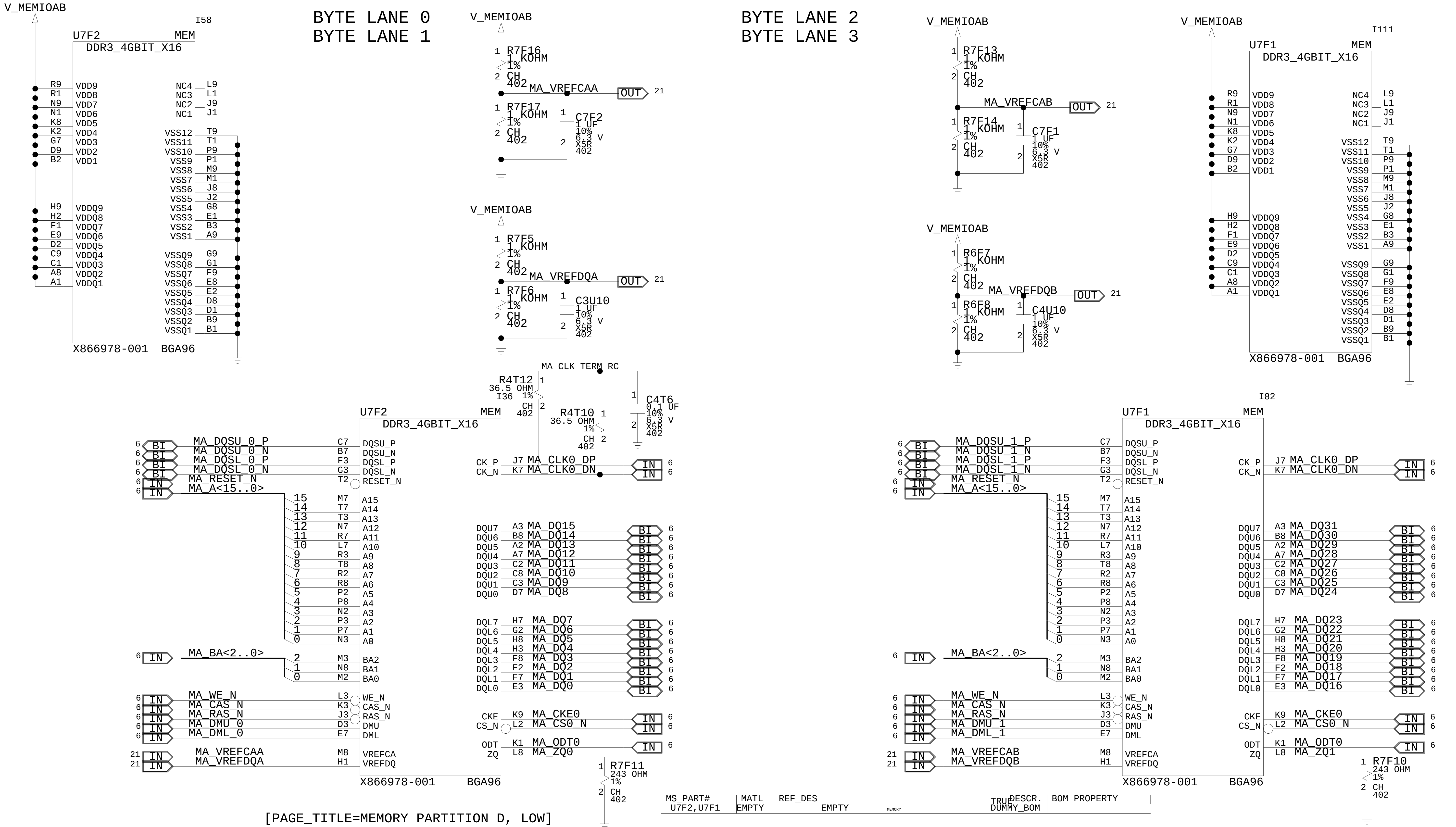
# MEMORY CHANNEL B, DECOUPLING + TERMINATION




[PAGE_TITLE=MEMORY PARTITION C, DECOUPLING, TERMINATION]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 20/72 | 20/72 | M | 1.0 |

# MEMORY CHANNEL A




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U7F2,U7F1 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

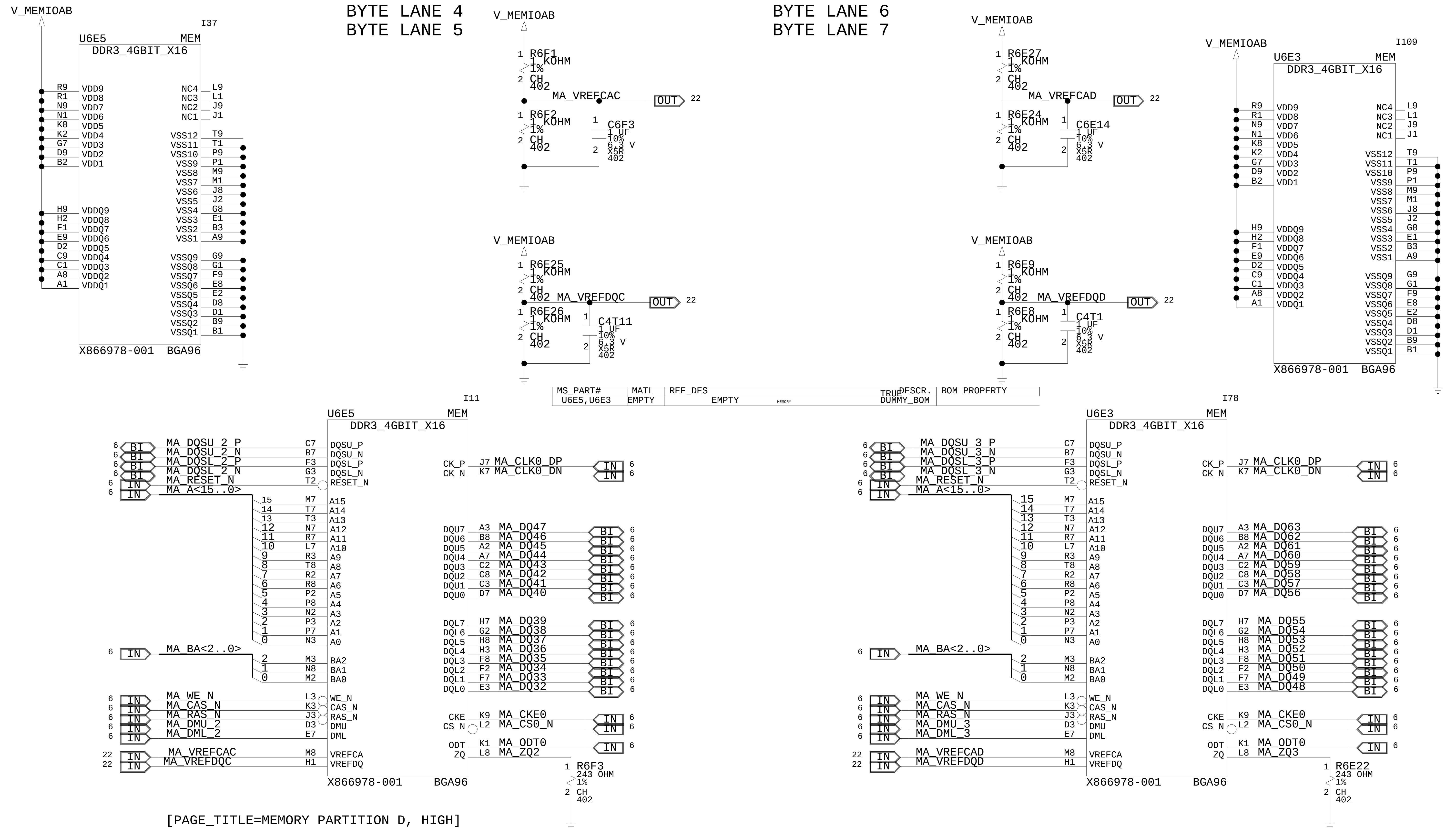
[PAGE_TITLE=MEMORY PARTITION D, LOW]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 21/72 | 21/72 | M | 1.0 |

# MEMORY CHANNEL A

BYTE LANE 4
BYTE LANE 5

BYTE LANE 6
BYTE LANE 7

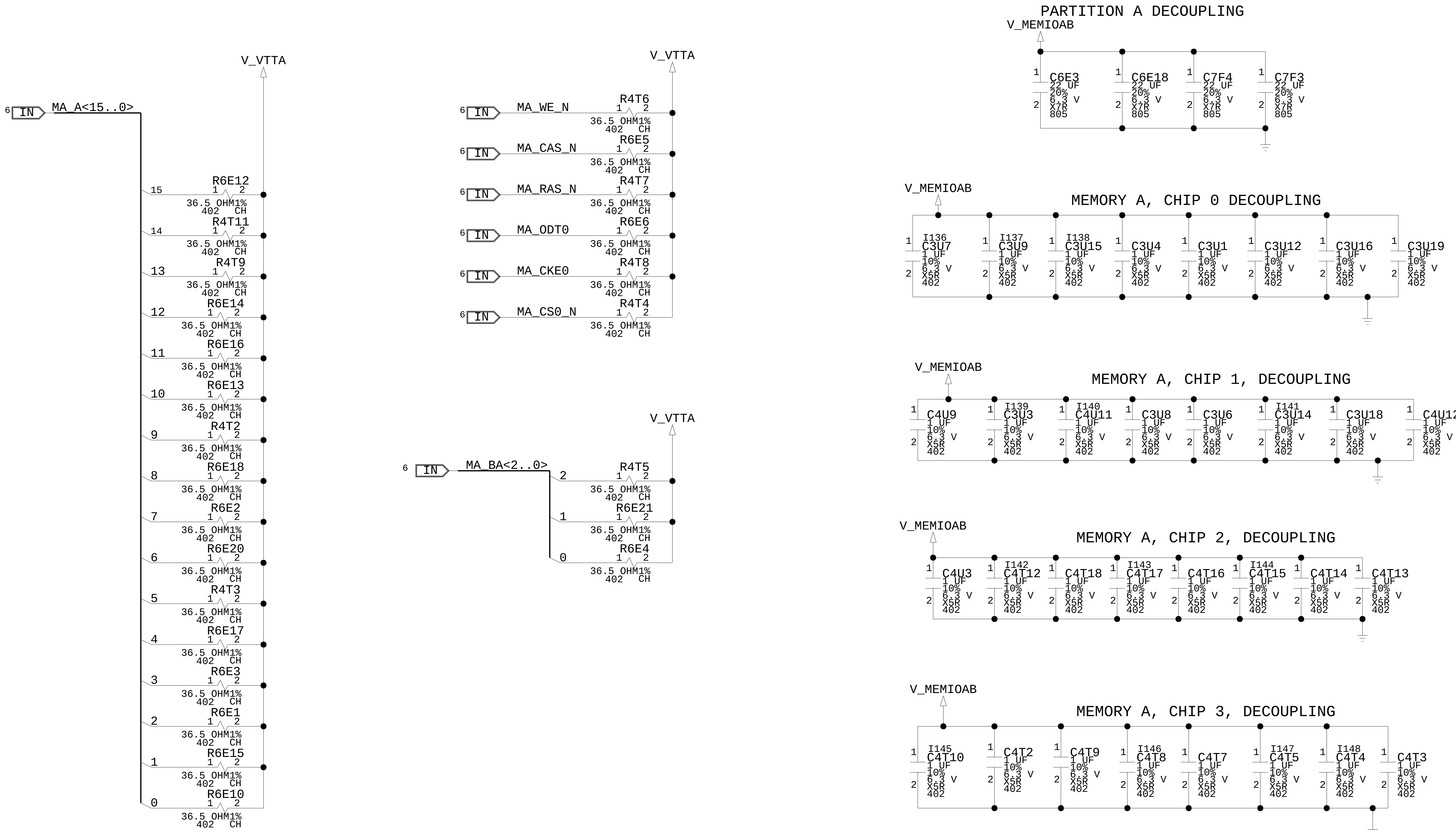



| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U6E5,U6E3 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

[PAGE_TITLE=MEMORY PARTITION D, HIGH]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 22/72 | 22/72 | M | 1.0 |

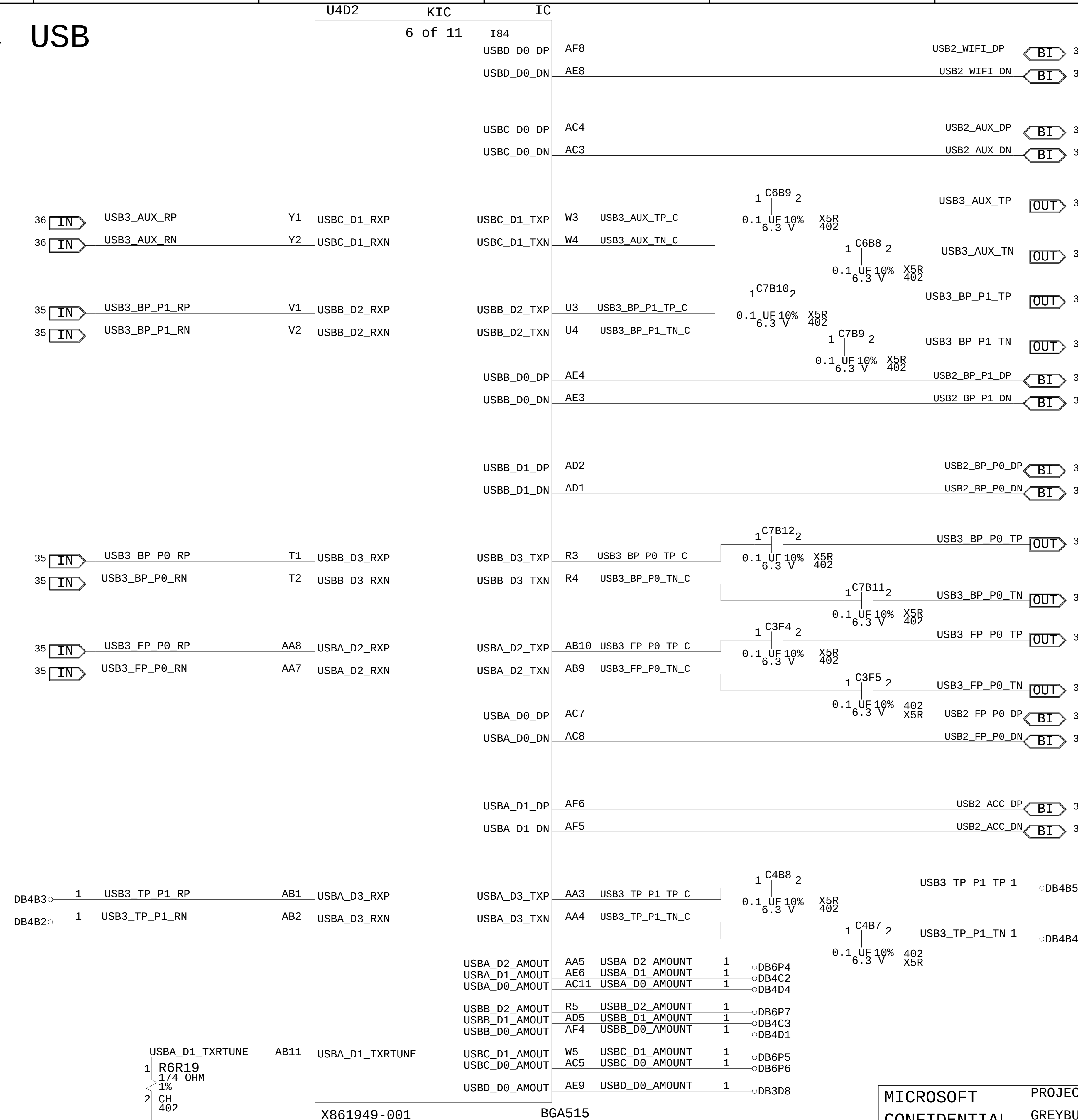
# MEMORY CHANNEL A, DECOUPLING + TERMINATION




[PAGE_TITLE=MEMORY PARTITION D, DECOUPLING, TERMINATION]

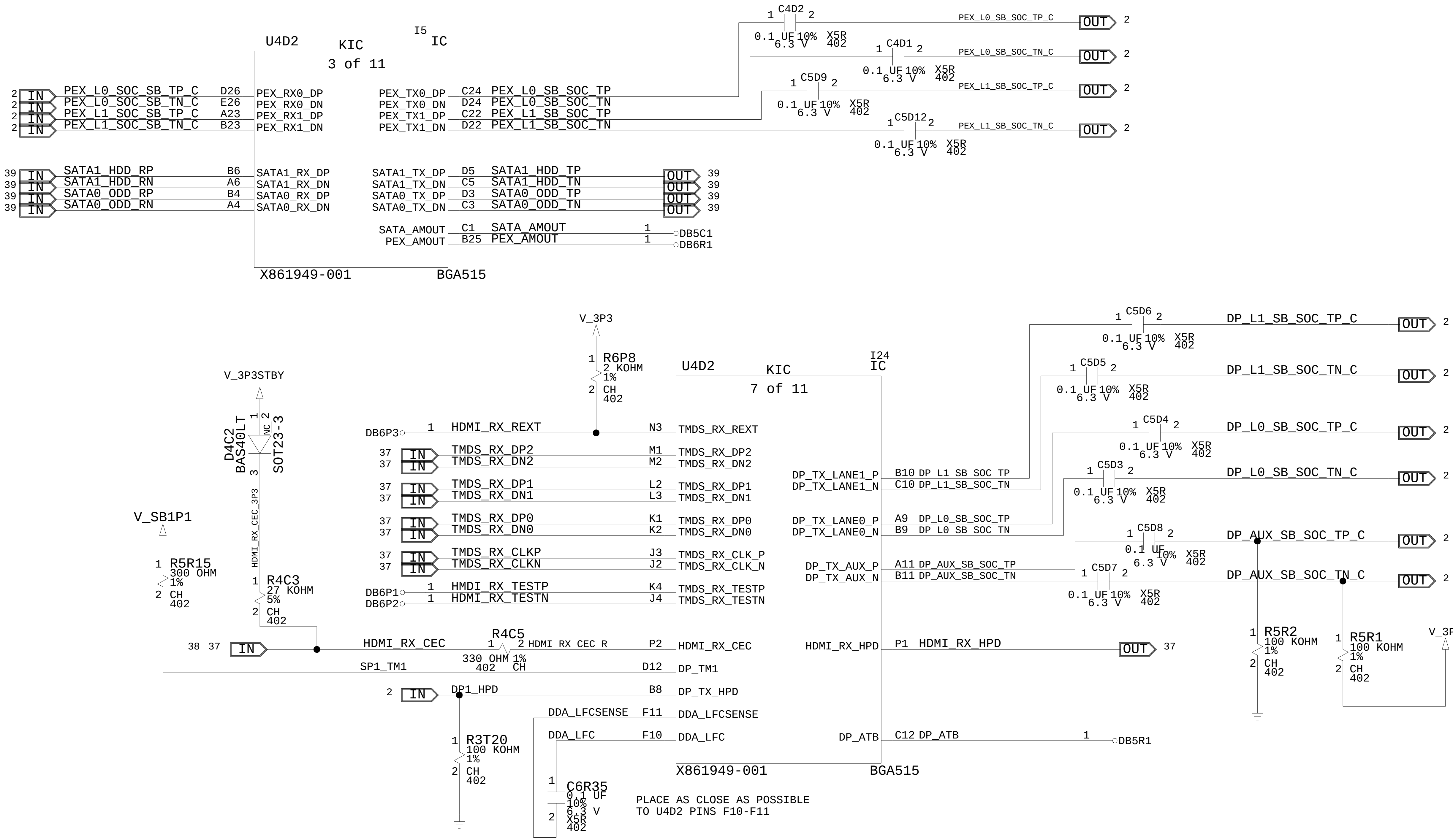
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 23/72 | 23/72 | M | 1.0 |

# KIC, USB




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 24/72 | 24/72 | M | 1.0 |

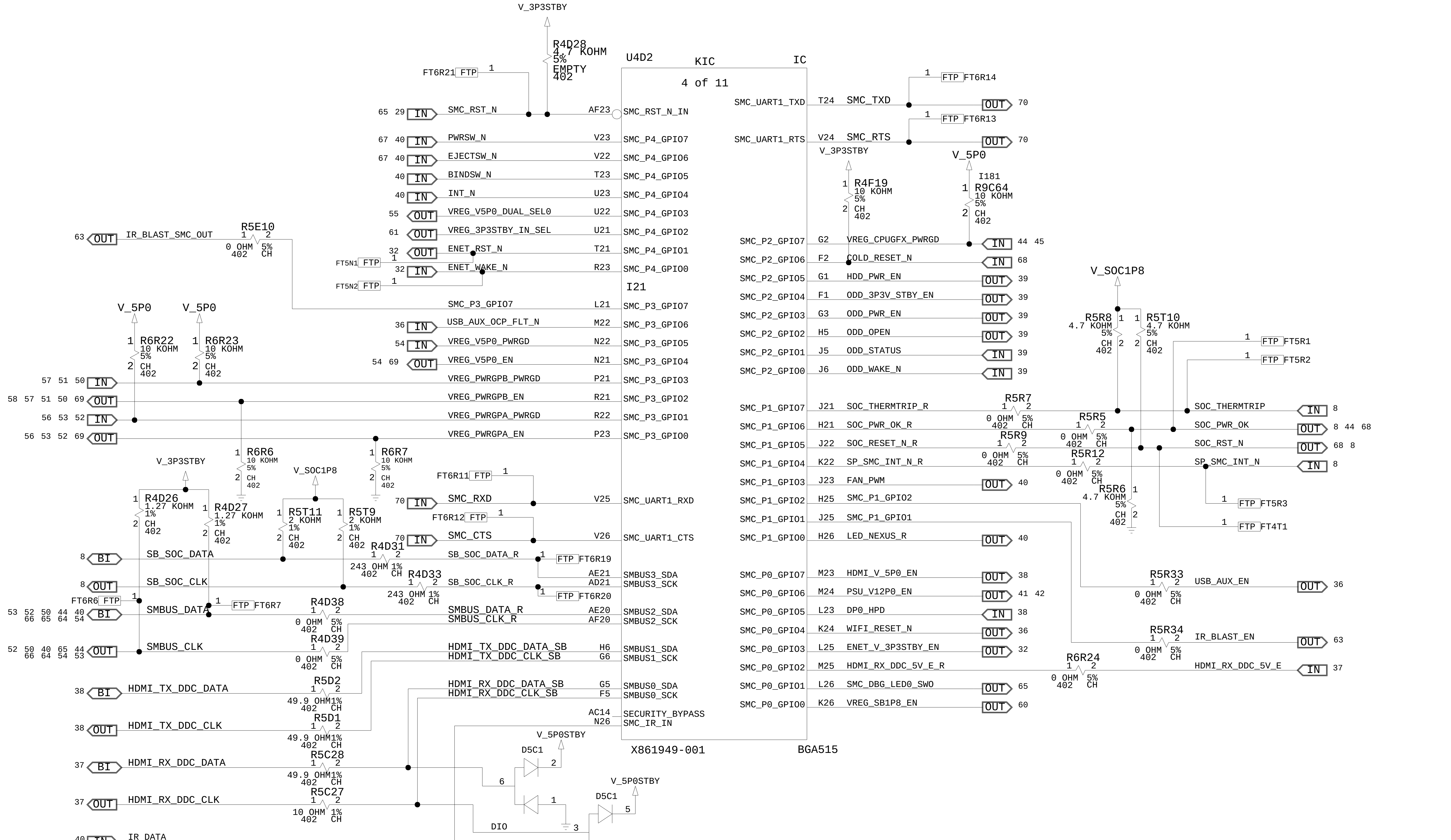
# KIC, PCIEX + SATA + VIDEO




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| U4D2 | EMPTY | EMPTY | TRUE DUMMY_BOM | |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 25/72 | 25/72 | M | 1.0 |

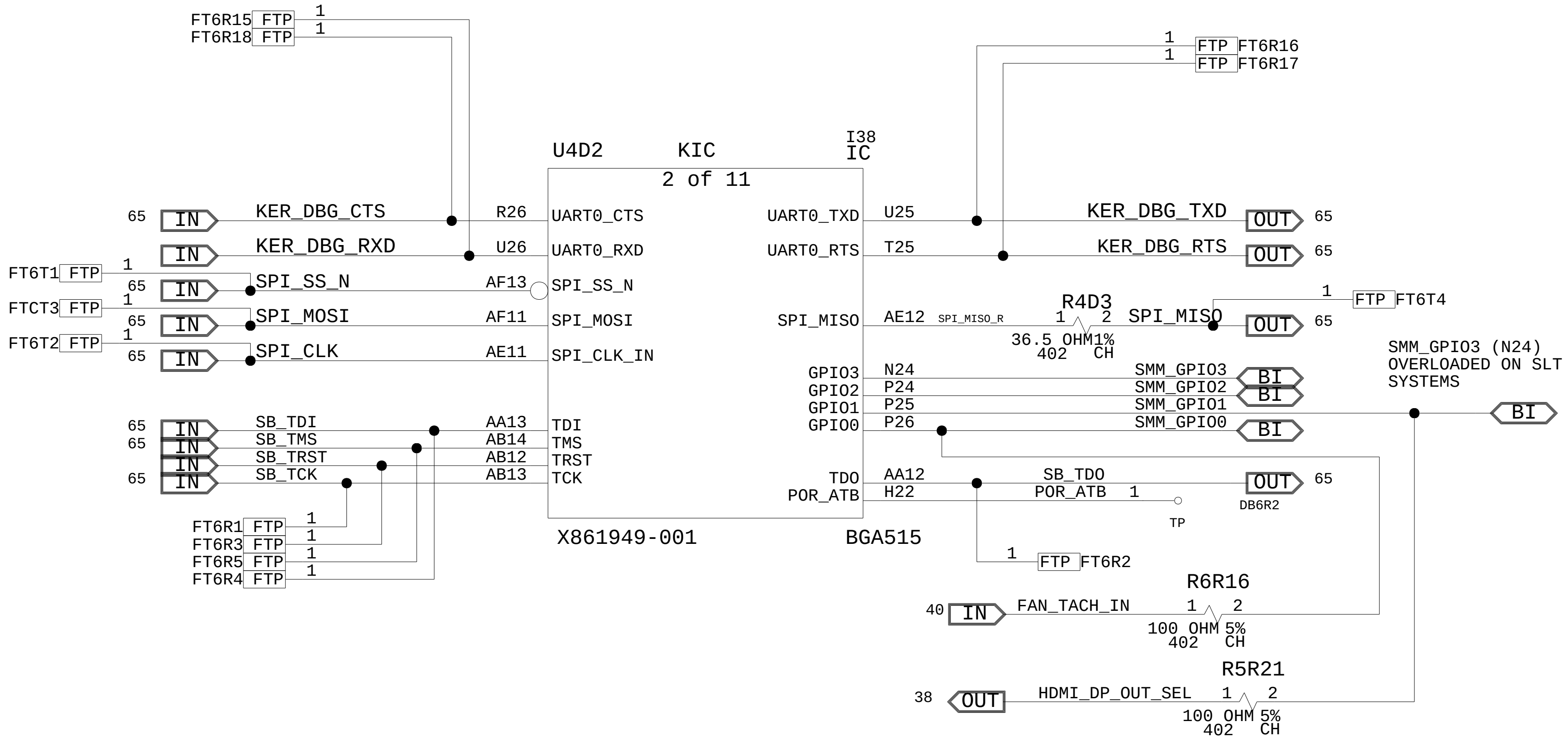
# KIC, SMC



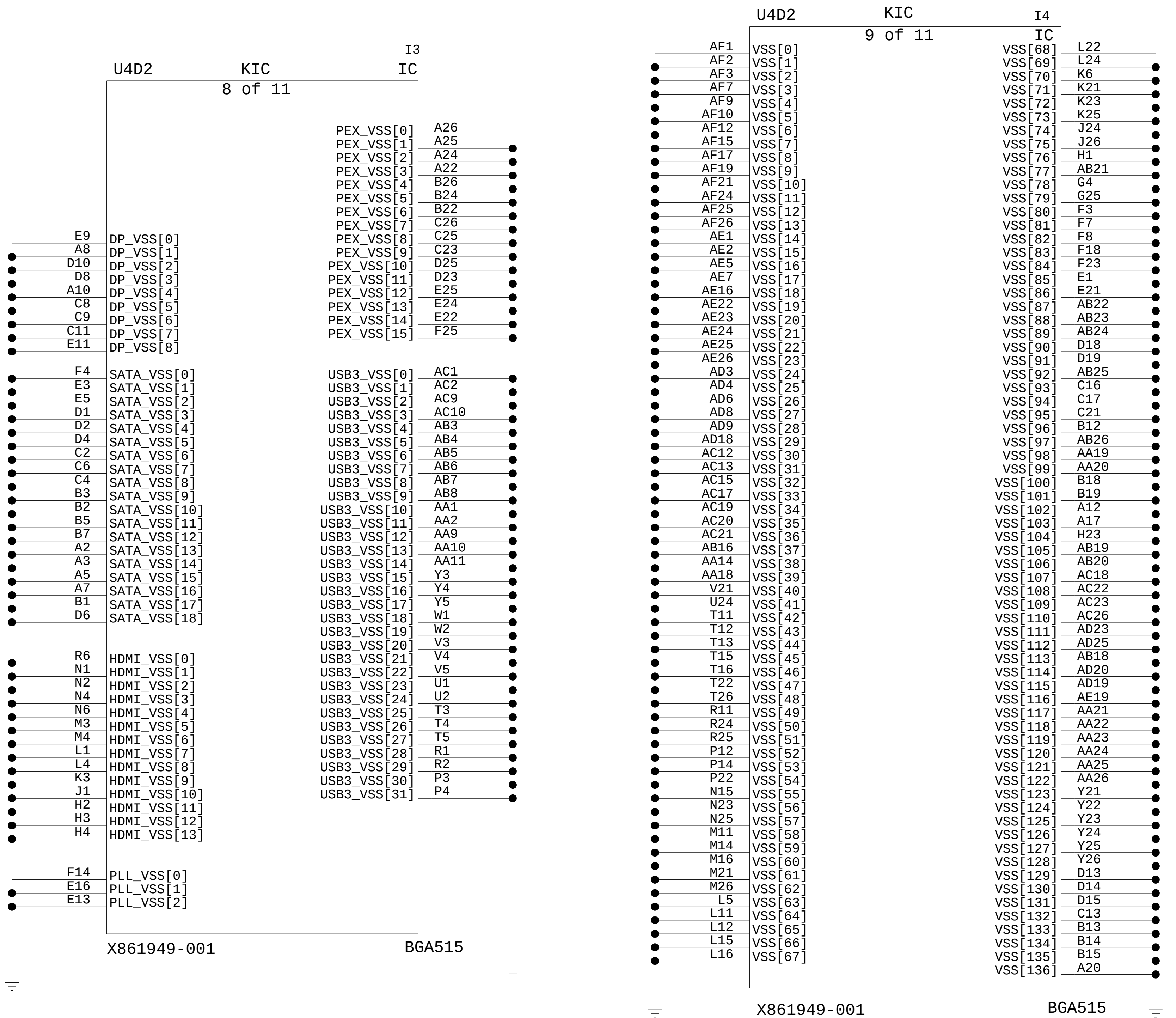

THESE SIGNALS ARE NOT COMPLETE- ENDING IS CLK OR P AND N

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 26/72 | 26/72 | M | 1.0 |

# KIC, FACET

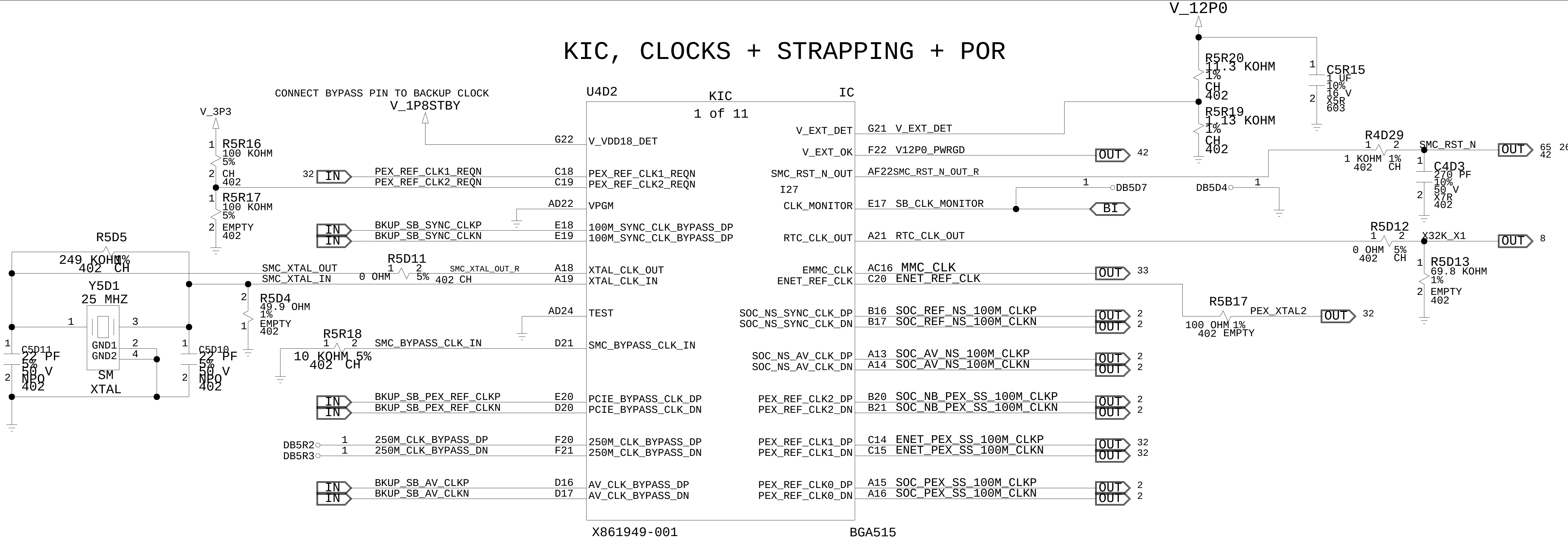



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 27/72 | 27/72 | M | 1.0 |

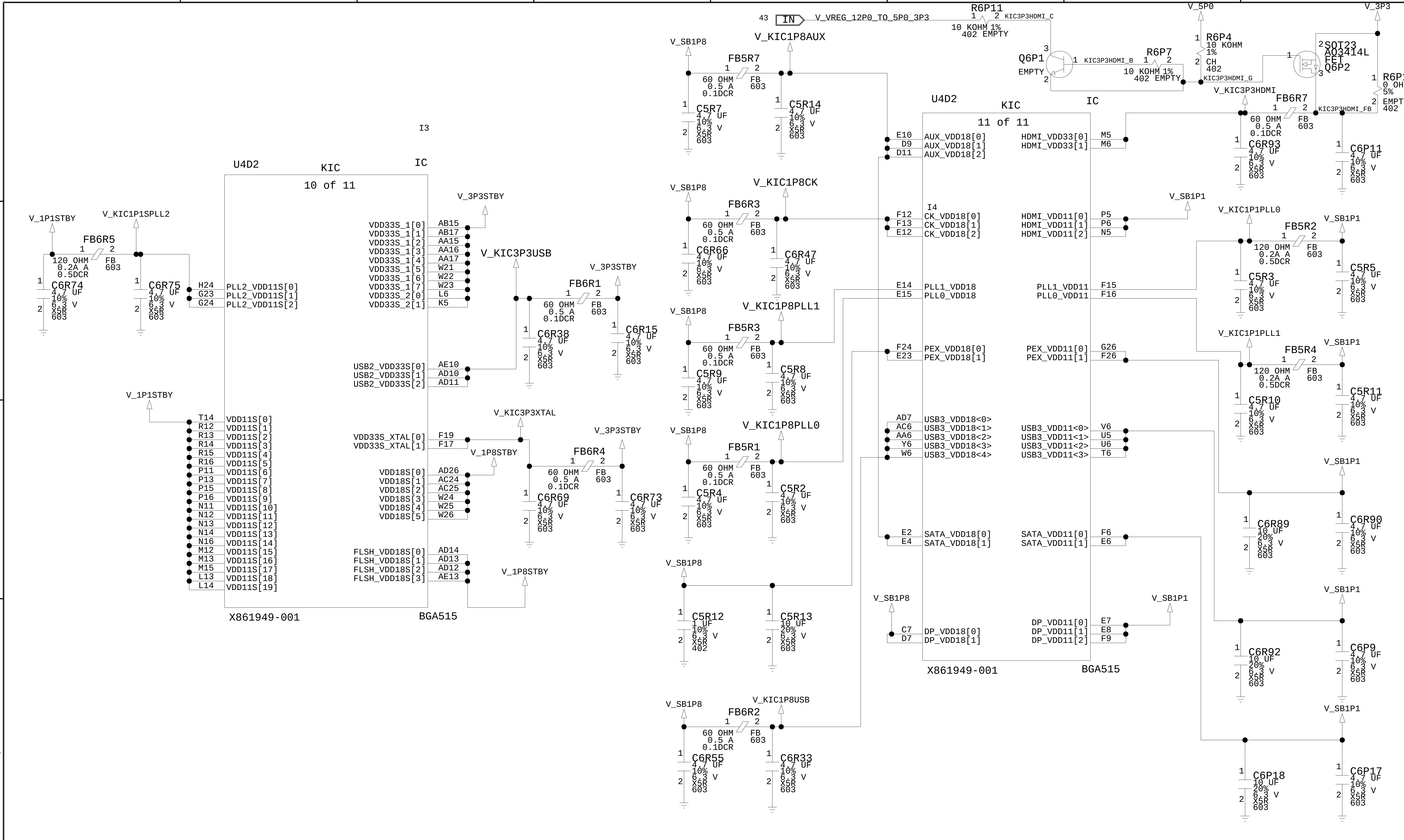



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 28/72 | 28/72 | M | 1.0 |

# KIC, CLOCKS + STRAPPING + POR

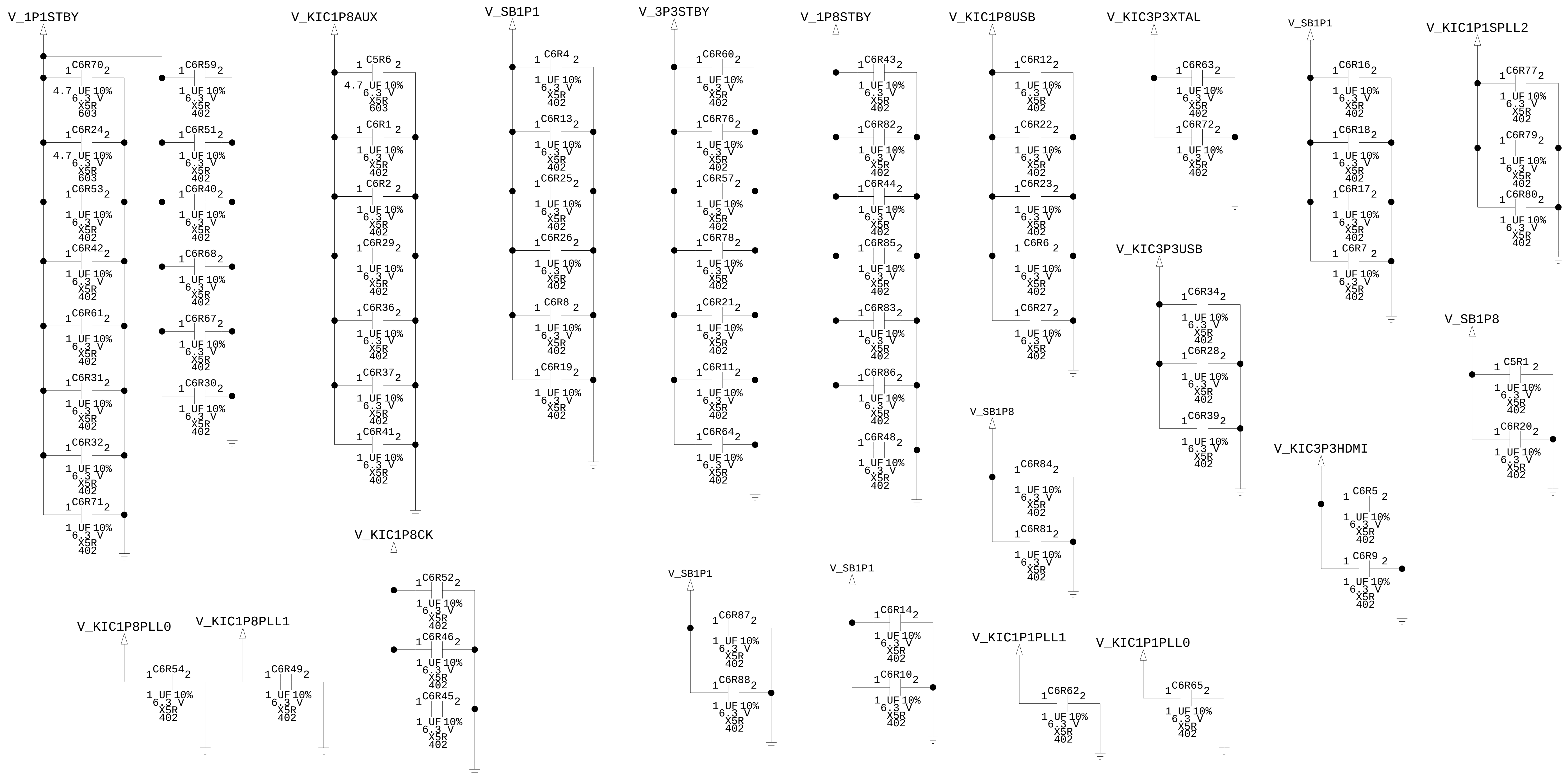



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 29/72 | 29/72 | M | 1.0 |



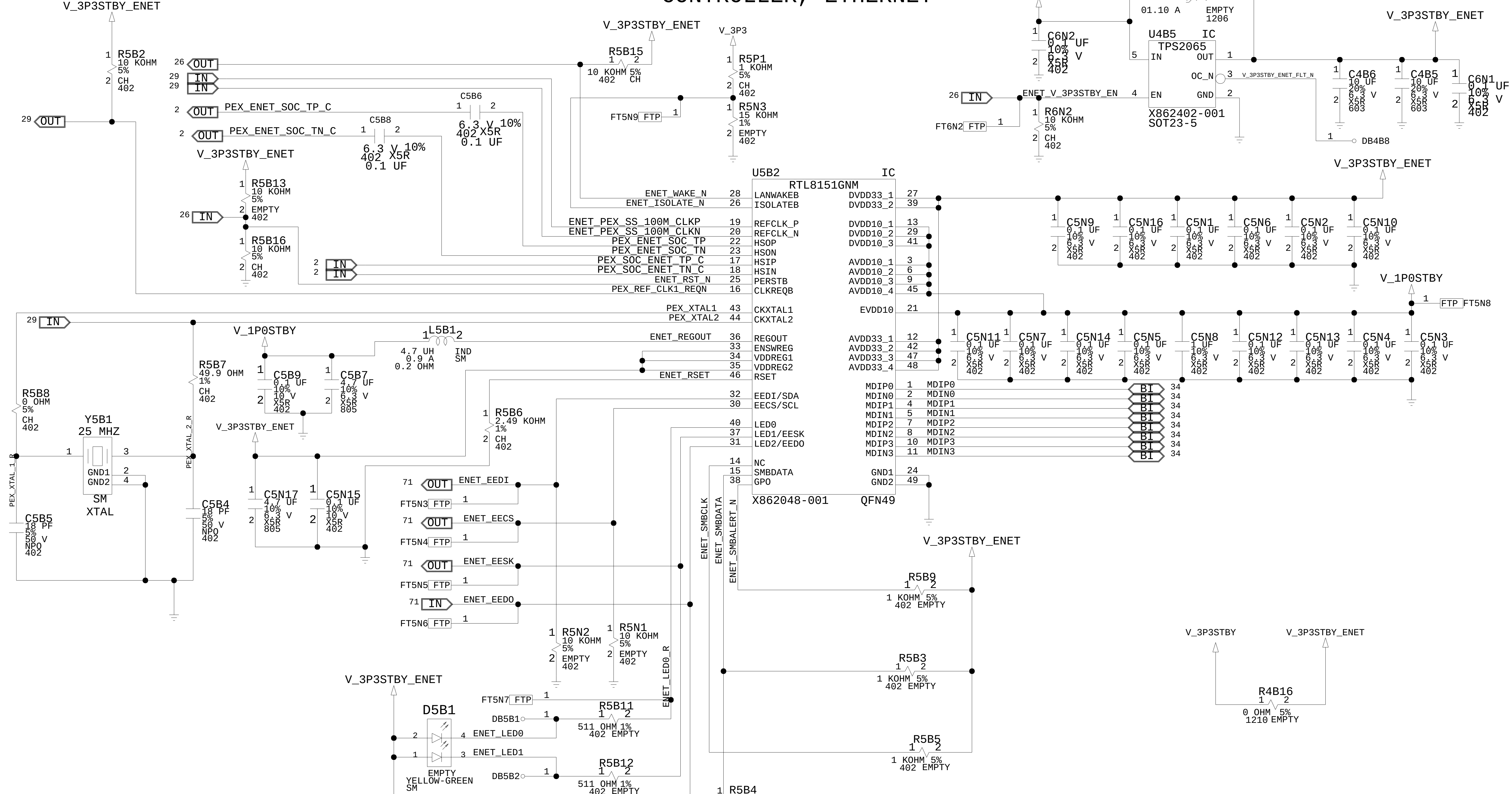

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 30/72 | 30/72 | M | 1.0 |

# KIC, DECOUPLING




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 31/72 | 31/72 | M | 1.0 |

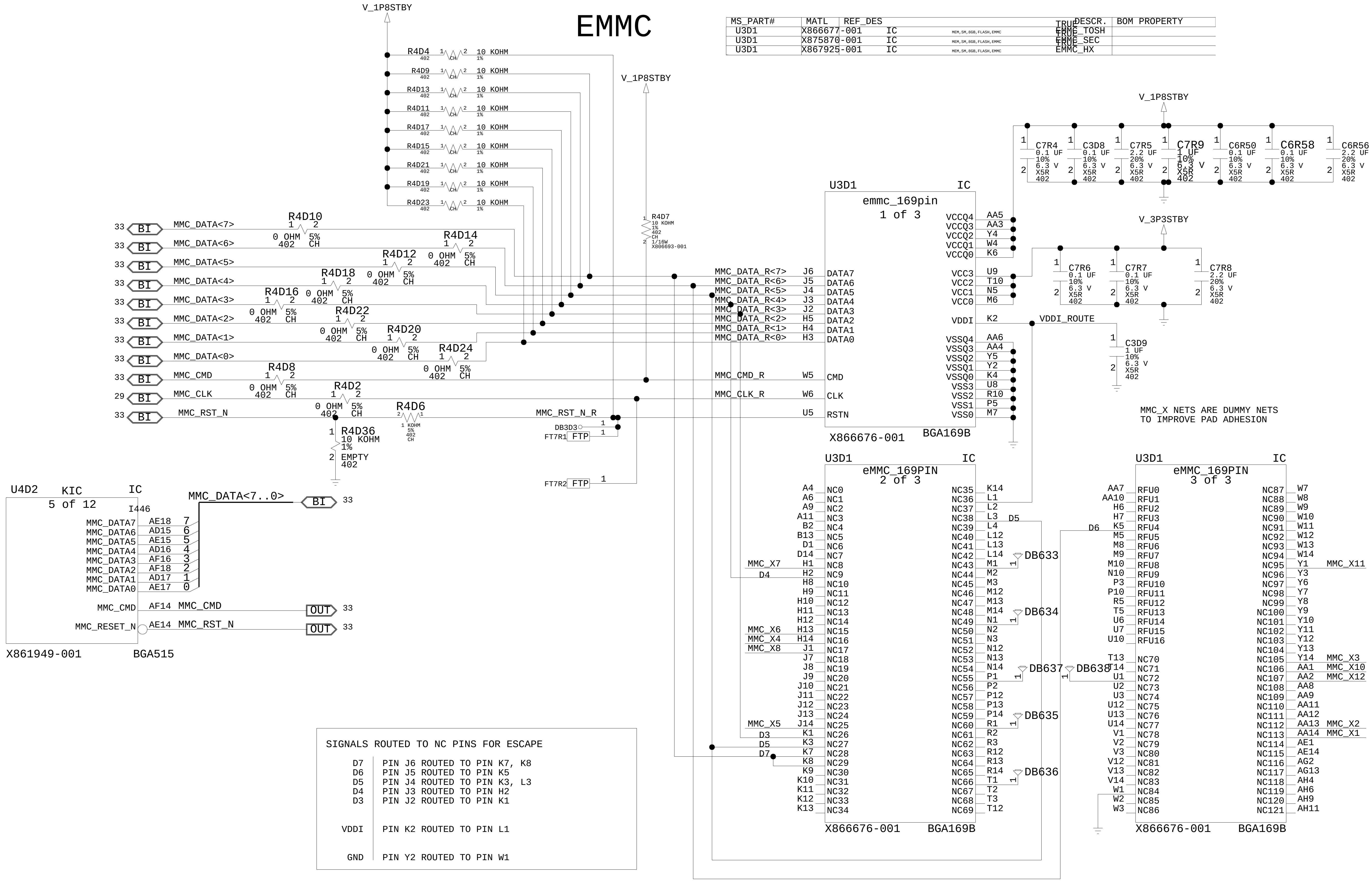
# CONTROLLER, ETHERNET




| EEPROM CONFIG (93C66) | CHIPSET | U5B1 | C5B3 | R5B3 | R5B4 | R5N2 | R5B9 | R5B5 R5N1 |
|---|---|---|---|---|---|---|---|---|
| GREYBULL WITH EEPROM | RTL8151GNM | STUFF | STUFF | STUFF | EMPTY | STUFF | EMPTY | EMPTY |
| GREYBULL W/O EEPROM | RTL8151GNM | EMPTY | EMPTY | EMPTY | EMPTY | EMPTY | EMPTY | EMPTY |

# EMMC

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| U3D1 | X866677-001 | IC | MEM,SM,8GB,FLASH,EMMC | TRUE EMMC_TOSH | |
| U3D1 | X875870-001 | IC | MEM,SM,8GB,FLASH,EMMC | TRUE EMMC_SEC | |
| U3D1 | X867925-001 | IC | MEM,SM,8GB,FLASH,EMMC | TRUE EMMC_HX | |

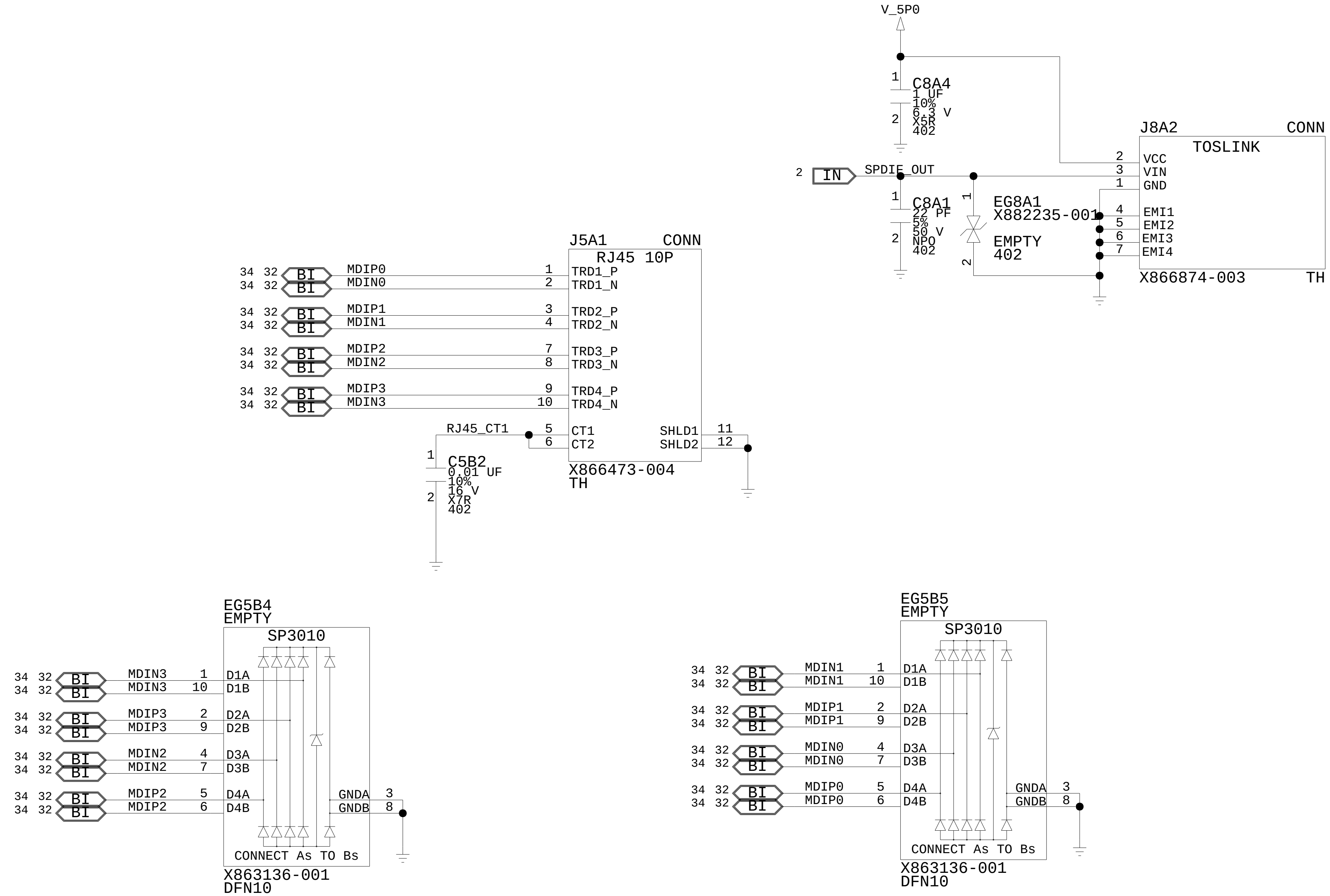



SIGNALS ROUTED TO NC PINS FOR ESCAPE

| | |
|---|---|
| D7 | PIN J6 ROUTED TO PIN K7, K8 |
| D6 | PIN J5 ROUTED TO PIN K5 |
| D5 | PIN J4 ROUTED TO PIN K3, L3 |
| D4 | PIN J3 ROUTED TO PIN H2 |
| D3 | PIN J2 ROUTED TO PIN K1 |
| VDDI | PIN K2 ROUTED TO PIN L1 |
| GND | PIN Y2 ROUTED TO PIN W1 |

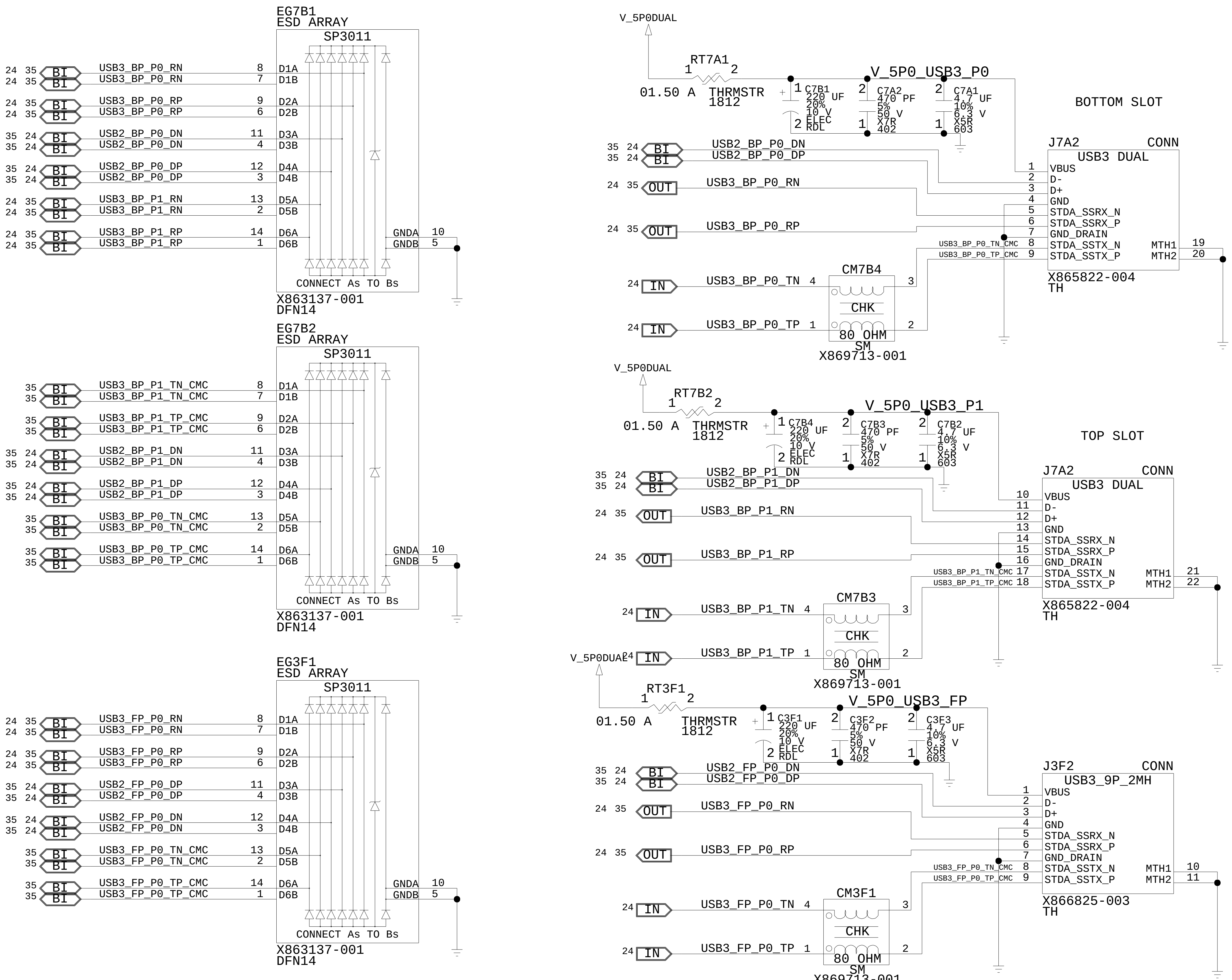
[PAGE_TITLE=EMMC]

# CONN, RJ45 + TOSLINK



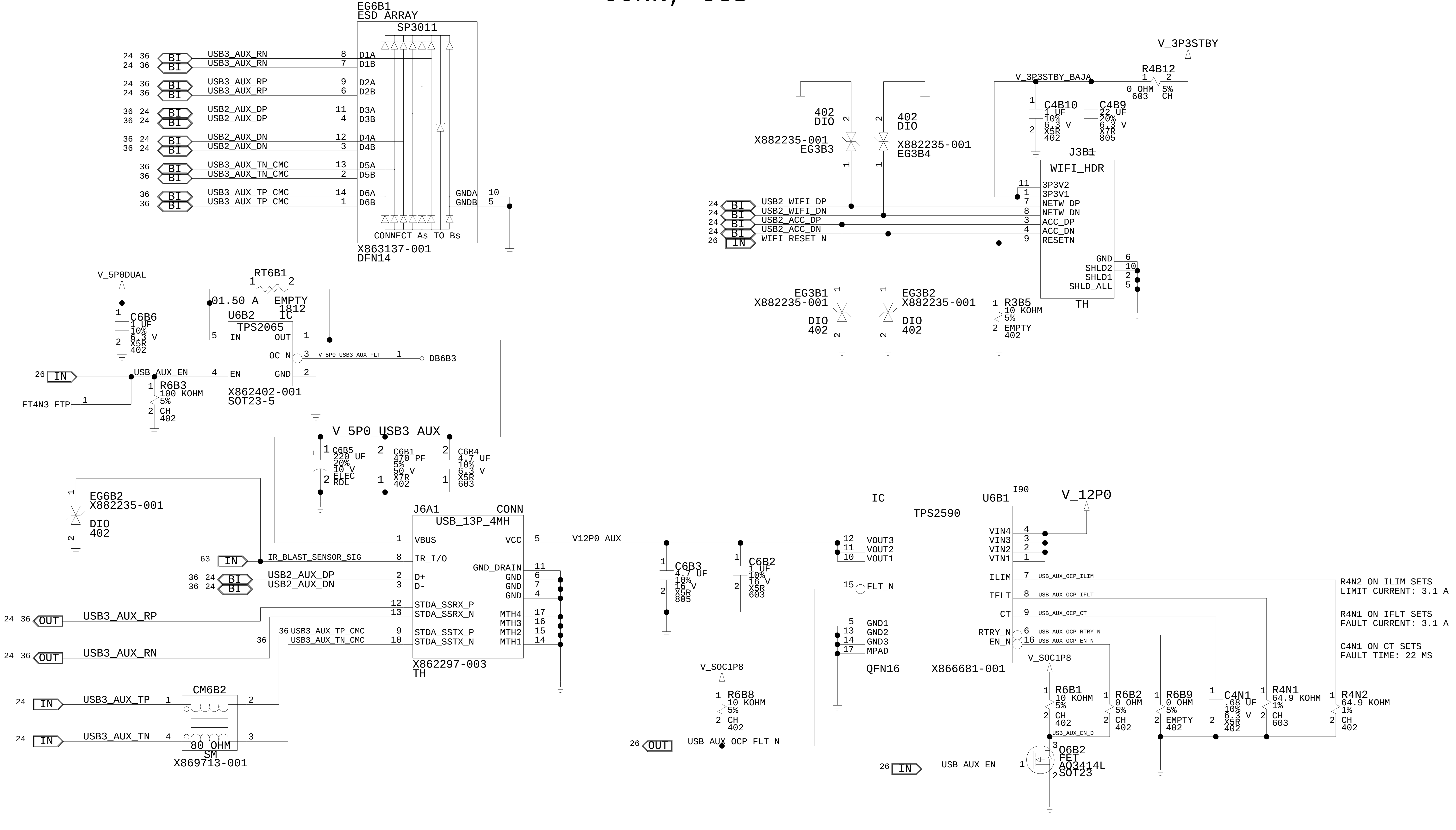

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 34/72 | 34/72 | M | 1.0 |

# CONN, USB



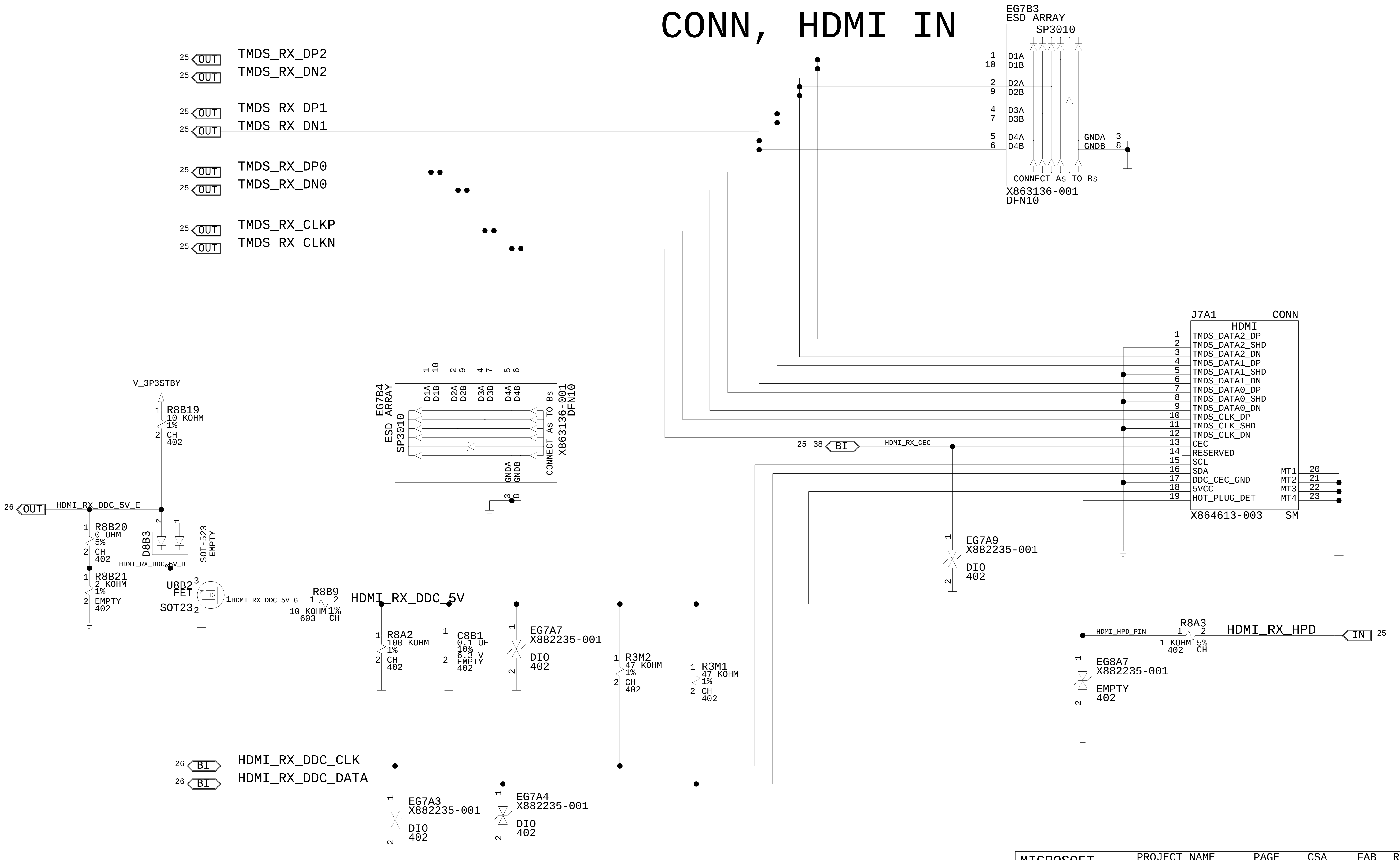

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 35/72 | 35/72 | M | 1.0 |

# CONN, USB



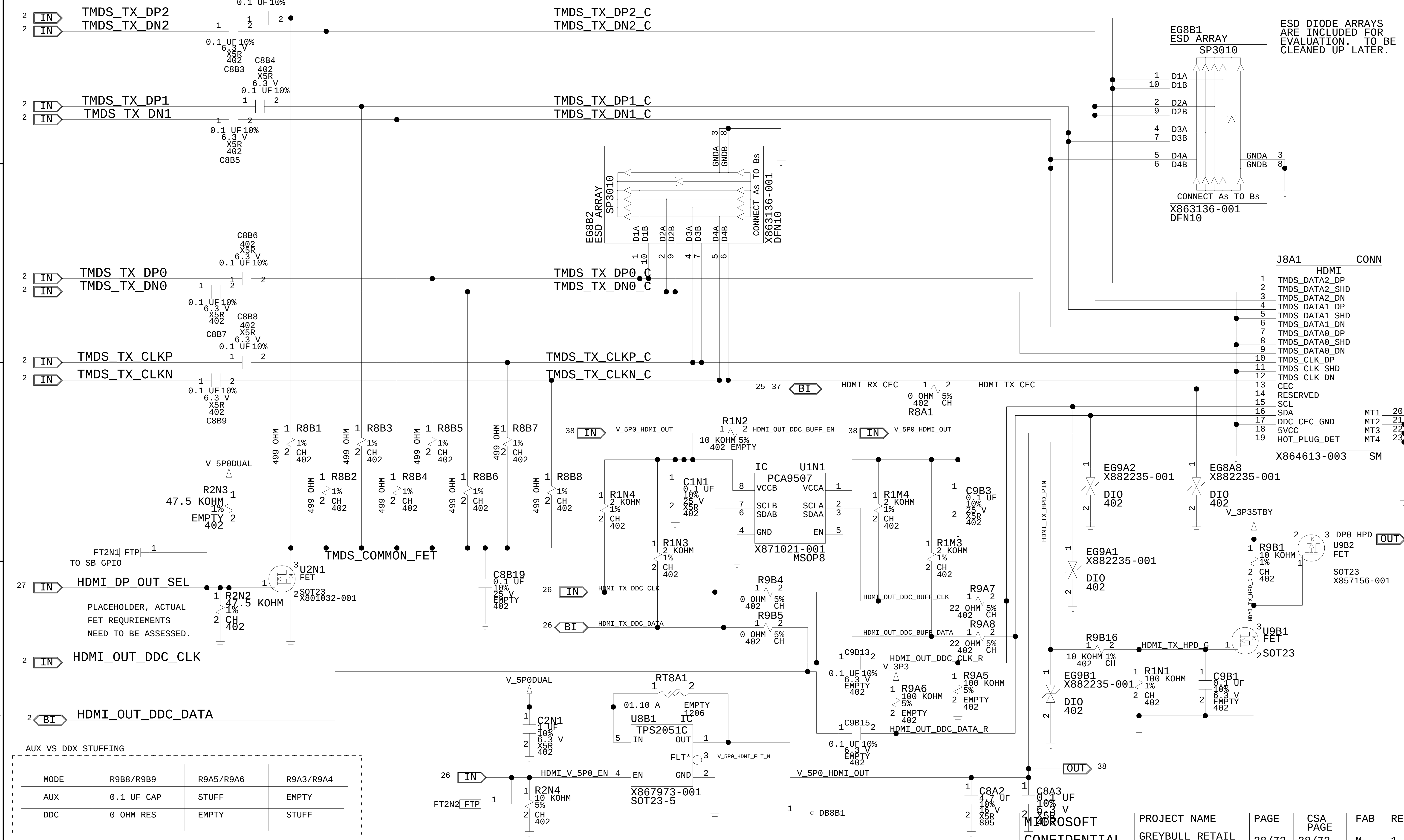

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 36/72 | 36/72 | M | 1.0 |

# CONN, HDMI IN




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 37/72 | 37/72 | M | 1.0 |

# CONN, HDMI OUT

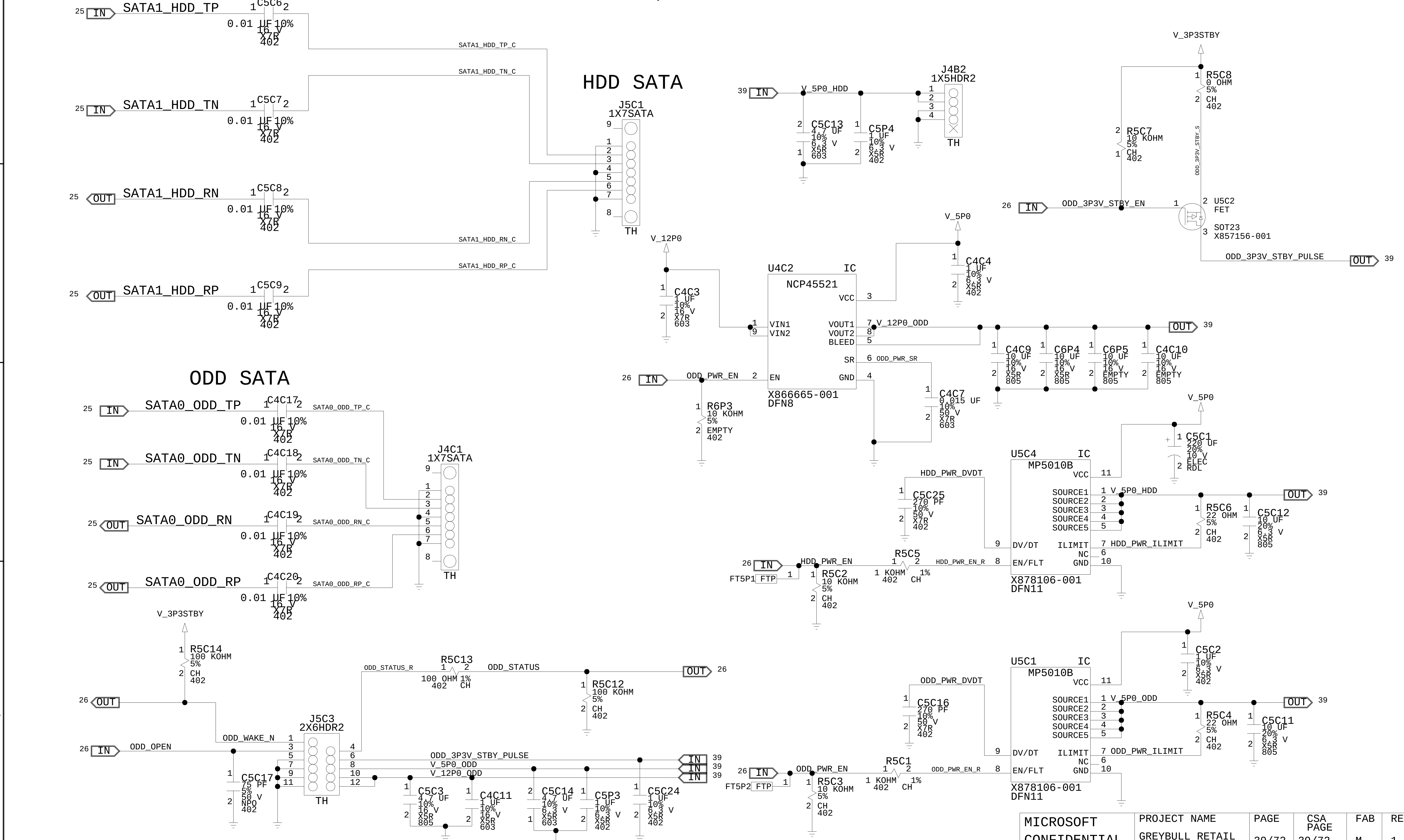



AUX VS DDX STUFFING

| MODE | R9B8/R9B9 | R9A5/R9A6 | R9A3/R9A4 |
|---|---|---|---|
| AUX | 0.1 UF CAP | STUFF | EMPTY |
| DDC | 0 OHM RES | EMPTY | STUFF |

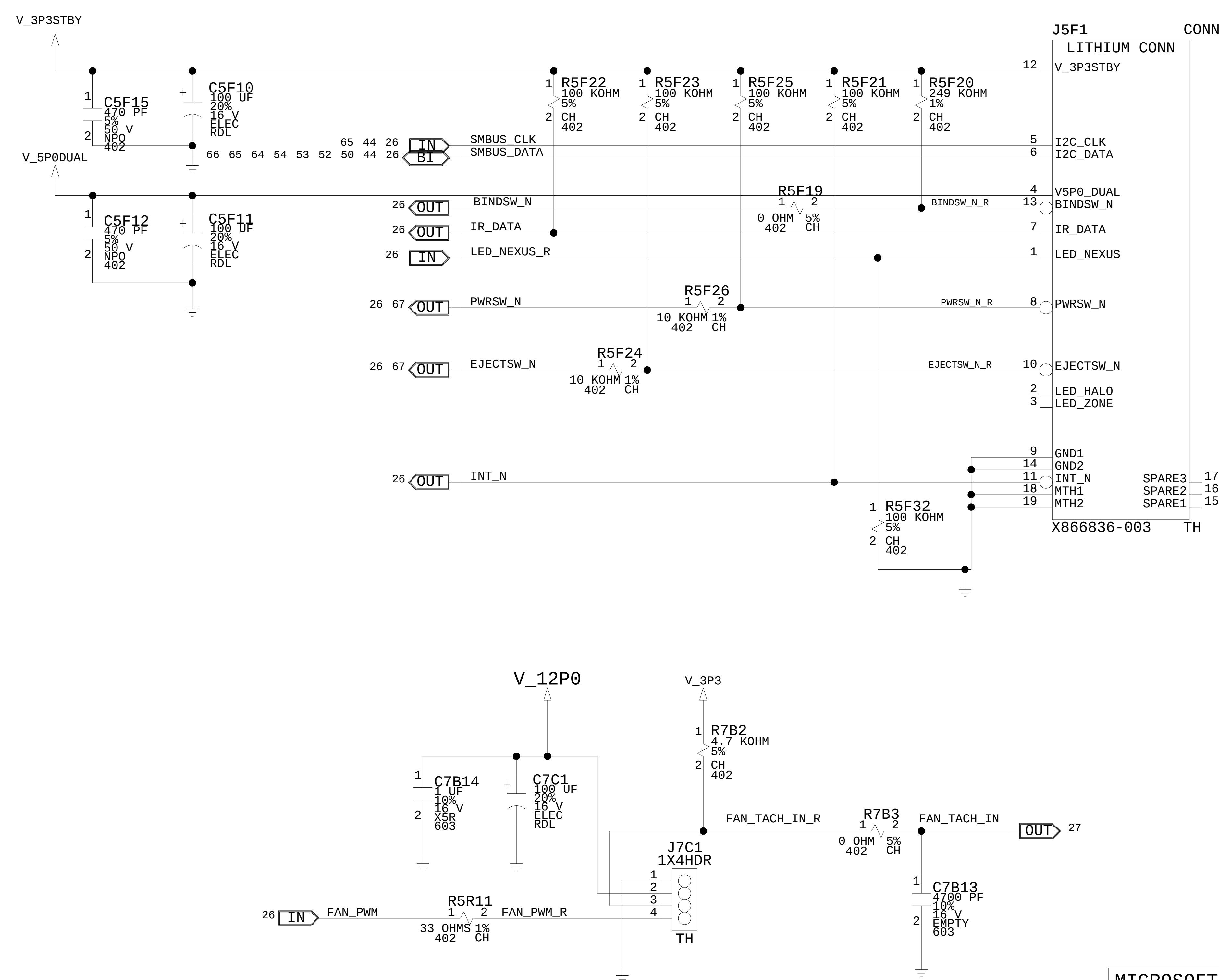
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 38/72 | 38/72 | M | 1.0 |

# CONN, ODD + HDD



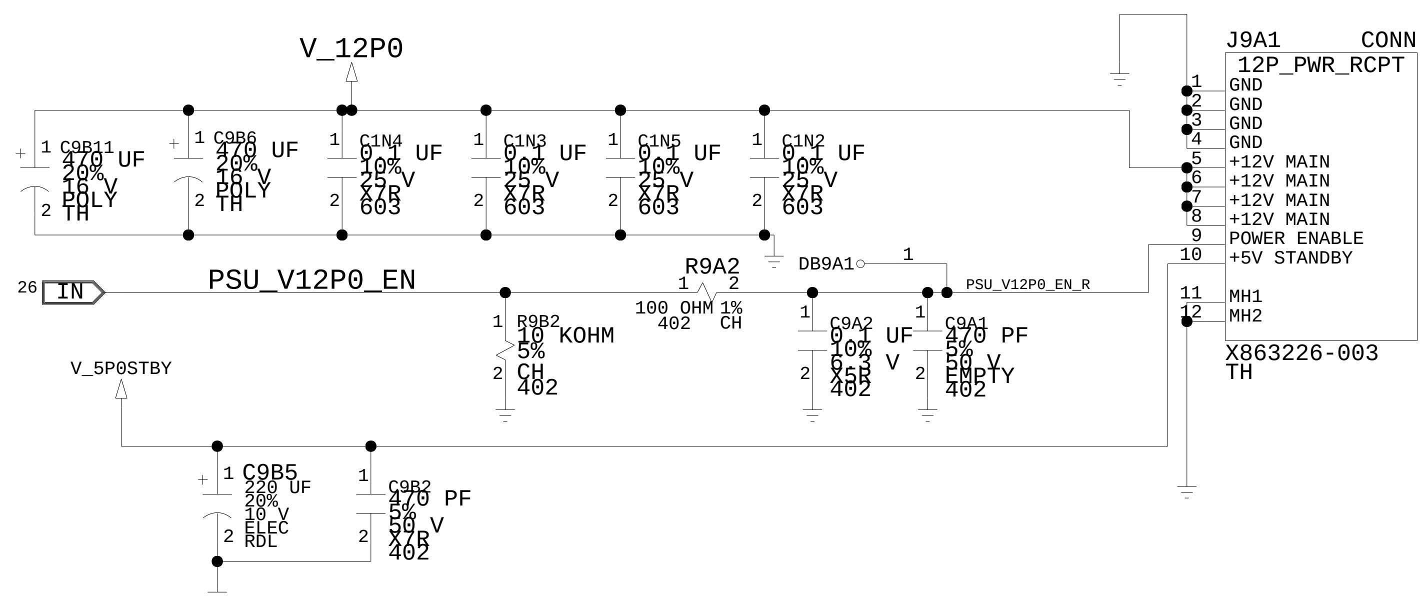

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 39/72 | 39/72 | M | 1.0 |

# CONN, LITHIUM + FAN



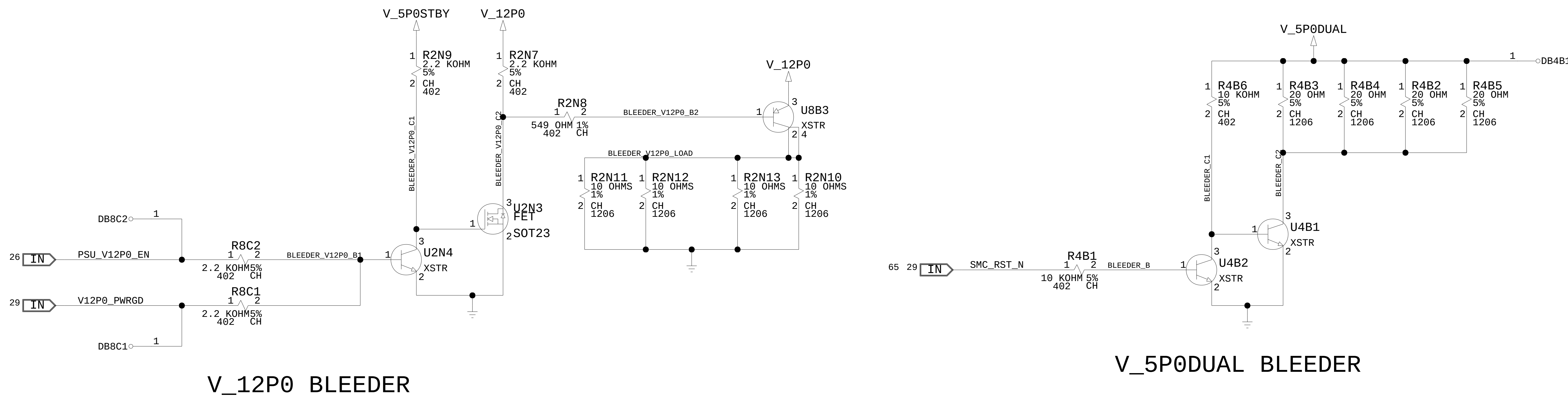

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 40/72 | 40/72 | M | 1.0 |

# CONN, PWR




| DRAWING | MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|---|
| Tue Jun 18 16:42:28 2003 | | GREYBULL_RETAIL | 41/72 | 41/72 | M | 1.0 |

# VREGS, BLEEDERS

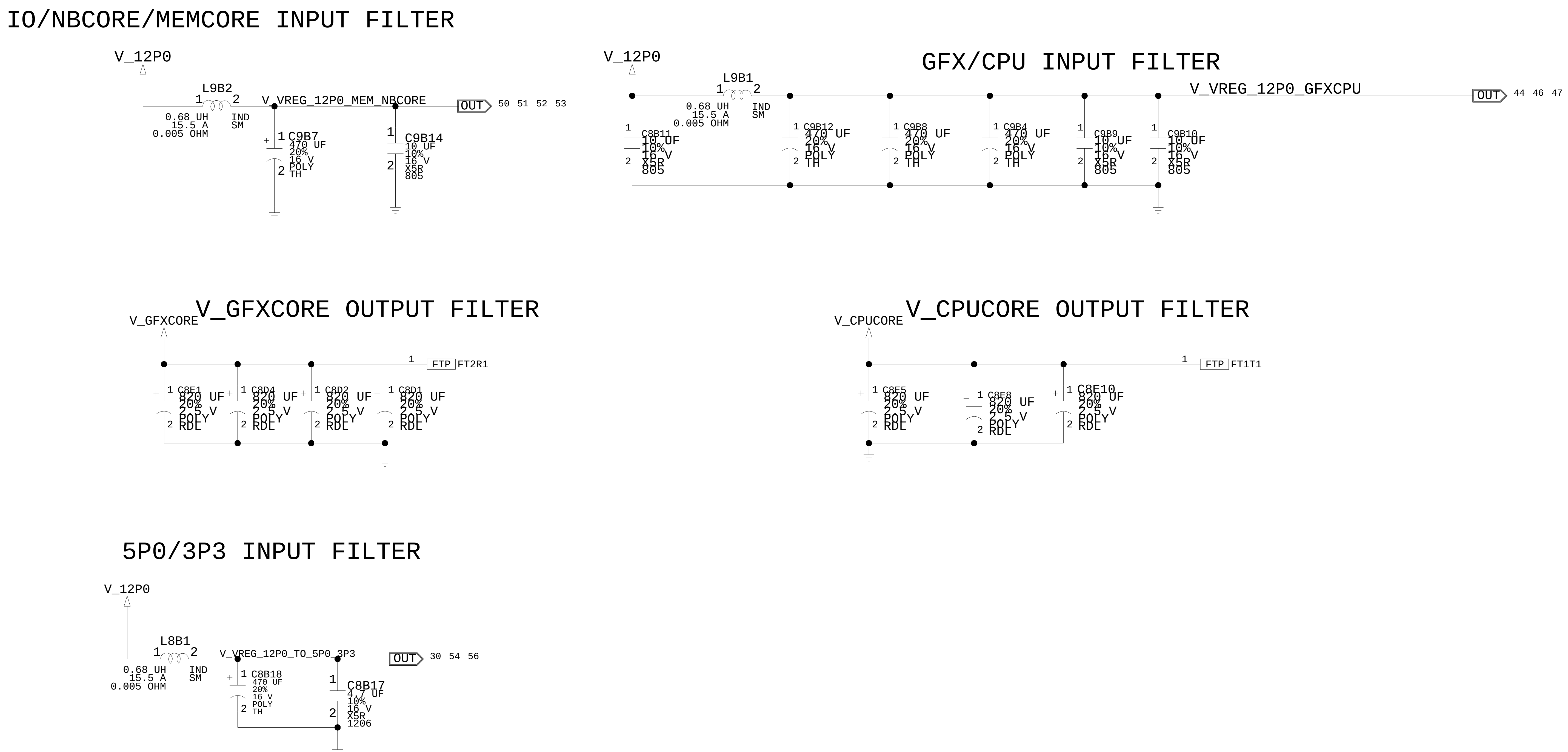



[PAGE_TITLE=VREGS, INPUT + OUTPUT FILTERS]

DRAWING
Tue Jun 18 16:42:20 2005

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 42/72 | 42/72 | M | 1.0 |

# VREGS, INPUT + OUTPUT FILTERS

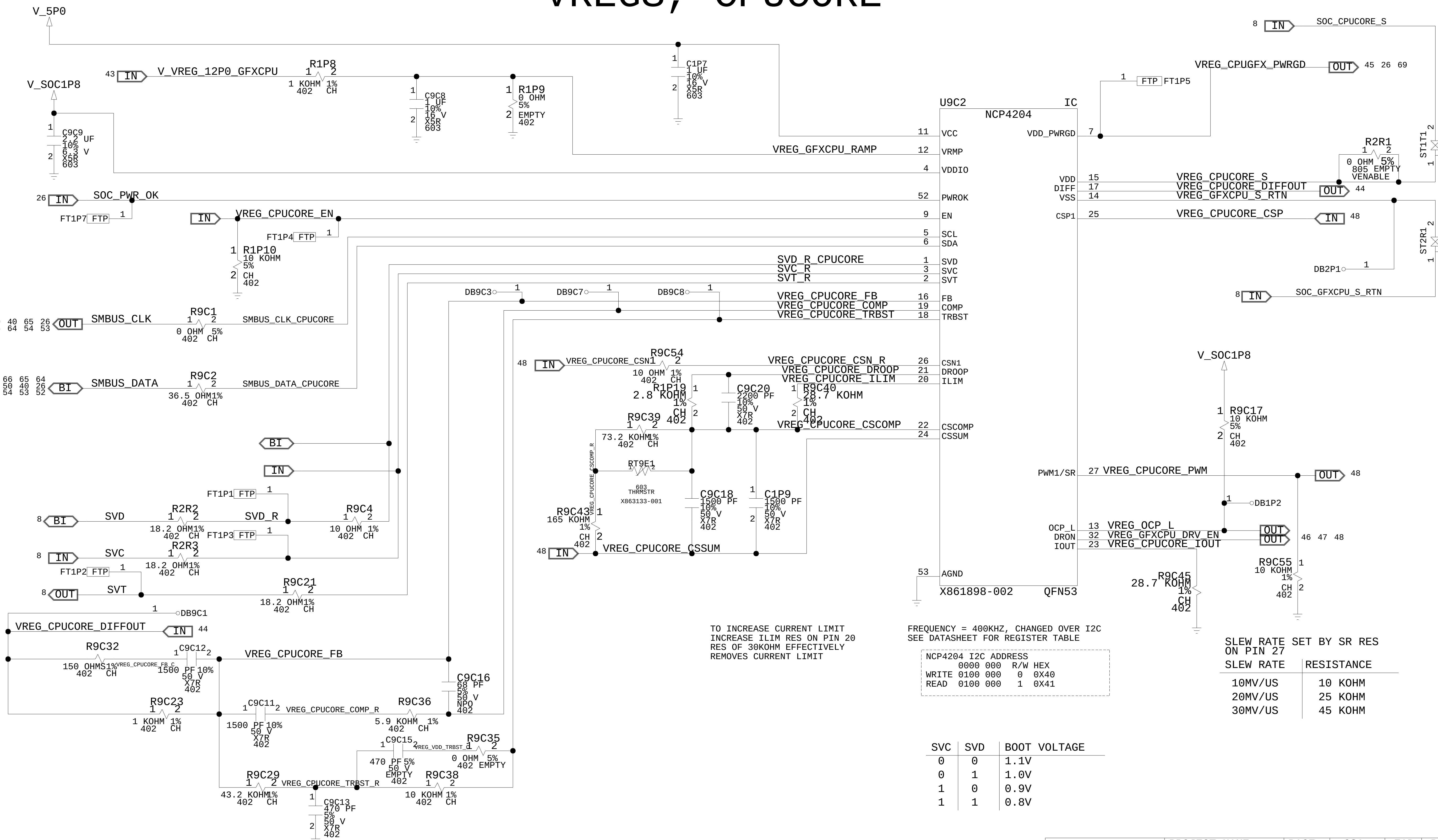



[PAGE_TITLE=VREGS, INPUT + OUTPUT FILTERS]

DRAWING
Tue Jun 18 16:42:25 2003

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 43/72 | 43/72 | M | 1.0 |

# VREGS, CPUCORE




TO INCREASE CURRENT LIMIT
INCREASE ILIM RES ON PIN 20
RES OF 30KOHM EFFECTIVELY
REMOVES CURRENT LIMIT

FREQUENCY = 400KHZ, CHANGED OVER I2C
SEE DATASHEET FOR REGISTER TABLE

NCP4204 I2C ADDRESS

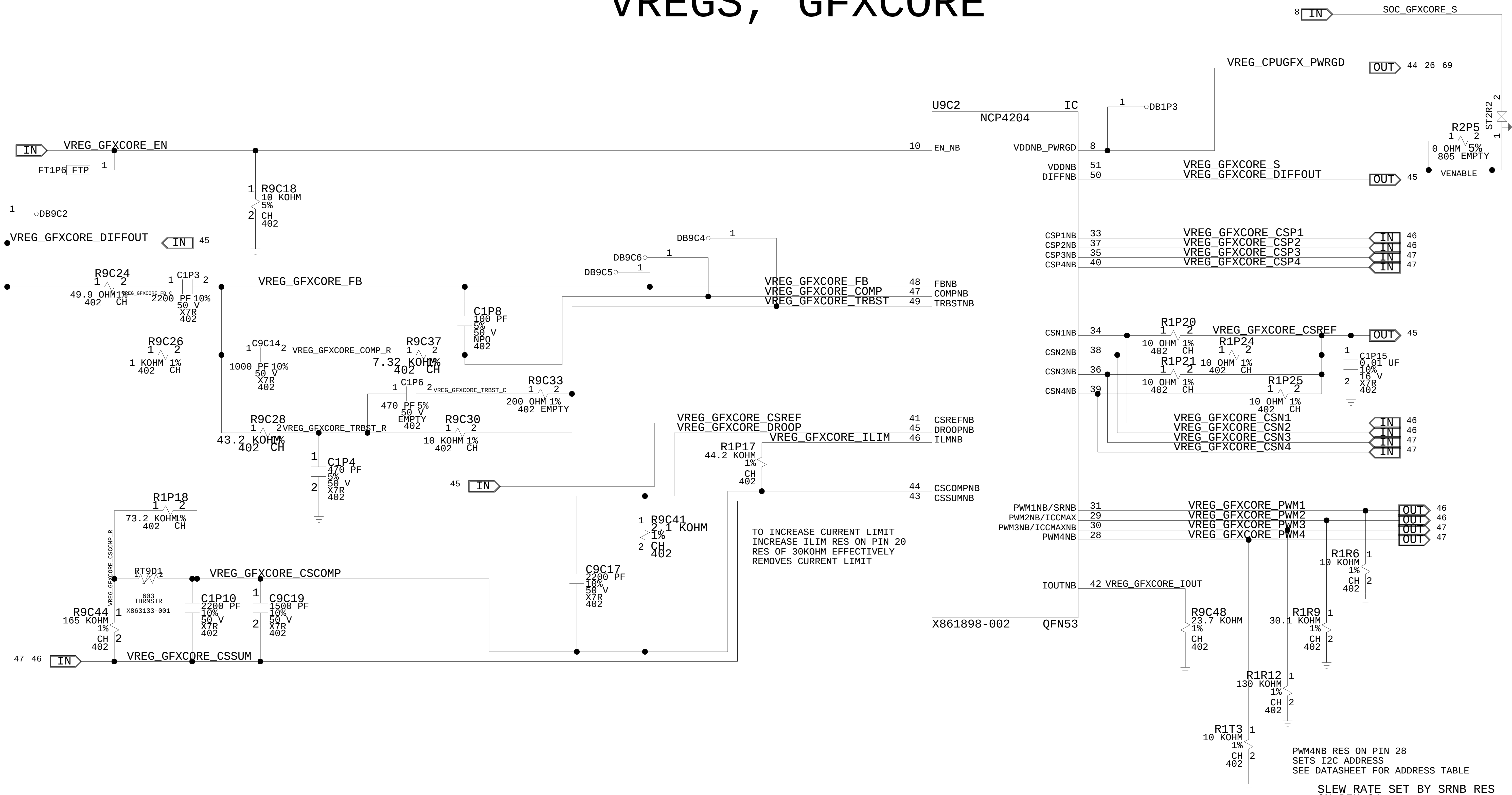
| | | R/W | HEX |
|---|---|---|---|
| | 0000 000 | | |
| WRITE | 0100 000 | 0 | 0X40 |
| READ | 0100 000 | 1 | 0X41 |

SLEW RATE SET BY SR RES
ON PIN 27

| SLEW RATE | RESISTANCE |
|---|---|
| 10MV/US | 10 KOHM |
| 20MV/US | 25 KOHM |
| 30MV/US | 45 KOHM |

| SVC | SVD | BOOT VOLTAGE |
|---|---|---|
| 0 | 0 | 1.1V |
| 0 | 1 | 1.0V |
| 1 | 0 | 0.9V |
| 1 | 1 | 0.8V |

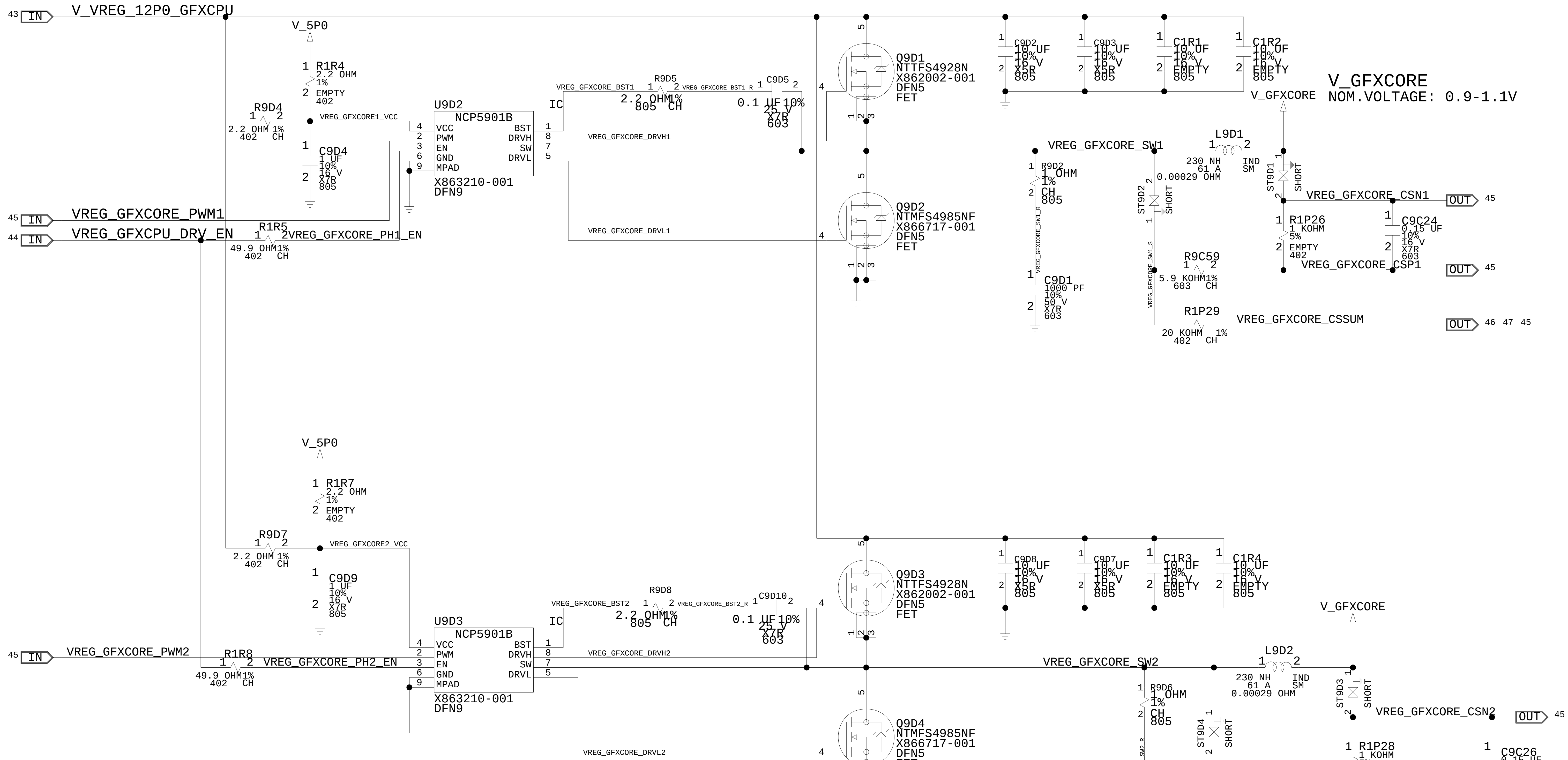
# VREGS, GFXCORE




SLEW RATE SET BY SRNB RES ON PIN 31

| SLEW RATE | RESISTANCE |
|---|---|
| 10MV/US | 10 KOHM |
| 20MV/US | 25 KOHM |
| 30MV/US | 45 KOHM |

# VREGS, GFXCORE OUTPUT PHASE 1 & 2




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY | |
|---|---|---|---|---|---|
| | | | | | TRUE |
| Q9D1,Q9D3 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | FET_CPUGFX_IRF | TRUE |
| Q9D1,Q9D3 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | FET_CPUGFX_TOS | TRUE |
| Q9D1,Q9D3 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | FET_CPUGFX_STM | TRUE |
| Q9D1,Q9D3 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | FET_CPUGFX_AOS | TRUE |
| Q9D2,Q9D4 | X875320-001 | FET | TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A | FET_CPUGFX_IRF | TRUE |
| Q9D2,Q9D4 | X875312-001 | FET | TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A | FET_CPUGFX_TOS | TRUE |
| Q9D2,Q9D4 | X878211-001 | FET | TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A | FET_CPUGFX_STM | |
| Q9D2,Q9D4 | X875317-001 | FET | TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A | FET_CPUGFX_AOS | |

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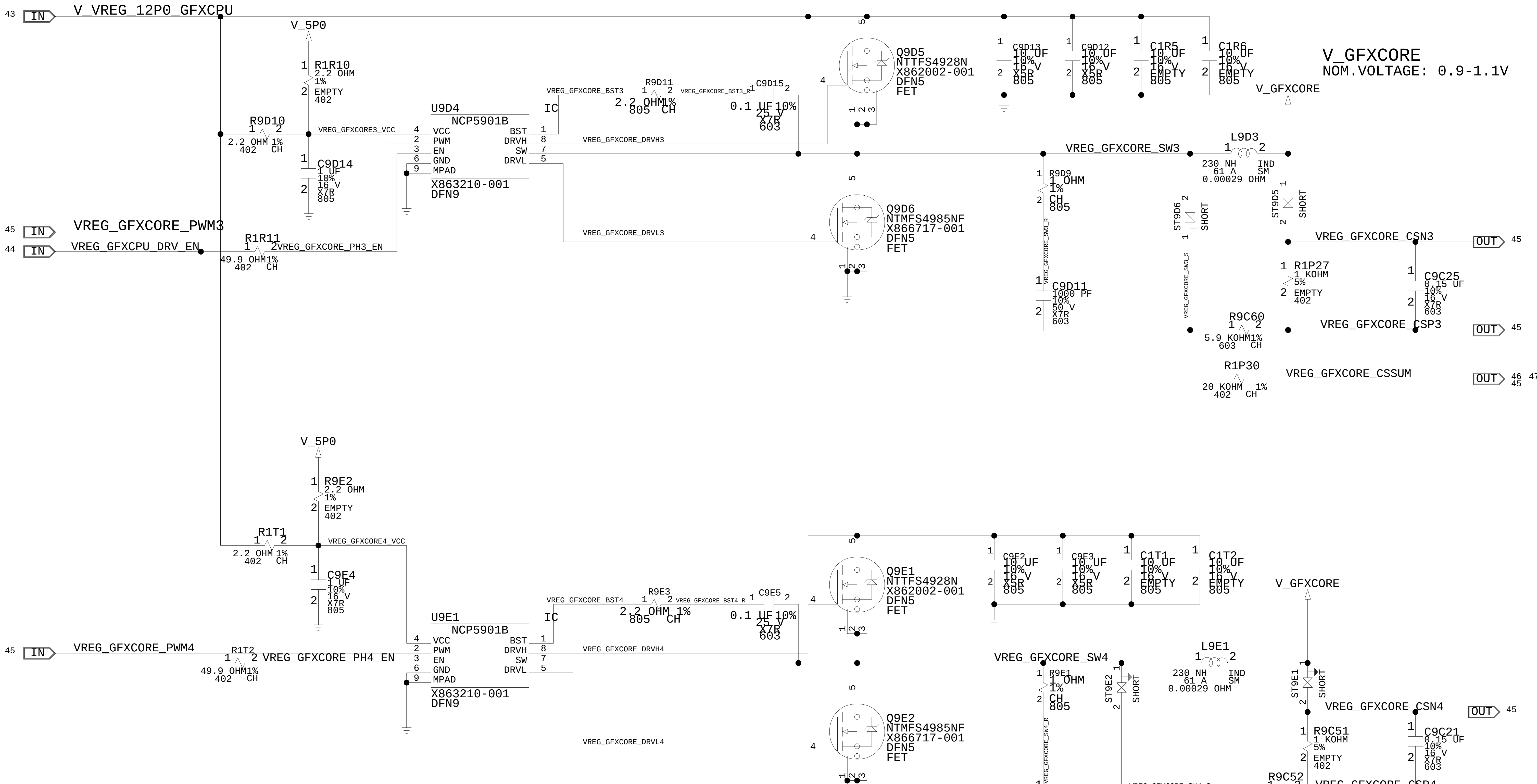
Tue Jun 18 16:42:25 2013

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 46/72 | 46/72 | M | 1.0 |

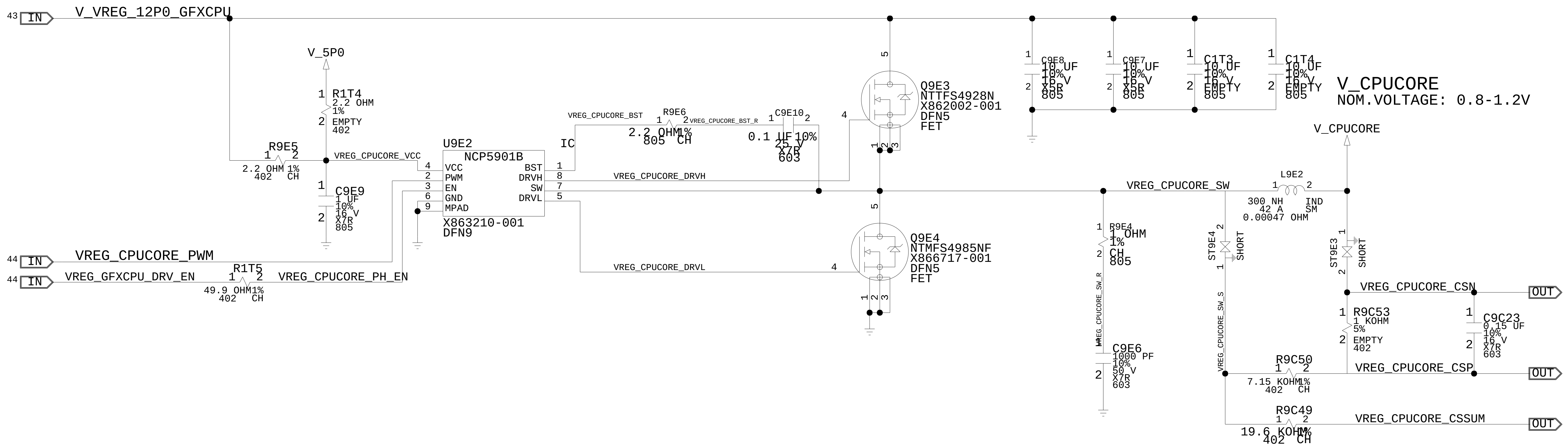
# VREGS, GFXCORE OUTPUT PHASE 3 & 4




| MS_PART# | MATL | REF_DES | | TRUE DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| Q9D5,Q9E1 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | FET_CPUGFX_IRF TRUE | |
| Q9D5,Q9E1 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | FET_CPUGFX_TOS TRUE | |
| Q9D5,Q9E1 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | FET_CPUGFX_STM TRUE | |
| Q9D5,Q9E1 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | FET_CPUGFX_AOS TRUE | |
| Q9D6,Q9E2 | X875320-001 | FET | TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A | FET_CPUGFX_IRF TRUE | |
| Q9D6,Q9E2 | X875312-001 | FET | TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A | FET_CPUGFX_TOS TRUE | |
| Q9D6,Q9E2 | X878211-001 | FET | TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A | FET_CPUGFX_STM TRUE | |
| Q9D6,Q9E2 | X875317-001 | FET | TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A | FET_CPUGFX_AOS | |

| DRAWING | MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|---|
| Tue Jun 18 16:42:25 2013 | | GREYBULL_RETAIL | 47/72 | 47/72 | M | 1.0 |

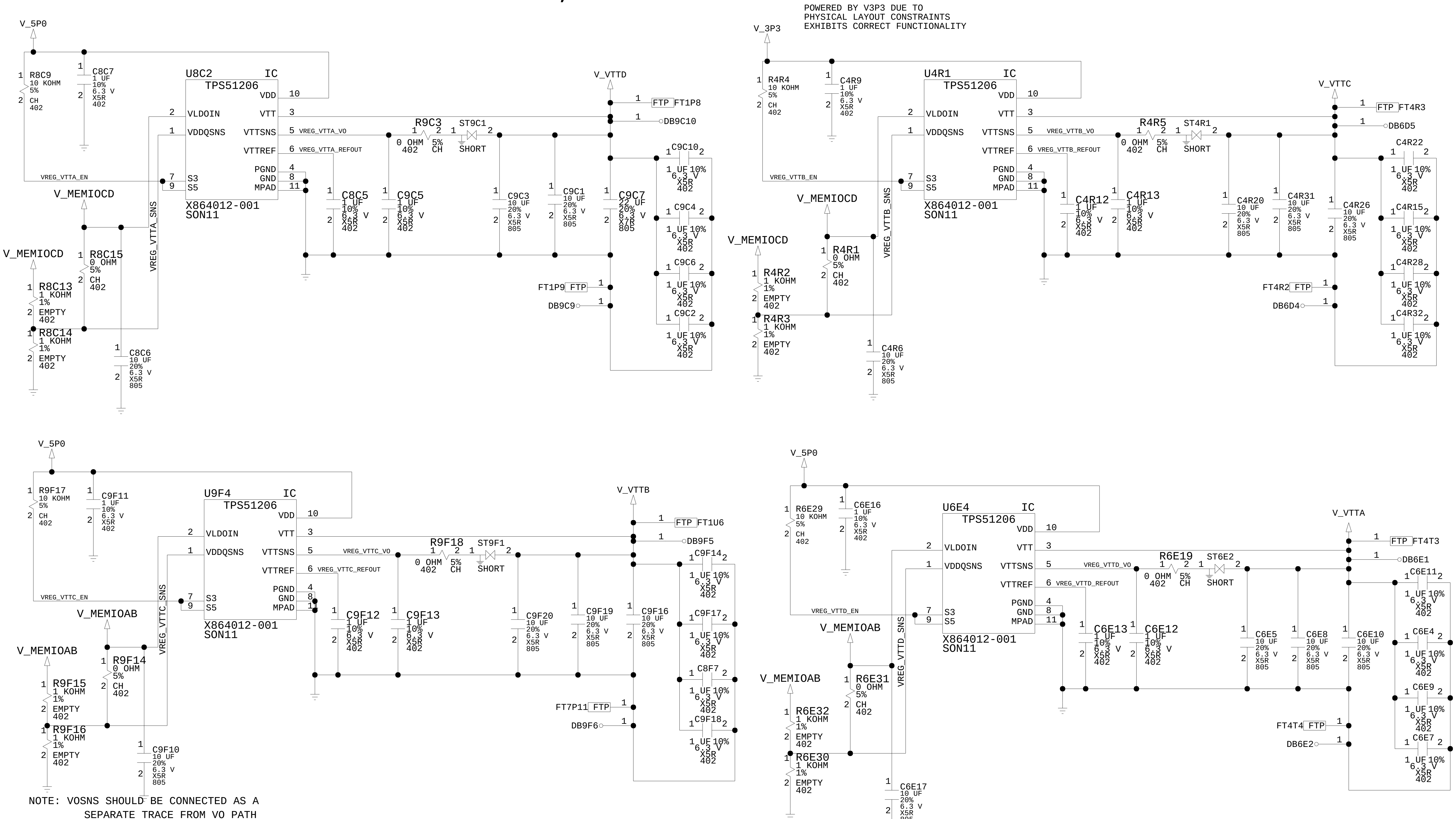
# VREGS, CPUCORE OUTPUT PHASE




| MS_PART# | MATL | REF_DES | DESCR. | | BOM PROPERTY |
|---|---|---|---|---|---|
| Q9E3 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | TRUE | FET_CPUGFX_IRF |
| Q9E3 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | TRUE | FET_CPUGFX_TOS |
| Q9E3 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | TRUE | FET_CPUGFX_STM |
| Q9E3 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | TRUE | FET_CPUGFX_AOS |
| Q9E4 | X875320-001 | FET | TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A | TRUE | FET_CPUGFX_IRF |
| Q9E4 | X875312-001 | FET | TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A | TRUE | FET_CPUGFX_TOS |
| Q9E4 | X878211-001 | FET | TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A | TRUE | FET_CPUGFX_STM |
| Q9E4 | X875317-001 | FET | TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A | TRUE | FET_CPUGFX_AOS |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 48/72 | 48/72 | M | 1.0 |

# VREGS, VTT TERMINATION




[PAGE_TITLE=VTT TERMINATION]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 49/72 | 49/72 | M | 1.0 |

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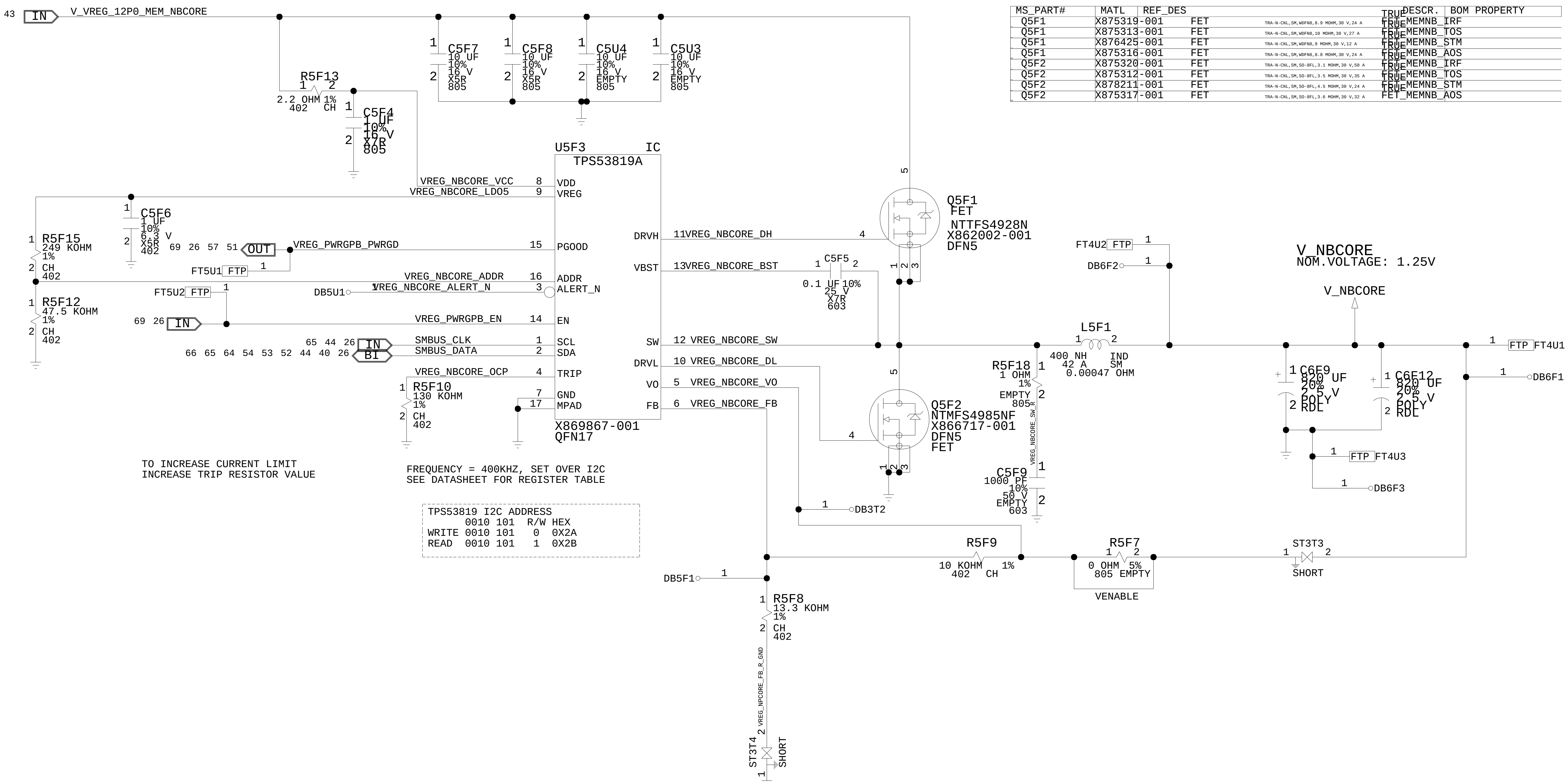
# VREGS, NBCORE




| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| Q5F1 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | TRUE | FET_MEMNB_IRF |
| Q5F1 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | TRUE | FET_MEMNB_TOS |
| Q5F1 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | TRUE | FET_MEMNB_STM |
| Q5F1 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | TRUE | FET_MEMNB_AOS |
| Q5F2 | X875320-001 | FET | TRA-N-CNL,SM,SO-8FL,3.1 MOHM,30 V,50 A | TRUE | FET_MEMNB_IRF |
| Q5F2 | X875312-001 | FET | TRA-N-CNL,SM,SO-8FL,3.5 MOHM,30 V,35 A | TRUE | FET_MEMNB_TOS |
| Q5F2 | X878211-001 | FET | TRA-N-CNL,SM,SO-8FL,4.5 MOHM,30 V,24 A | TRUE | FET_MEMNB_STM |
| Q5F2 | X875317-001 | FET | TRA-N-CNL,SM,SO-8FL,3.6 MOHM,30 V,32 A | TRUE | FET_MEMNB_AOS |

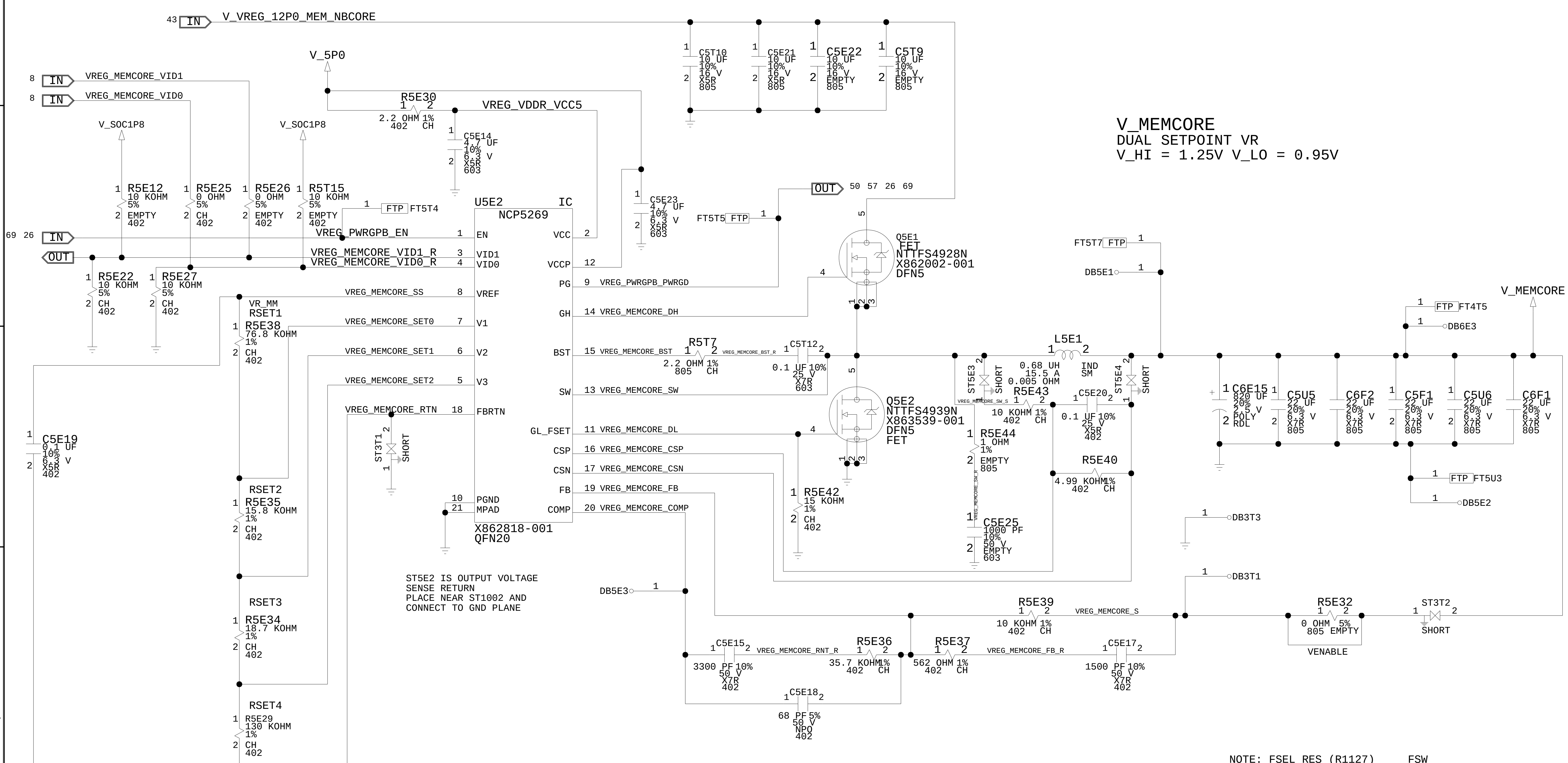
TO INCREASE CURRENT LIMIT
INCREASE TRIP RESISTOR VALUE

FREQUENCY = 400KHZ, SET OVER I2C
SEE DATASHEET FOR REGISTER TABLE

| TPS53819 I2C ADDRESS | | | |
|---|---|---|---|
| | 0010 101 | R/W | HEX |
| WRITE | 0010 101 | 0 | 0X2A |
| READ | 0010 101 | 1 | 0X2B |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 50/72 | 50/72 | M | 1.0 |

# VREGS, MEMCORE

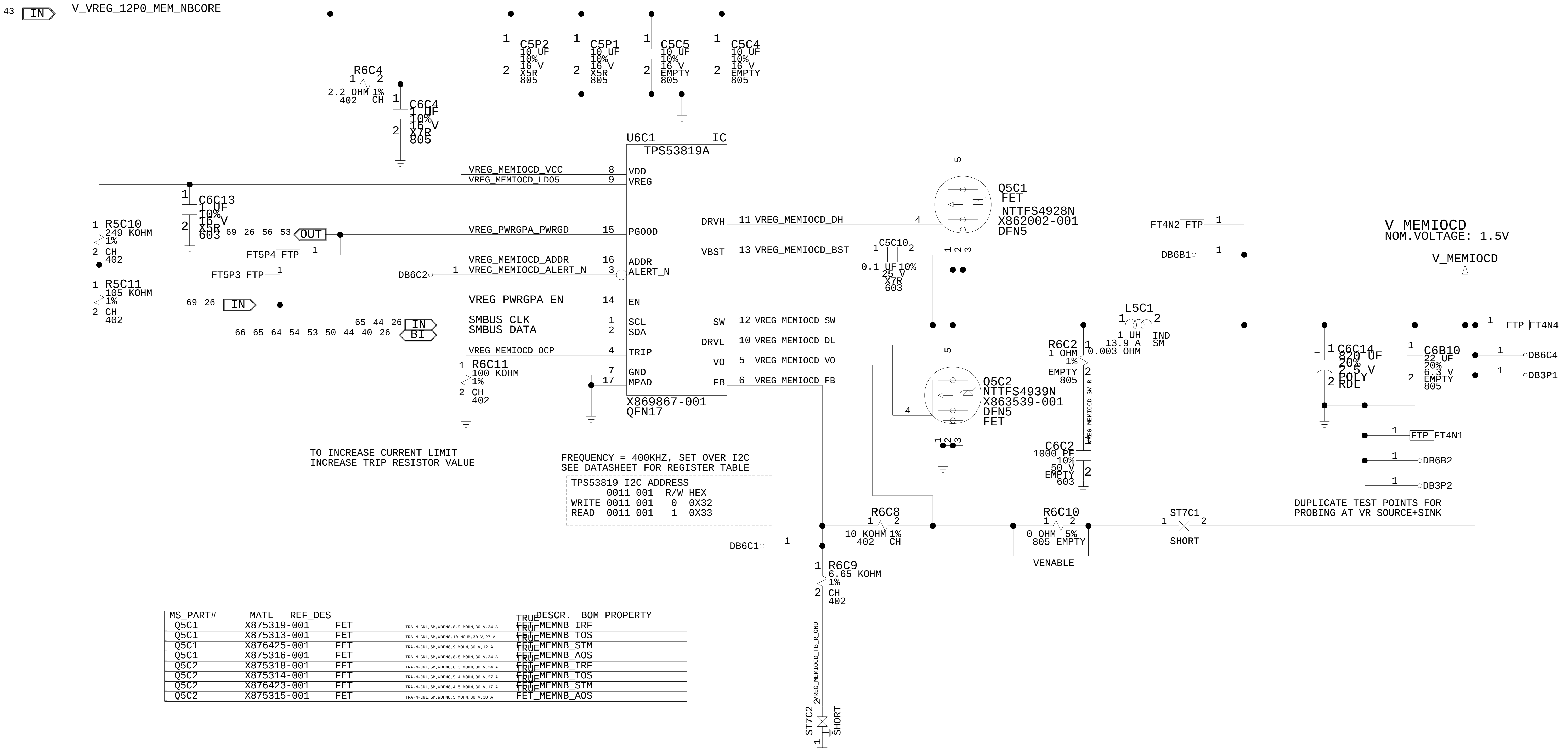



ST5E2 IS OUTPUT VOLTAGE
SENSE RETURN
PLACE NEAR ST1002 AND
CONNECT TO GND PLANE

NOTE: FSEL RES (R1127) FSW
2K 300KHZ
6K 400KHZ
15K 600KHZ

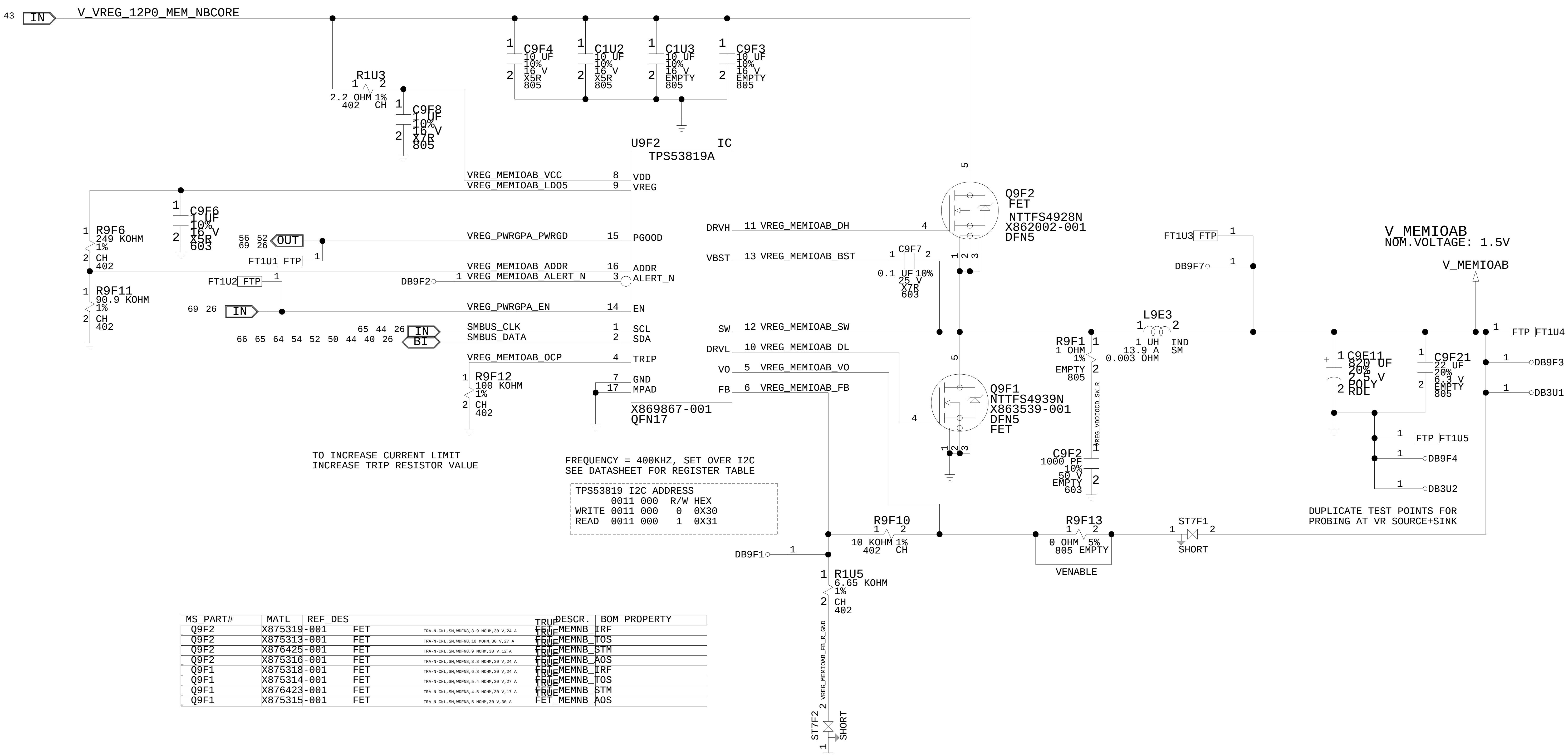
| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| Q5E1 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | TRUE FET_MEMNB_IRF |
| Q5E1 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | TRUE FET_MEMNB_TOS |
| Q5E1 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | TRUE FET_MEMNB_STM |
| Q5E1 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | TRUE FET_MEMNB_AOS |
| Q5E2 | X875318-001 | FET | TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A | TRUE FET_MEMNB_IRF |
| Q5E2 | X875314-001 | FET | TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A | TRUE FET_MEMNB_TOS |
| Q5E2 | X876423-001 | FET | TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A | TRUE FET_MEMNB_STM |
| Q5E2 | X875315-001 | FET | TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A | TRUE FET_MEMNB_AOS |

# VREG, MEMIOCD




| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| Q5C1 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | TRUE FET_MEMNB_IRF | |
| Q5C1 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | TRUE FET_MEMNB_TOS | |
| Q5C1 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | TRUE FET_MEMNB_STM | |
| Q5C1 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | TRUE FET_MEMNB_AOS | |
| Q5C2 | X875318-001 | FET | TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A | TRUE FET_MEMNB_IRF | |
| Q5C2 | X875314-001 | FET | TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A | TRUE FET_MEMNB_TOS | |
| Q5C2 | X876423-001 | FET | TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A | TRUE FET_MEMNB_STM | |
| Q5C2 | X875315-001 | FET | TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A | FET_MEMNB_AOS | |

# VREG, MEMIOAB




TO INCREASE CURRENT LIMIT
INCREASE TRIP RESISTOR VALUE

FREQUENCY = 400KHZ, SET OVER I2C
SEE DATASHEET FOR REGISTER TABLE

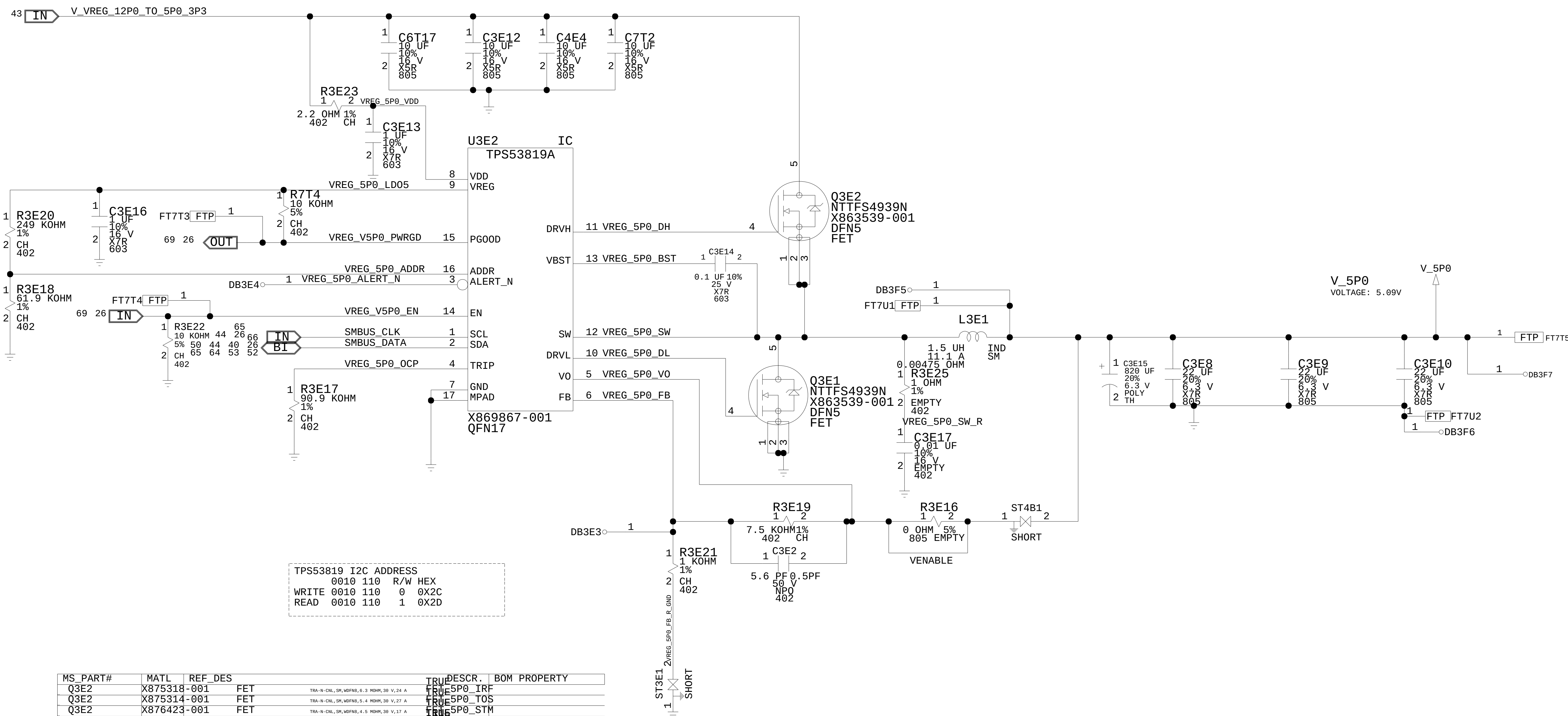
TPS53819 I2C ADDRESS

| | | R/W | HEX |
|---|---|---|---|
| | 0011 000 | R/W | HEX |
| WRITE | 0011 000 | 0 | 0X30 |
| READ | 0011 000 | 1 | 0X31 |

DUPLICATE TEST POINTS FOR
PROBING AT VR SOURCE+SINK

| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| Q9F2 | X875319-001 | FET | TRA-N-CNL,SM,WDFN8,8.9 MOHM,30 V,24 A | TRUE | FET_MEMNB_IRF |
| Q9F2 | X875313-001 | FET | TRA-N-CNL,SM,WDFN8,10 MOHM,30 V,27 A | TRUE | FET_MEMNB_TOS |
| Q9F2 | X876425-001 | FET | TRA-N-CNL,SM,WDFN8,9 MOHM,30 V,12 A | TRUE | FET_MEMNB_STM |
| Q9F2 | X875316-001 | FET | TRA-N-CNL,SM,WDFN8,8.8 MOHM,30 V,24 A | TRUE | FET_MEMNB_AOS |
| Q9F1 | X875318-001 | FET | TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A | TRUE | FET_MEMNB_IRF |
| Q9F1 | X875314-001 | FET | TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A | TRUE | FET_MEMNB_TOS |
| Q9F1 | X876423-001 | FET | TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A | TRUE | FET_MEMNB_STM |
| Q9F1 | X875315-001 | FET | TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A | TRUE | FET_MEMNB_AOS |

# VREGS, V5P0

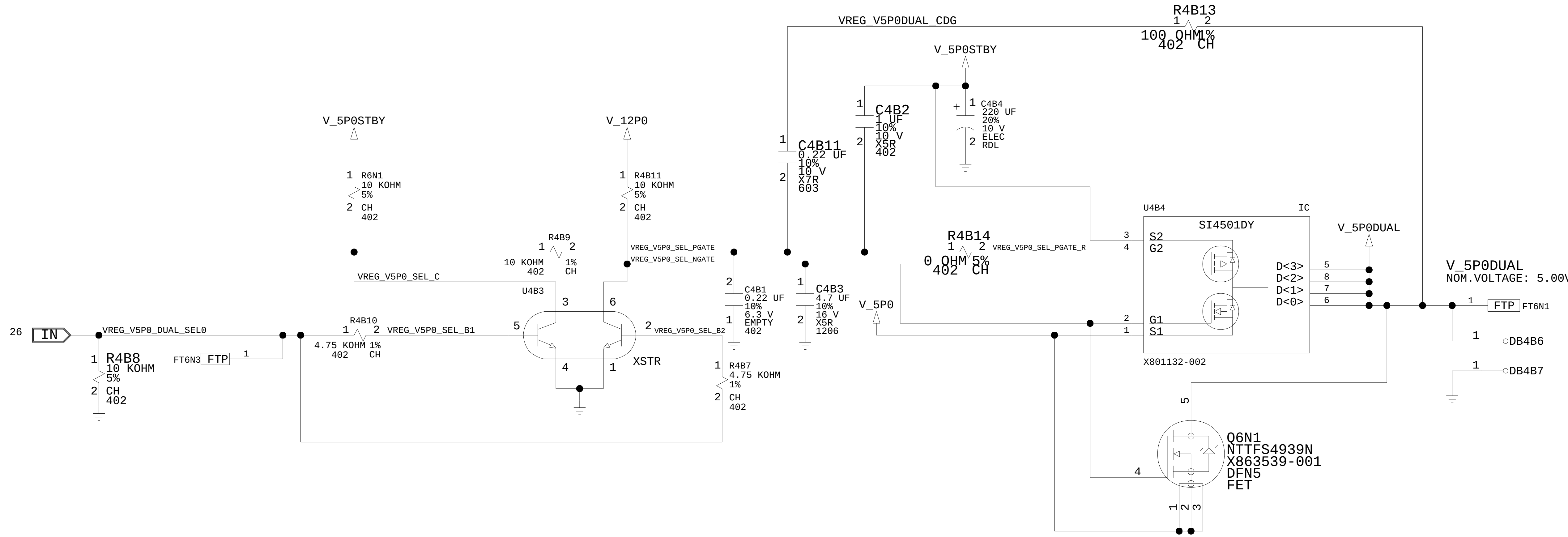



| MS_PART# | MATL | REF_DES | | DESCR. | BOM PROPERTY |
|---|---|---|---|---|---|
| Q3E2 | X875318-001 | FET | TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A | FET_5P0_IRF | |
| Q3E2 | X875314-001 | FET | TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A | FET_5P0_TOS | |
| Q3E2 | X876423-001 | FET | TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A | FET_5P0_STM | |
| Q3E2 | X875315-001 | FET | TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A | FET_5P0_AOS | |
| Q3E1 | X875318-001 | FET | TRA-N-CNL,SM,WDFN8,6.3 MOHM,30 V,24 A | FET_5P0_IRF | |
| Q3E1 | X875314-001 | FET | TRA-N-CNL,SM,WDFN8,5.4 MOHM,30 V,27 A | FET_5P0_TOS | |
| Q3E1 | X876423-001 | FET | TRA-N-CNL,SM,WDFN8,4.5 MOHM,30 V,17 A | FET_5P0_STM | |
| Q3E1 | X875315-001 | FET | TRA-N-CNL,SM,WDFN8,5 MOHM,30 V,30 A | FET_5P0_AOS | |

[PAGE_TITLE=VREGS, V5P0]

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 54/72 | 54/72 | M | 1.0 |

# VREGS, V5P0 DUAL




| VREG_V5P0_DUAL_SEL0 | VREG_5P0_SEL NGATE/PGATE | V_5P0DUAL |
|---|---|---|
| HIGH | LOW | V_5P0STBY |
| LOW | HIGH | V_5P0 |

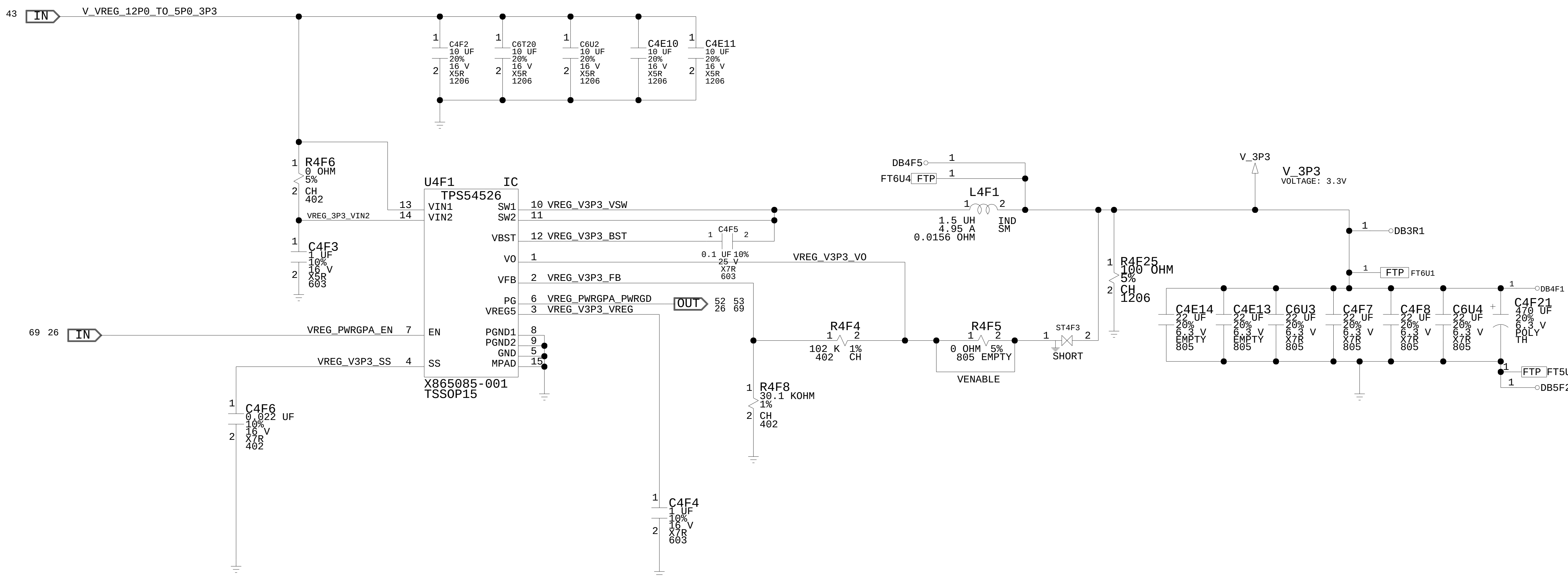
[PAGE_TITLE=VREGS, V5P0 DUAL]

DRAWING
Tue Jun 18 16:42:27 2013



| PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|
| GREYBULL_RETAIL | 55/72 | 55/72 | M | 1.0 |

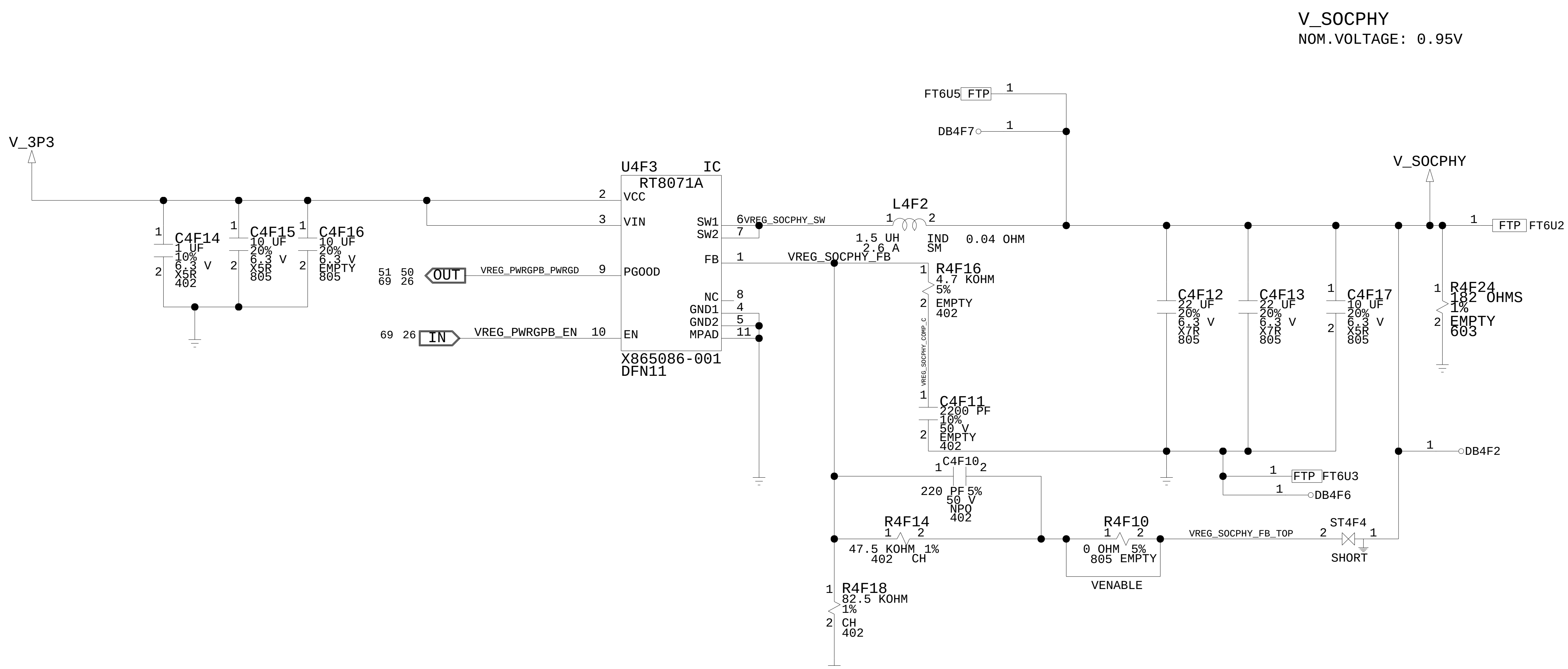
# VREGS, V3P3




[PAGE_TITLE=VREGS, V3P3]

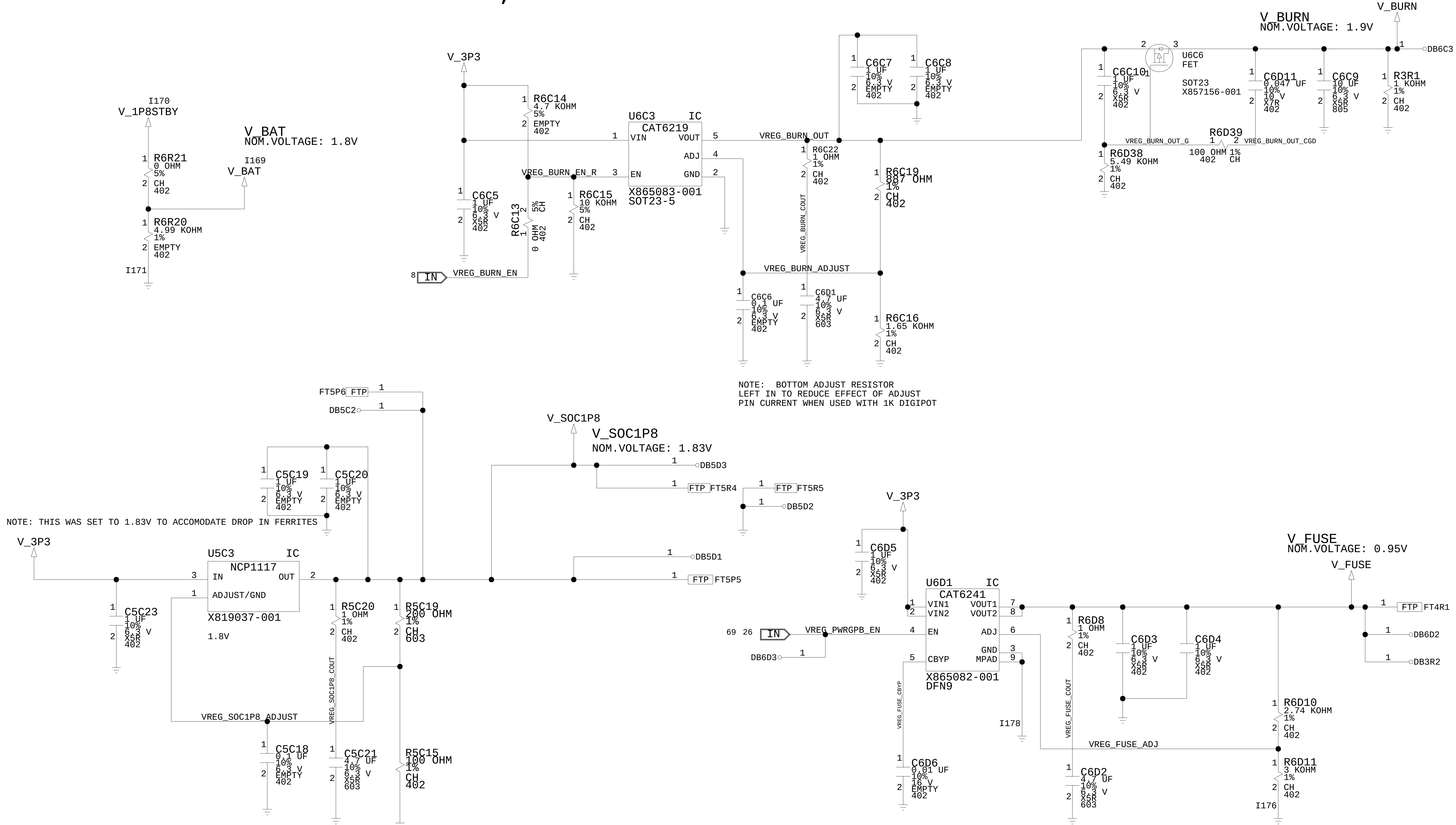
| | | | | | |
|---|---|---|---|---|---|
| MICROSOFT CONFIDENTIAL | PROJECT NAME GREYBULL_RETAIL | PAGE 56/72 | CSA PAGE 56/72 | FAB M | REV 1.0 |

# VREGS, VSOCPHY



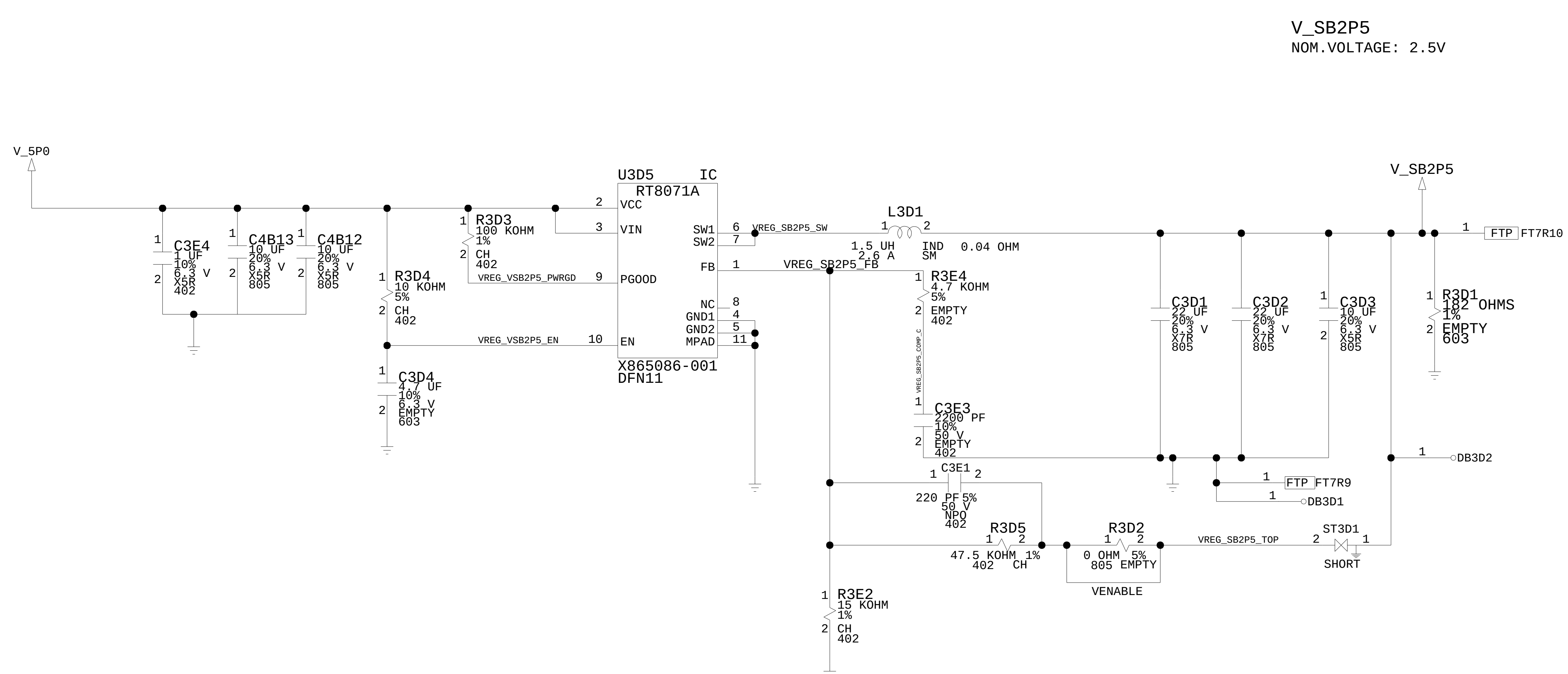

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 57/72 | 57/72 | M | 1.0 |

# VREGS, VSOCPLL + VBURN + VFUSE + VBAT



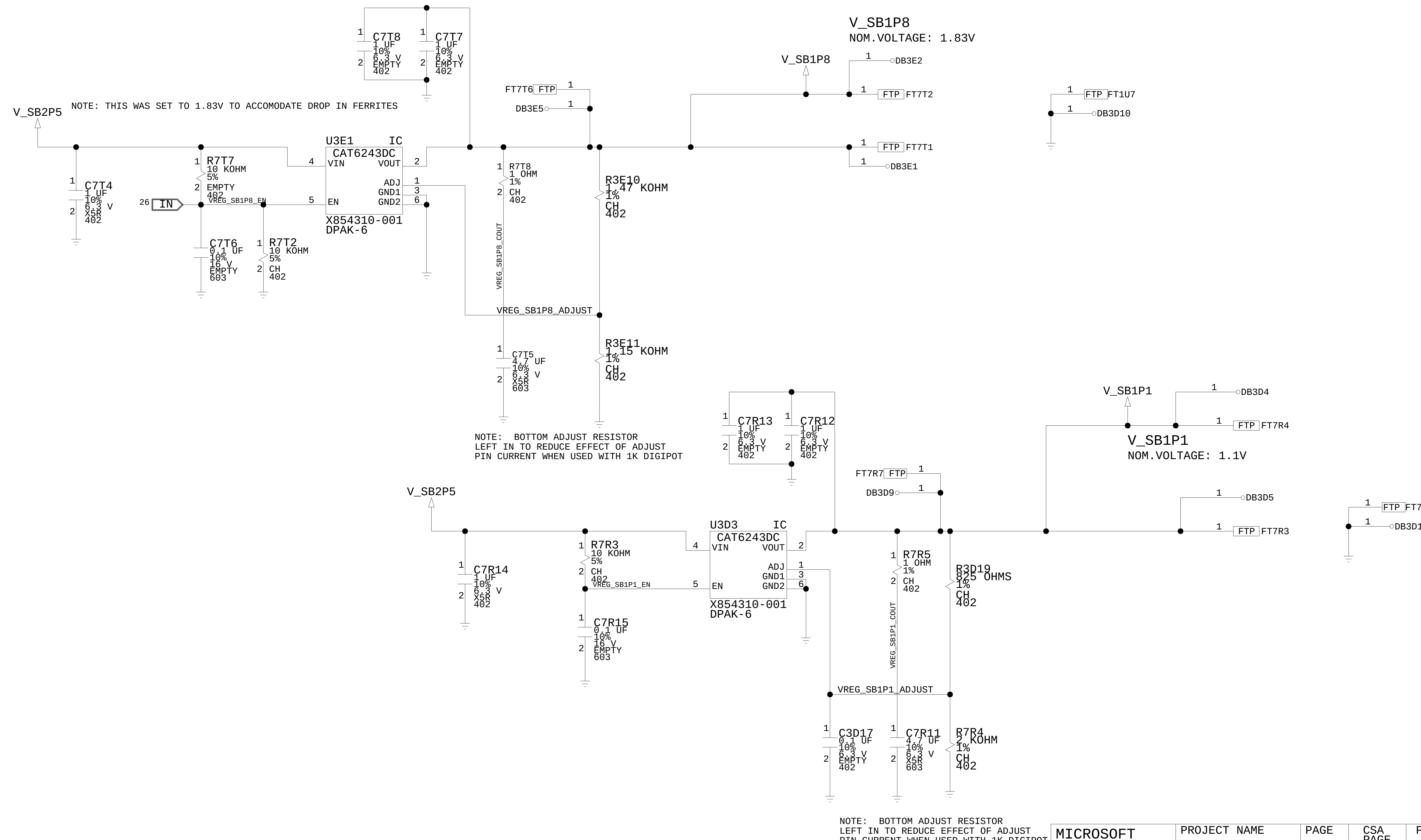

| MICROSOFT CONFIDENTIAL | PROJECT NAME GREYBULL_RETAIL | PAGE 58/72 | CSA PAGE 58/72 | FAB M | REV 1.0 |
|---|---|---|---|---|---|

# VREGS, VSB2P5




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 59/72 | 59/72 | M | 1.0 |

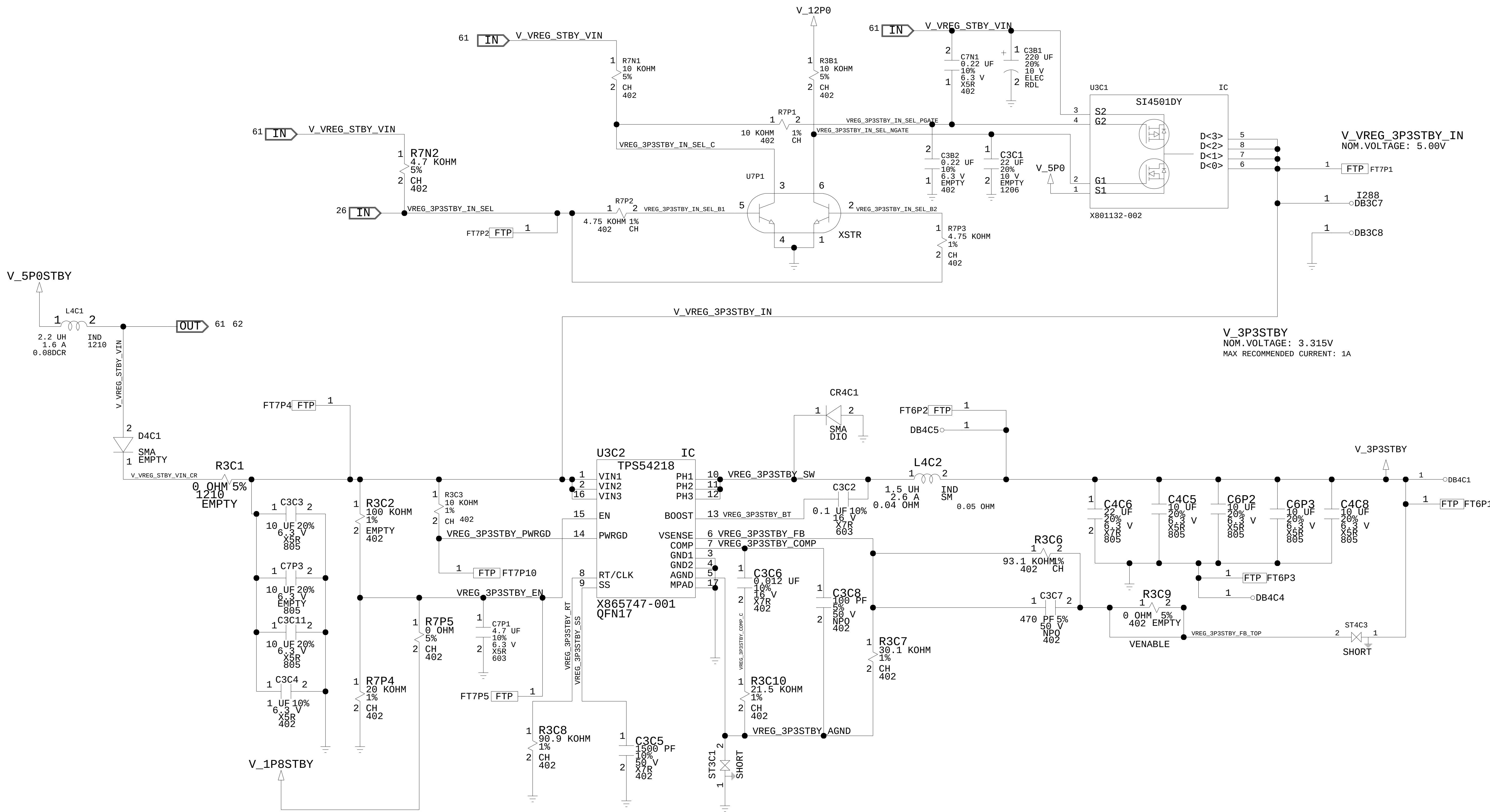
# VREGS, VSB1P8PLL + VSB1P8IO + VSBCORE + VSB1P1PLL




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 60/72 | 60/72 | M | 1.0 |

# VREGS,STANDBY SWITCHERS 3P3

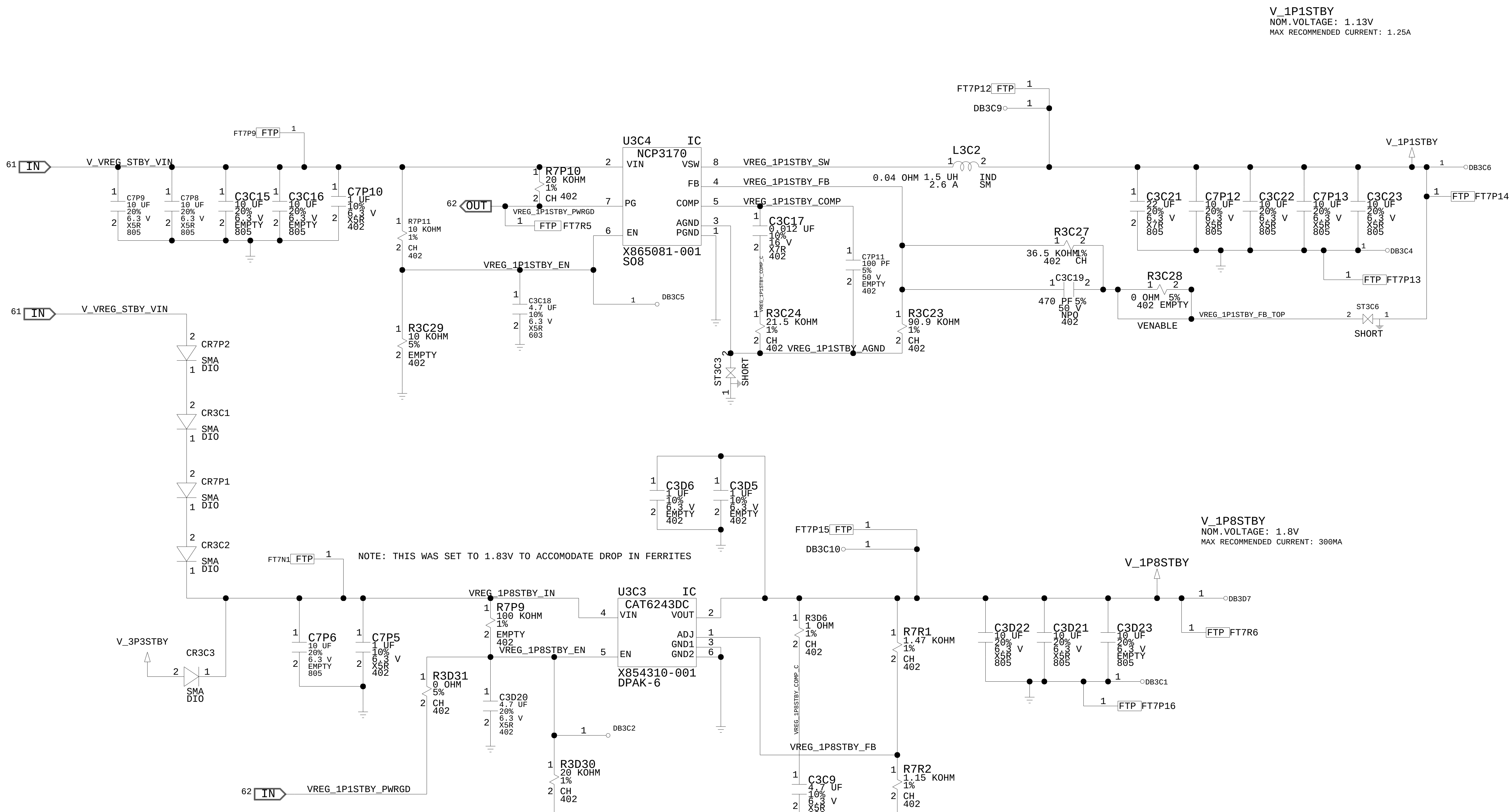
| VREG_3P3STBY_IN_SEL | VREG_3P3STBY_IN_SEL NGATE/PGATE | VREG_3P3STBY_VCC |
|---|---|---|
| HIGH | LOW | V_5P0STBY |
| LOW | HIGH | V_5P0 |




[PAGE_TITLE=VREGS, STANDBY SWITCHERS]

R5A5 SHOULD BE 182KOHM 0402, WAITING ON COMPONENT TB ADDED IN TC

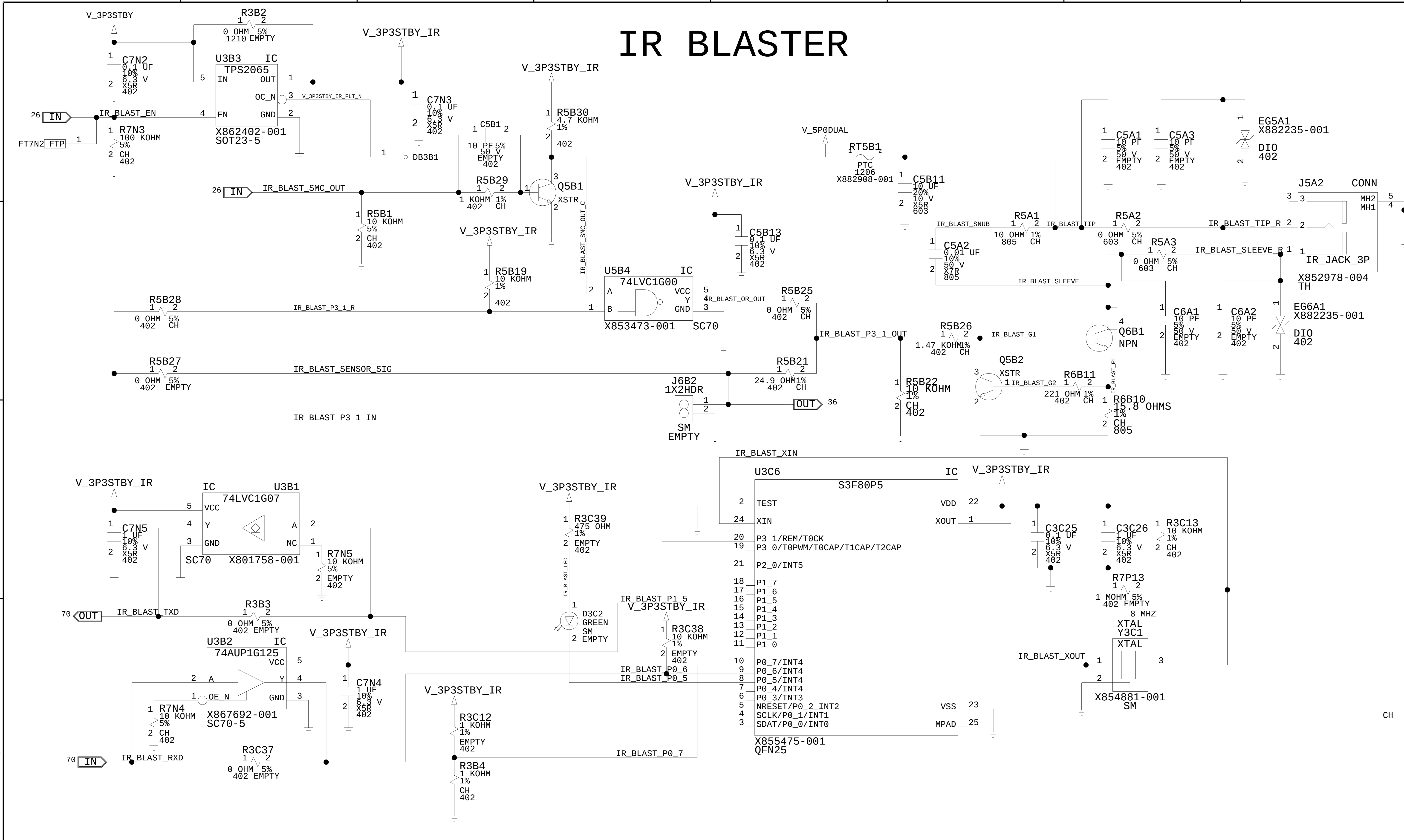
# VREGS,STANDBY SWITCHERS 1P1 + 1P8




[PAGE_TITLE=VREGS, STANDBY SWITCHERS]

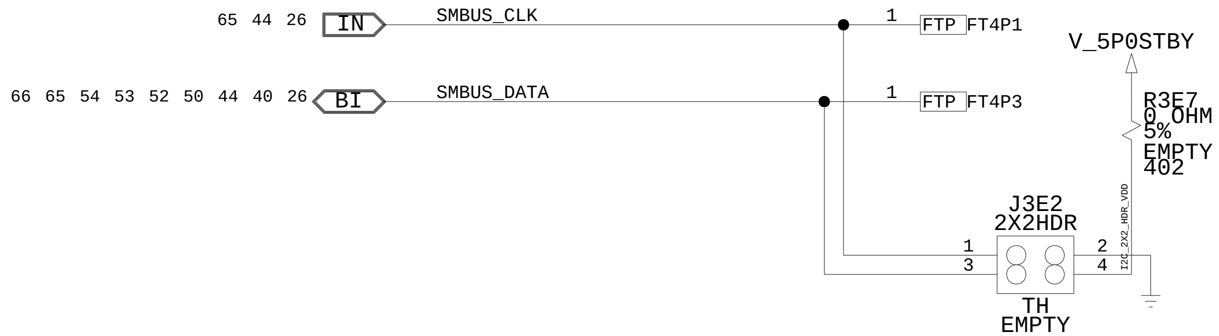
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 62/72 | 62/72 | M | 1.0 |

# IR BLASTER



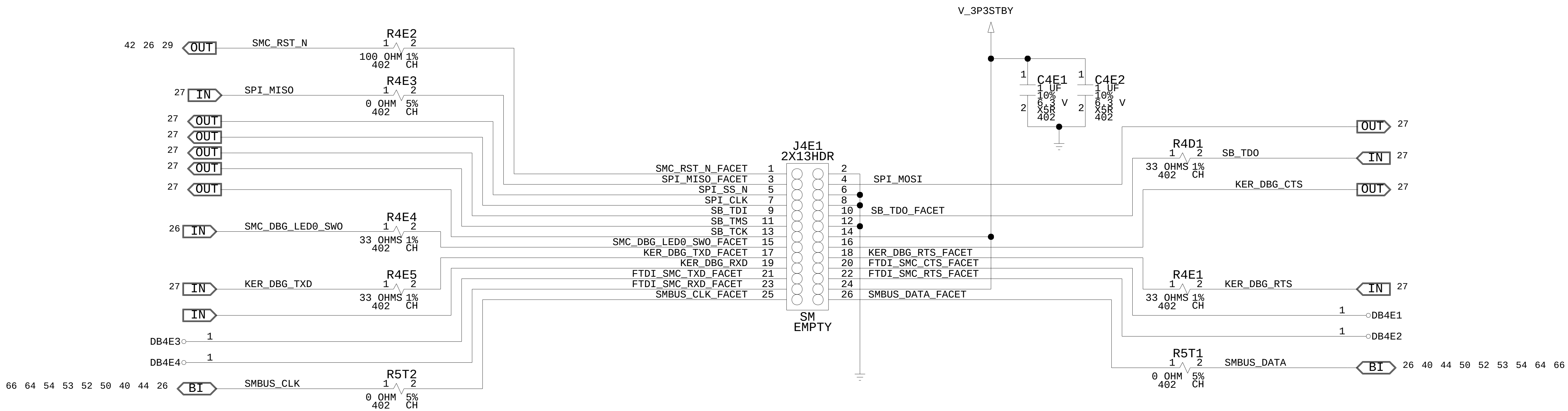

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 63/72 | 63/72 | M | 1.0 |

# I2C



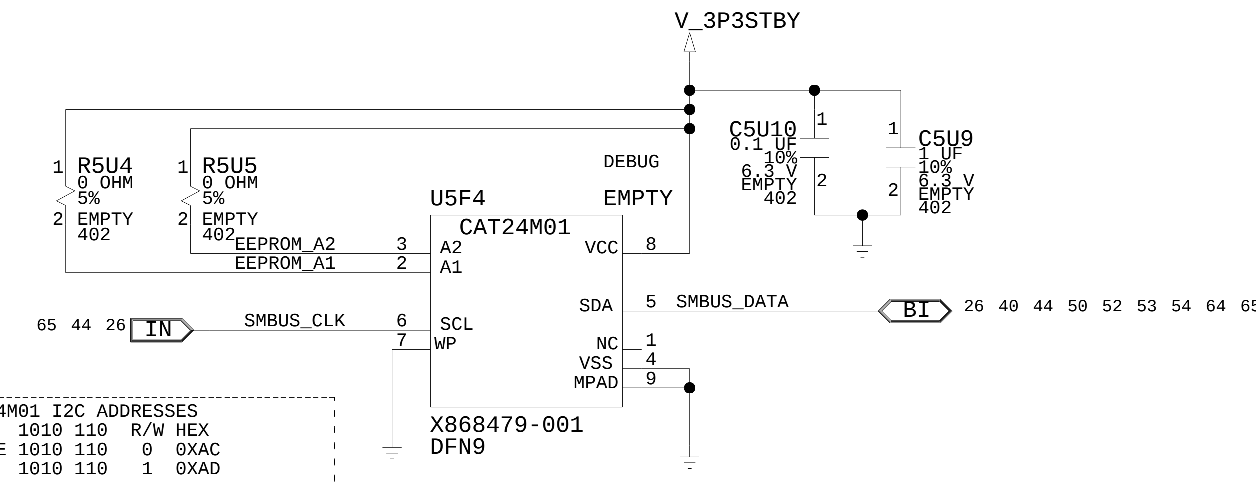

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 64/72 | 64/72 | M | 1.0 |

# FACET, FTDI




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 65/72 | 65/72 | M | 1.0 |

# FACET, FTDI+MISC



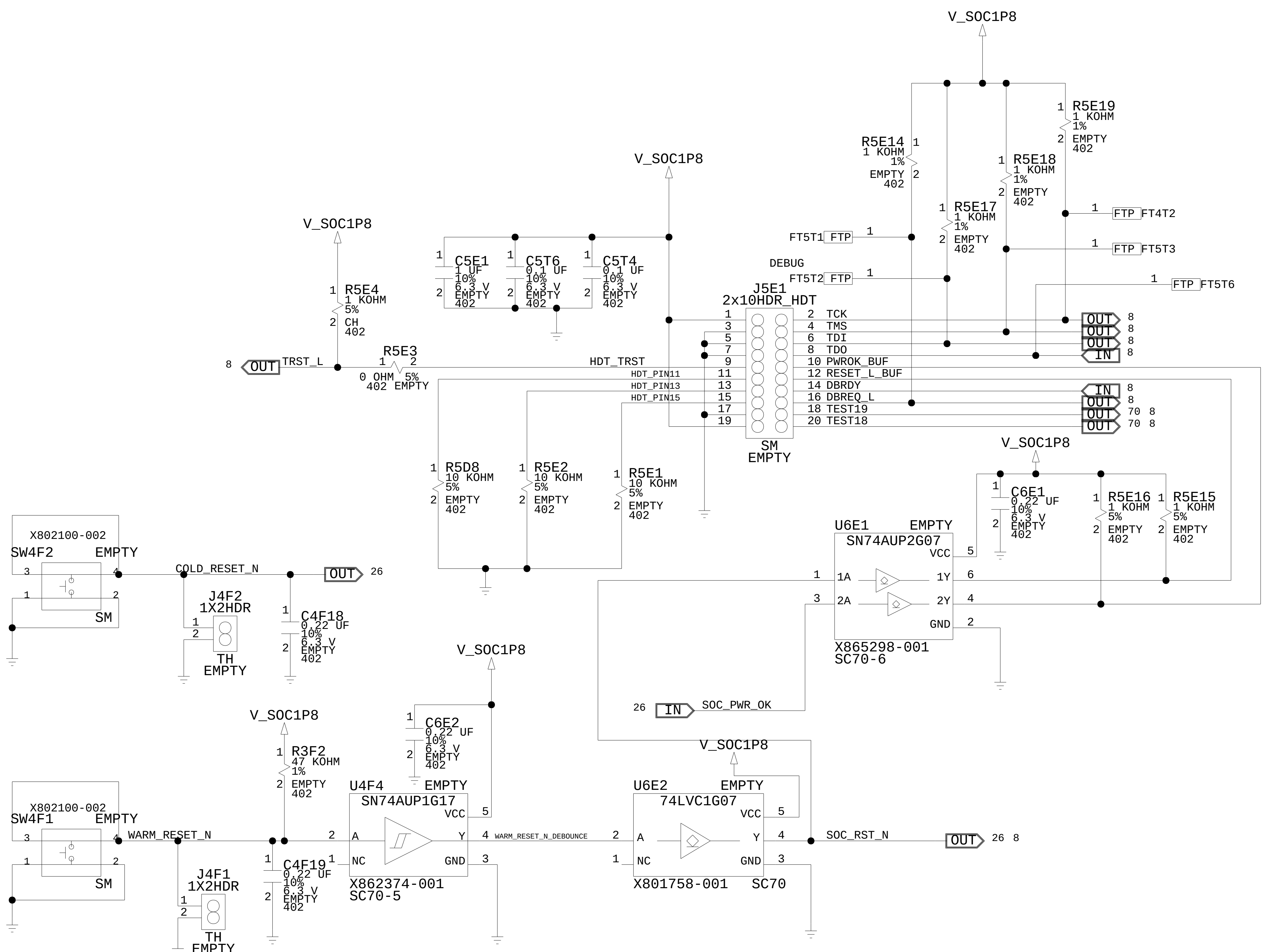

CAT24M01 I2C ADDRESSES

| | | | | |
|---|---|---|---|---|
| | 1010 | 110 | R/W | HEX |
| WRITE | 1010 | 110 | 0 | 0XAC |
| READ | 1010 | 110 | 1 | 0XAD |
| | 1010 | 111 | R/W | HEX |
| WRITE | 1010 | 111 | 0 | 0XAE |
| READ | 1010 | 111 | 1 | 0XAF |

# CONN, SWITCHES

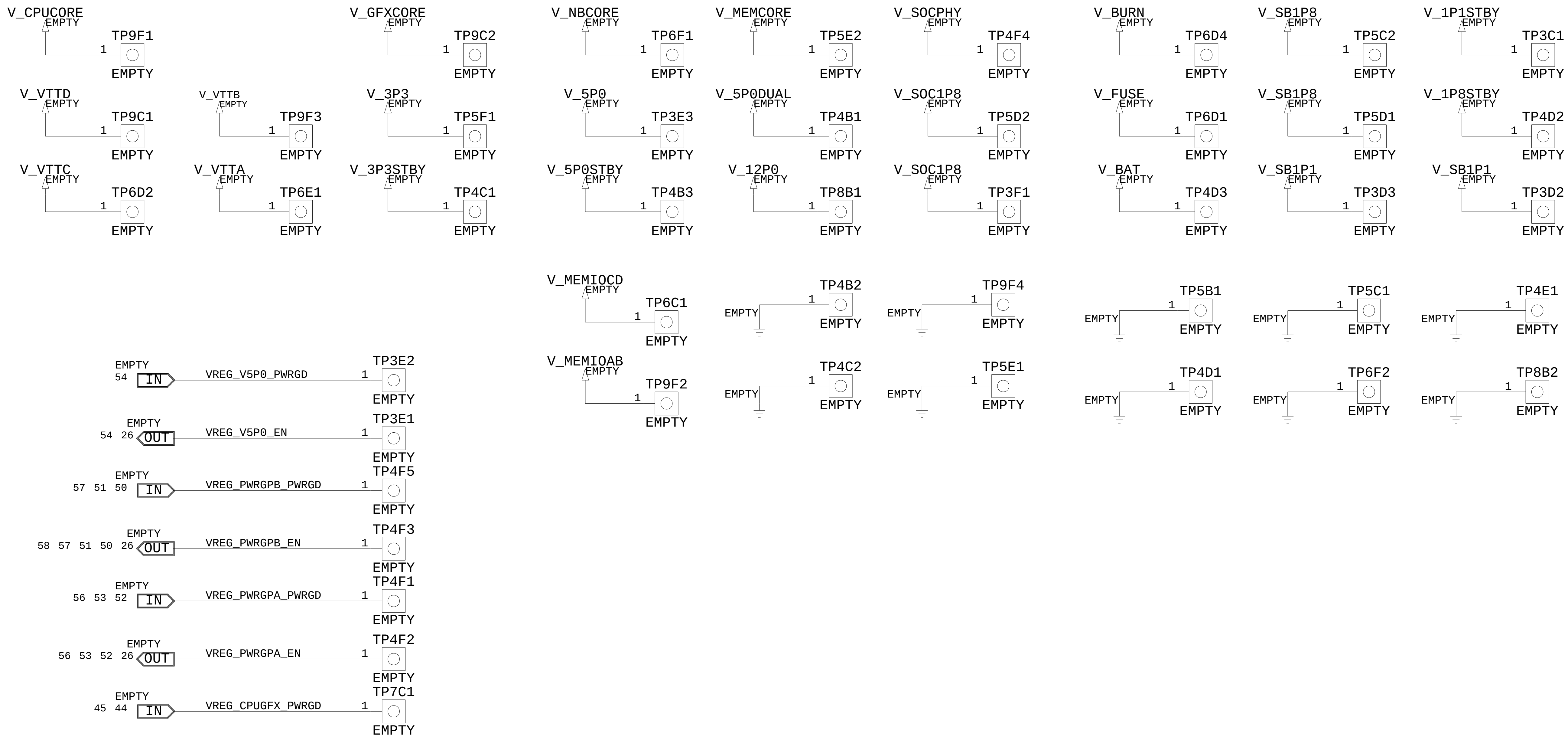

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 67/72 | 67/72 | M | 1.0 |

# CONN, HDT



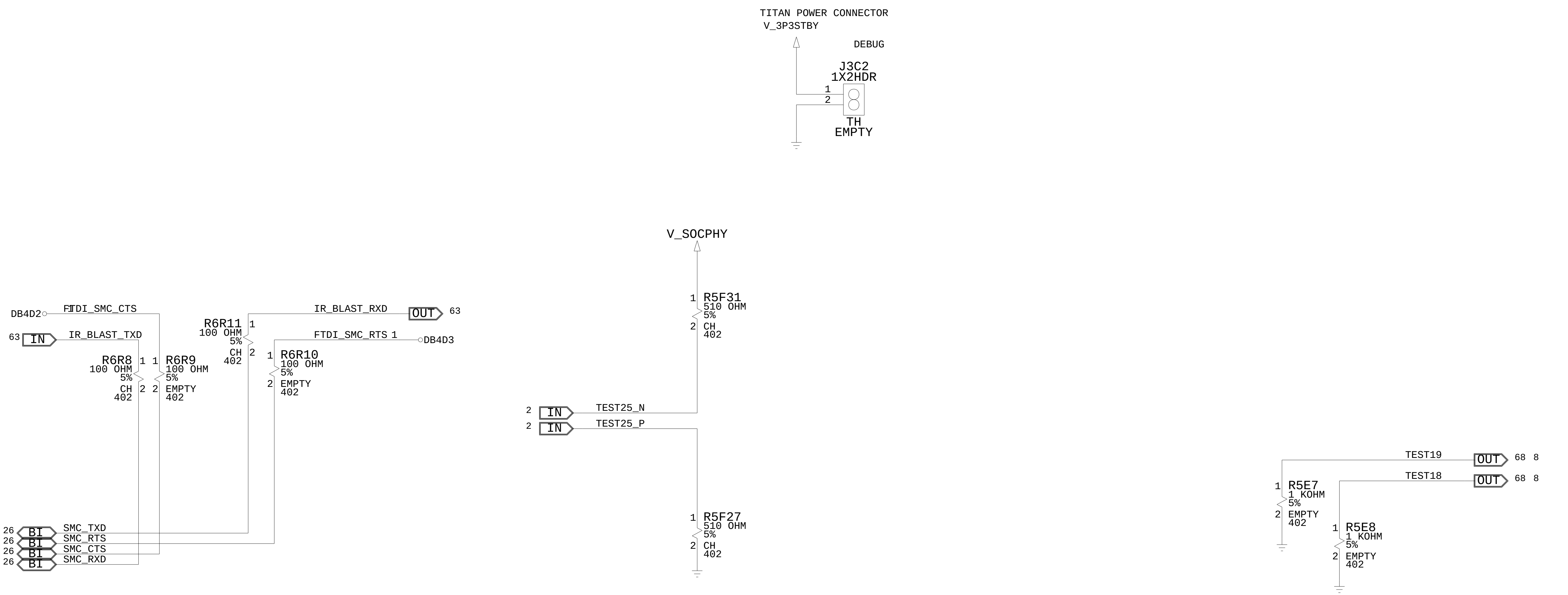

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 68/72 | 68/72 | M | 1.0 |

# DEBUG, VR HEADERS AND TEST POINTS



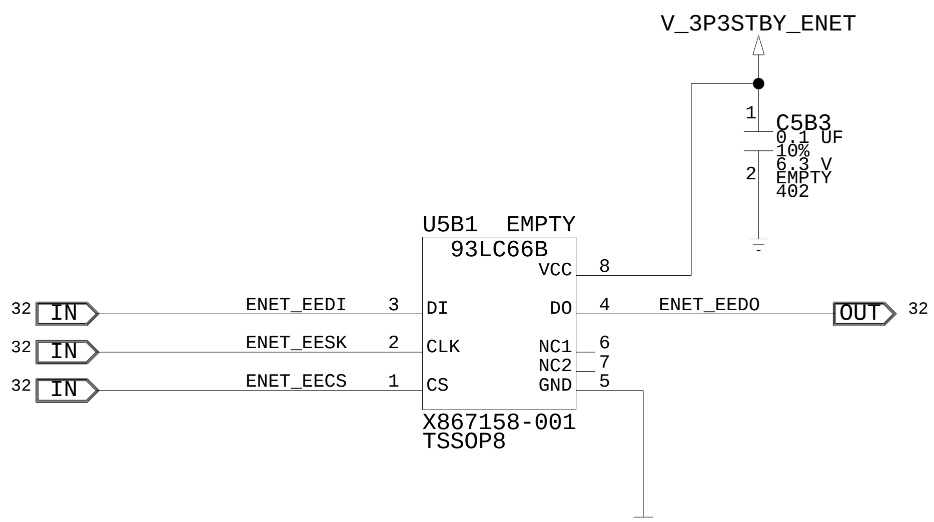

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 69/72 | 69/72 | M | 1.0 |

# DEBUG, CONNECTORS




| MICROSOFT CONFIDENTIAL | PROJECT NAME GREYBULL_RETAIL | PAGE 70/72 | CSA PAGE 70/72 | REV 1.0 |
|---|---|---|---|---|

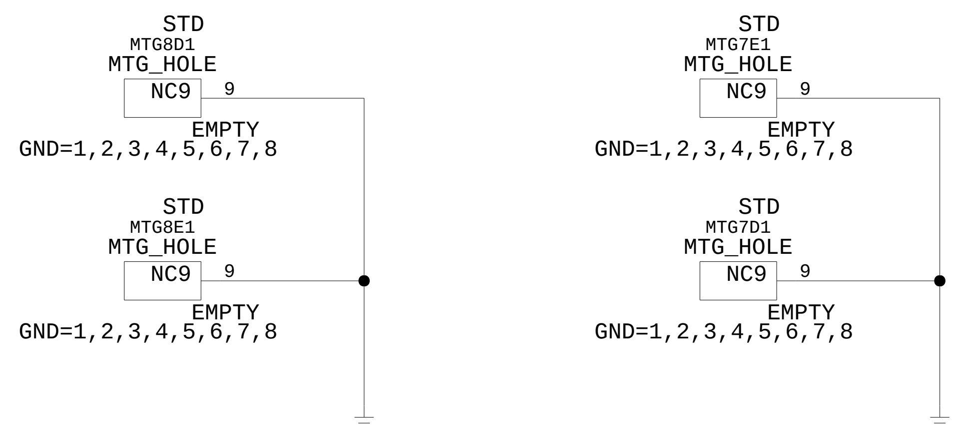
# DEBUG, ENET EEPROM + MISC



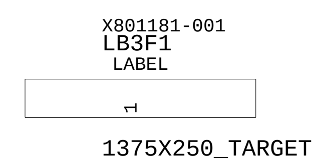

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 71/72 | 71/72 | M | 1.0 |

# LABELS AND MOUNTING

HEAT SINK MOUNTING HOLES




INTELLIGENT SERIAL NUMBER TARGET




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | REV |
|---|---|---|---|---|---|
| | GREYBULL_RETAIL | 72/72 | 72/72 | M | 1.0 |